Abstract Data Type

stack

preliminaries

- queues allow adding of items only at the rear and removal of items only at the front
- some problems require limited access to the data structure, much like the queue
  - adding of items only at the back/rear
  - removal of items only at the back/rear



preliminaries

- some real-world scenarios with similar behavior
  - stack of plates in a buffet
  - laying off of employees
  - dead-end when trapped in a maze
  - barbecue skewers/banana cue
  - tennis balls in a container
  - one-car wide garages



preliminaries



preliminaries



preliminaries



preliminaries



stack

- last-in-first-out (LIFO)
  - adds and removes items at the top
- operations
  - push
  - pop
  - top

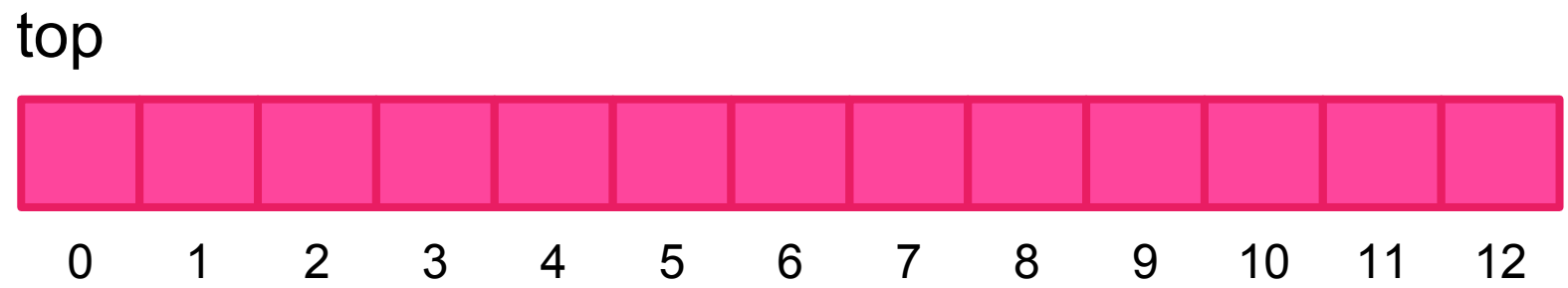


stack

- implementations
  - array
  - vector
  - linked list
    - singly-linked
    - doubly-linked

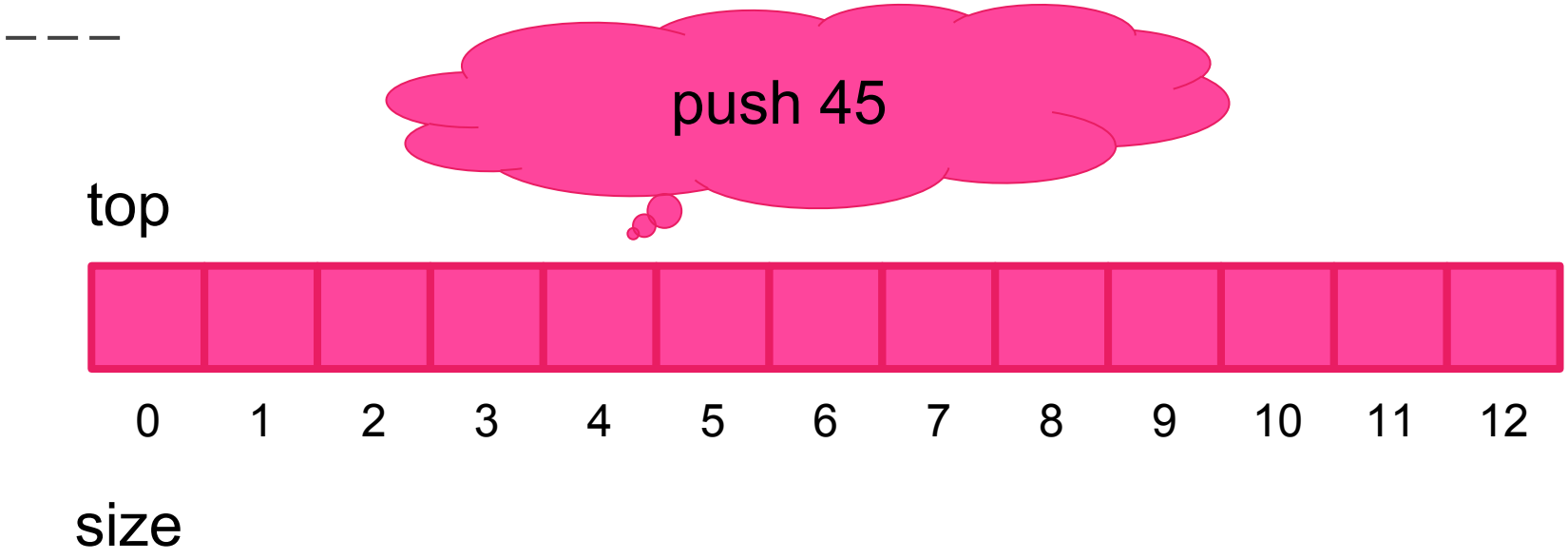


array implementation

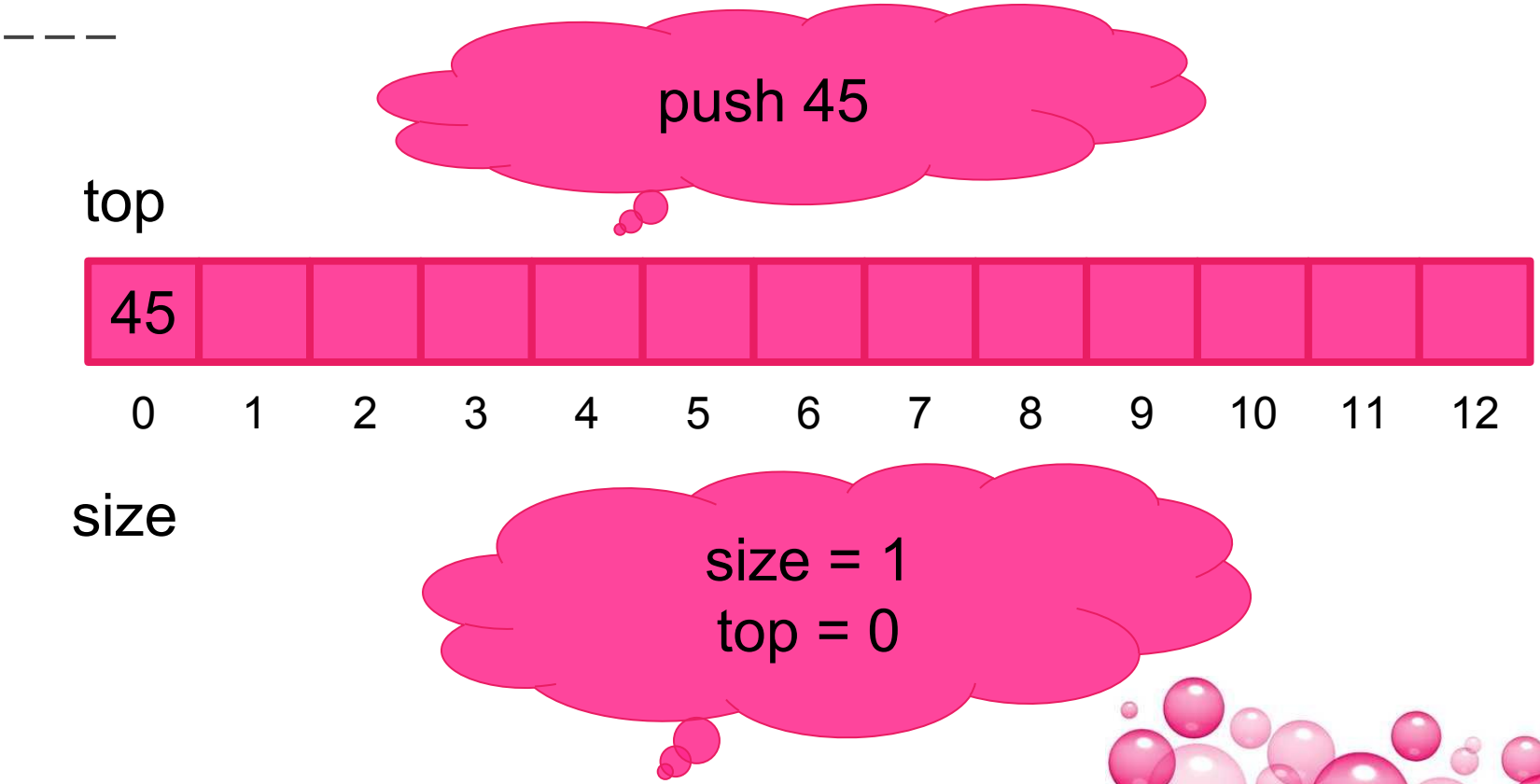


initial values of
top and size?

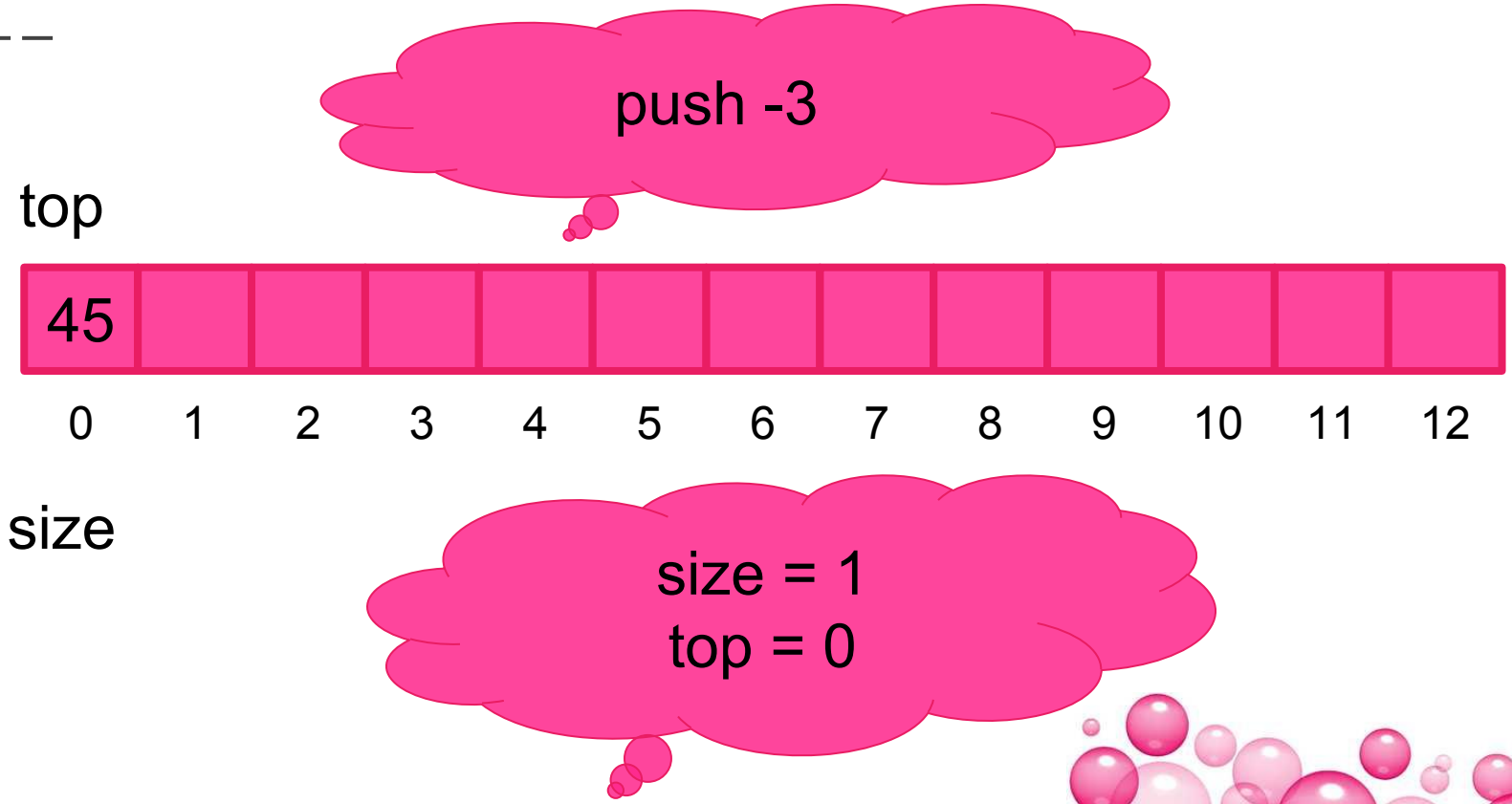
array implementation - push (empty case)



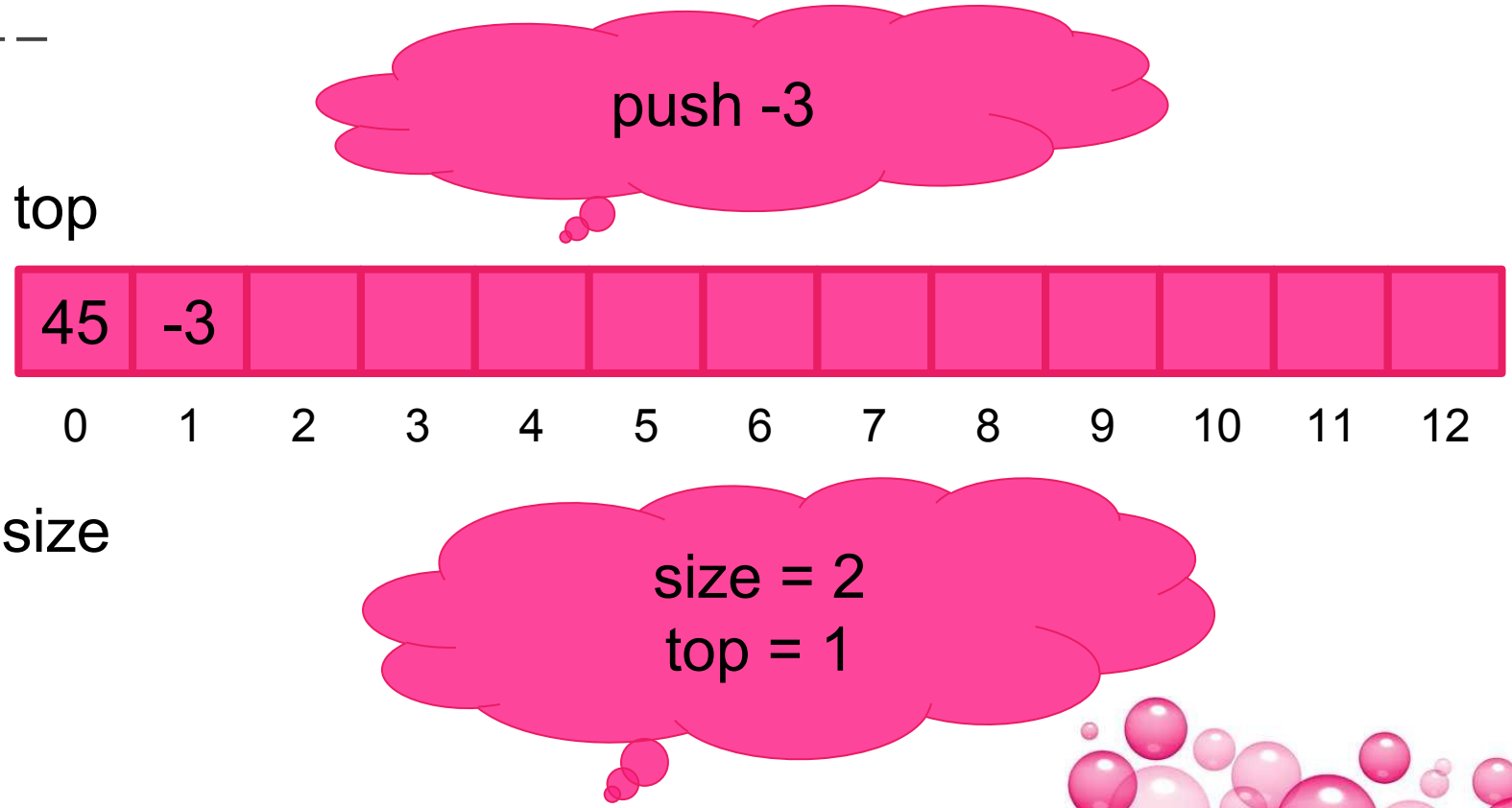
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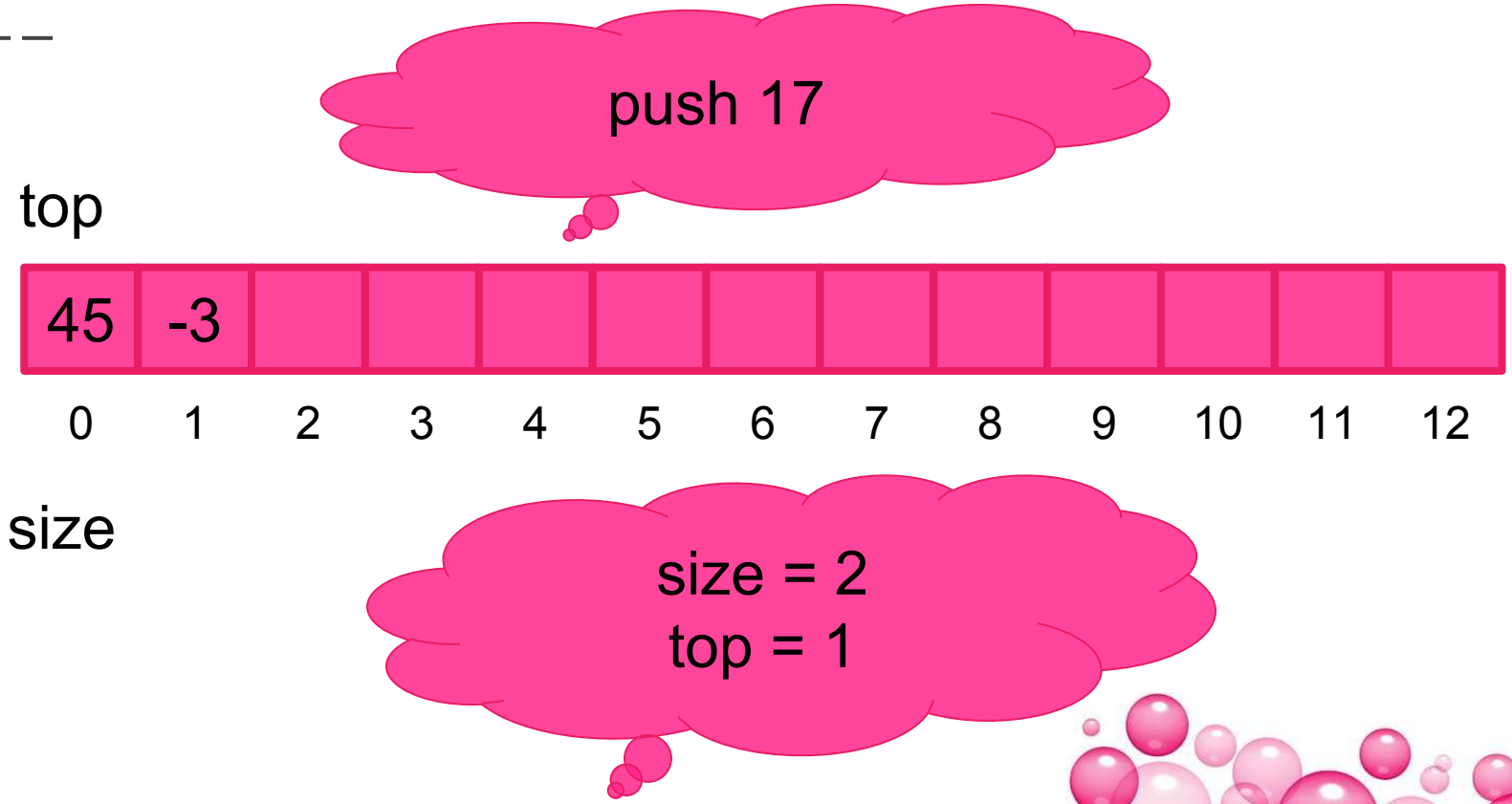
array implementation - push



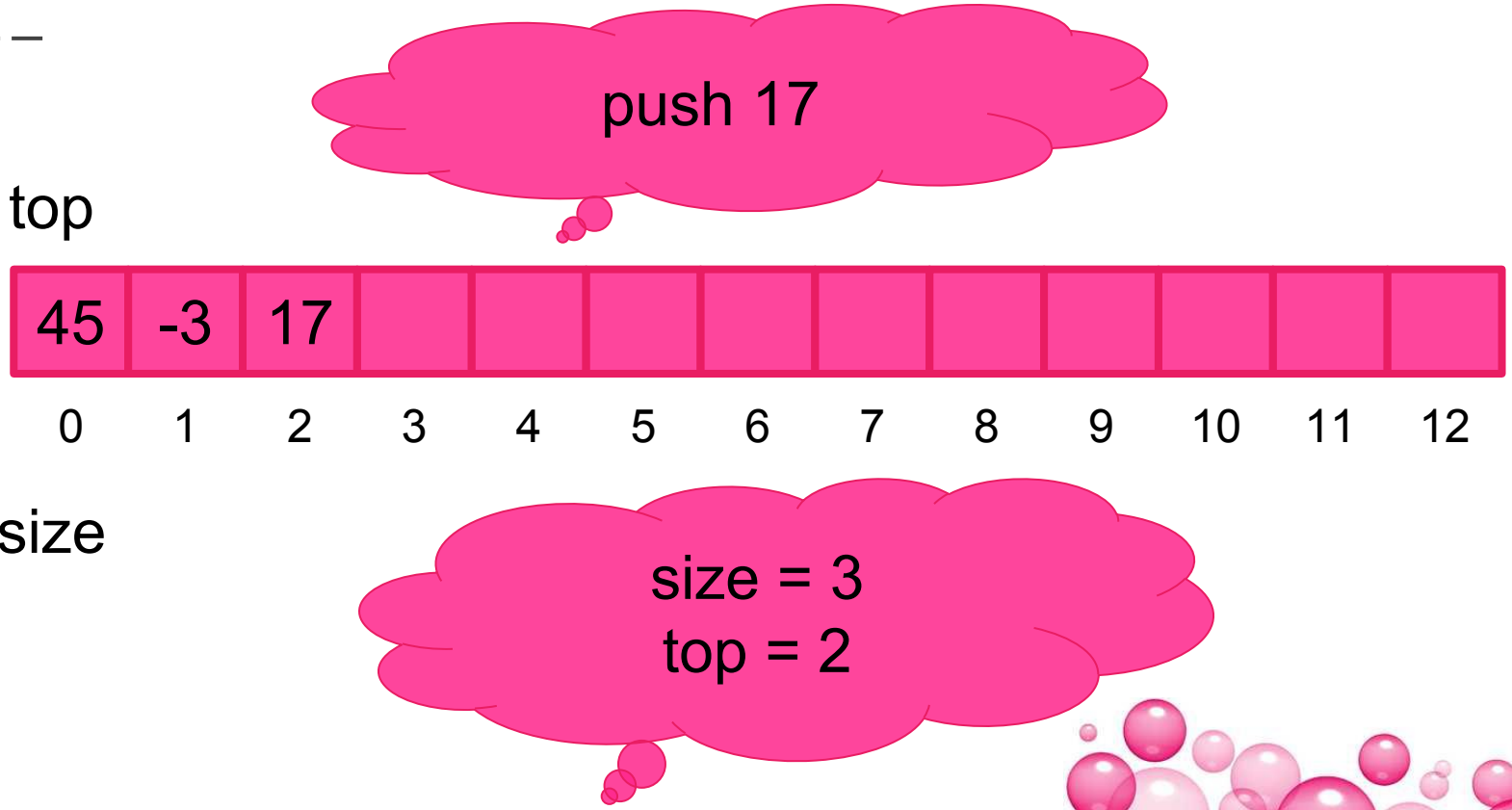
array implementation - push



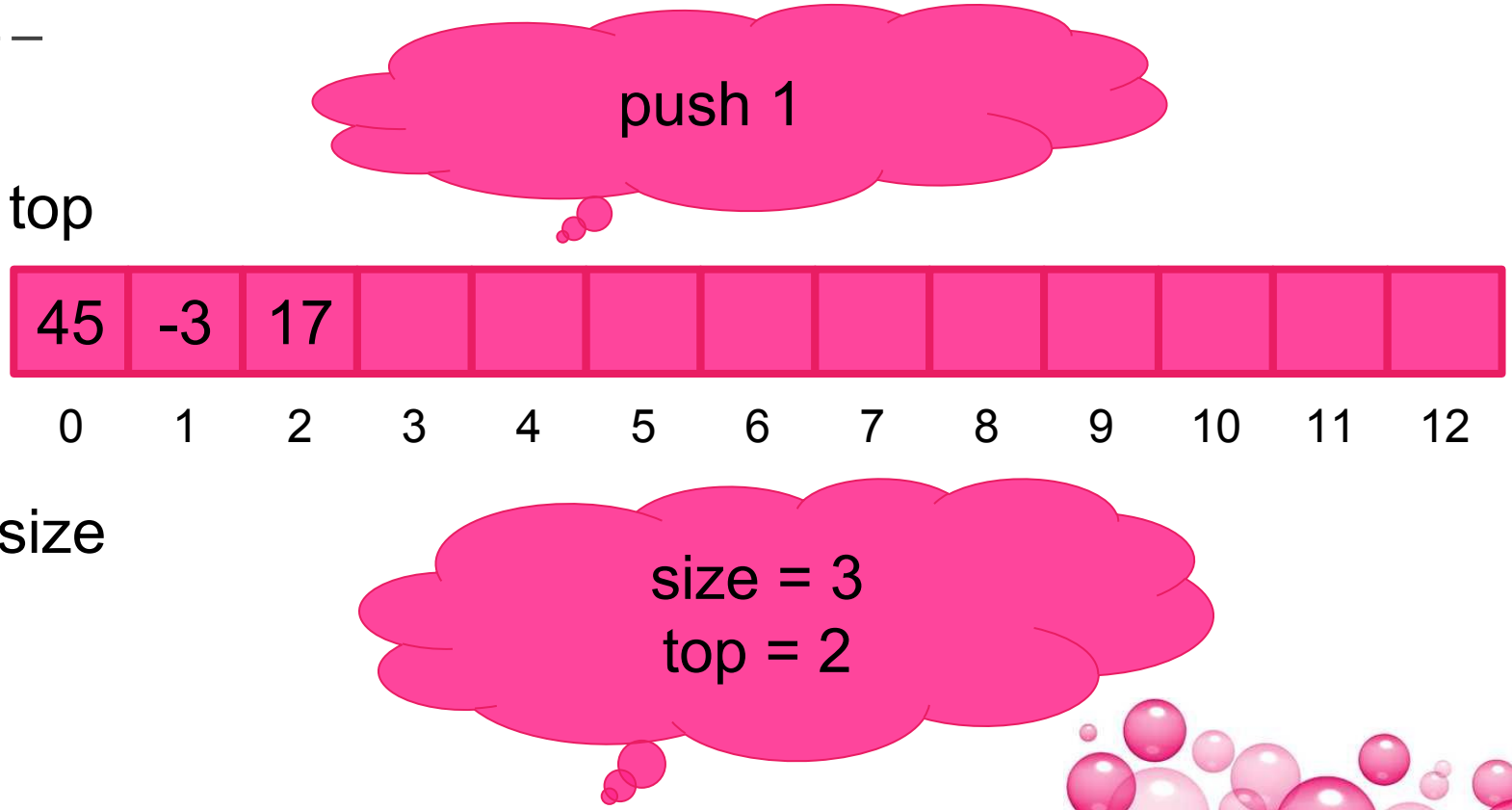
array implementation - push



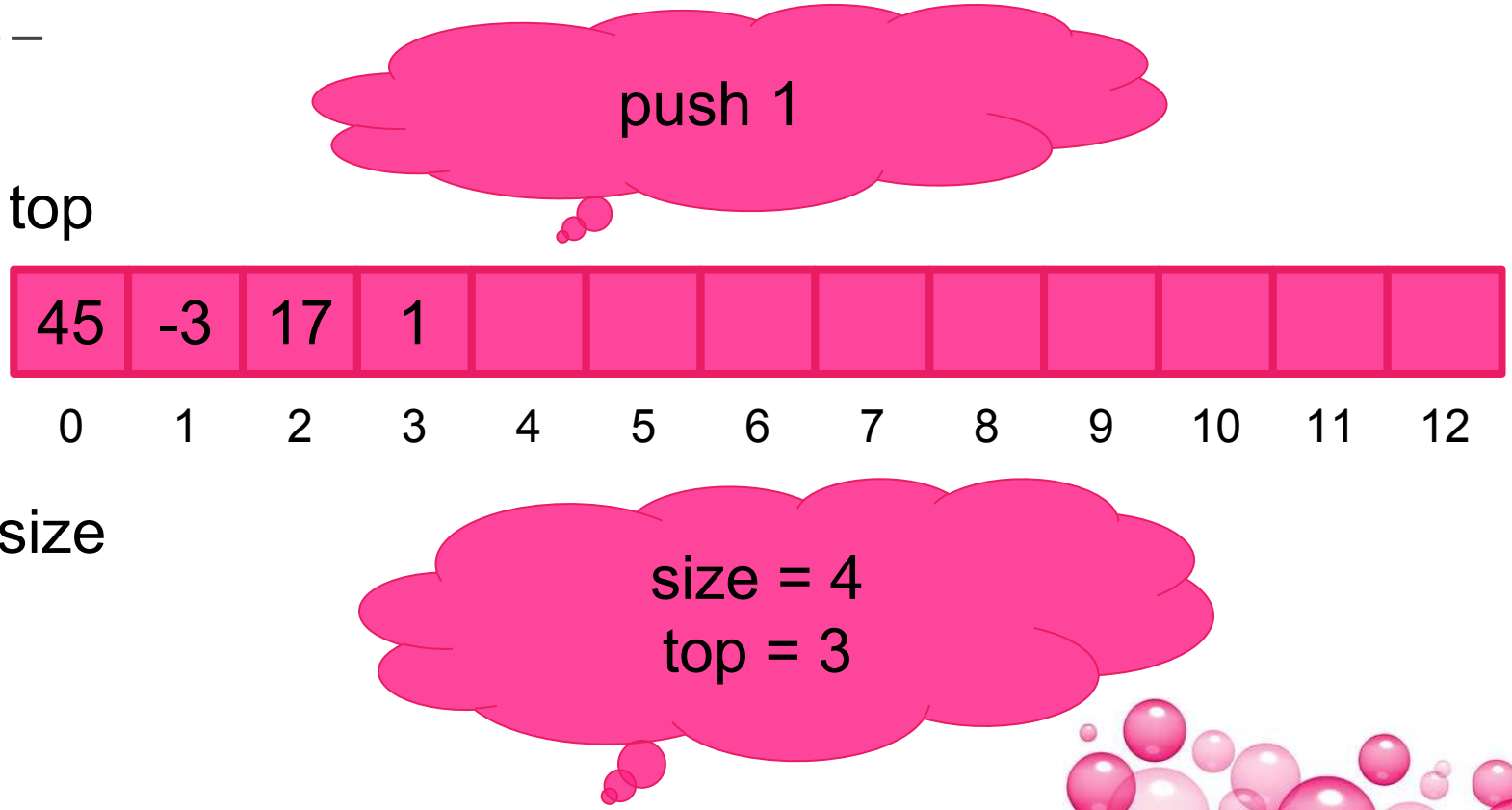
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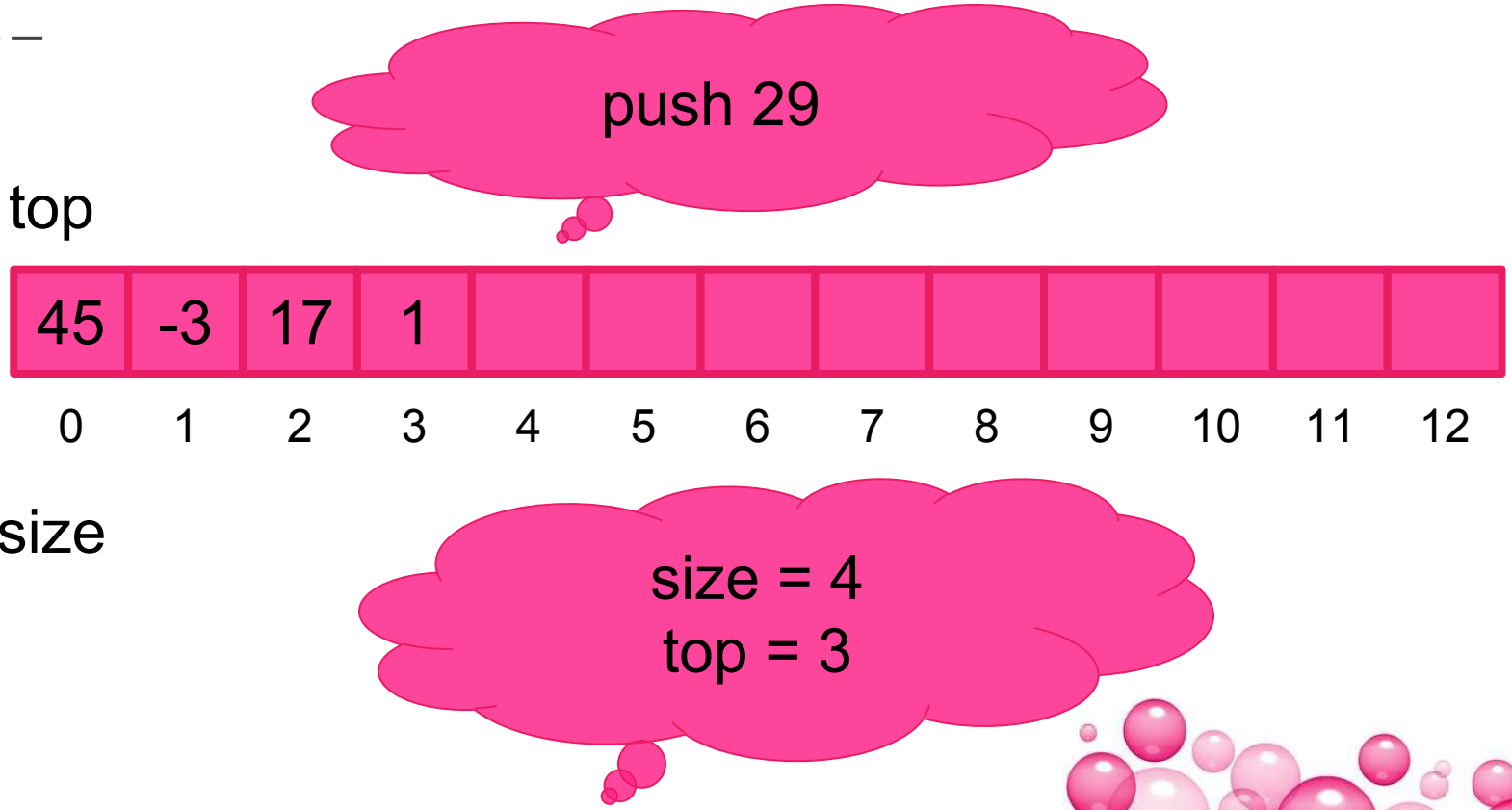
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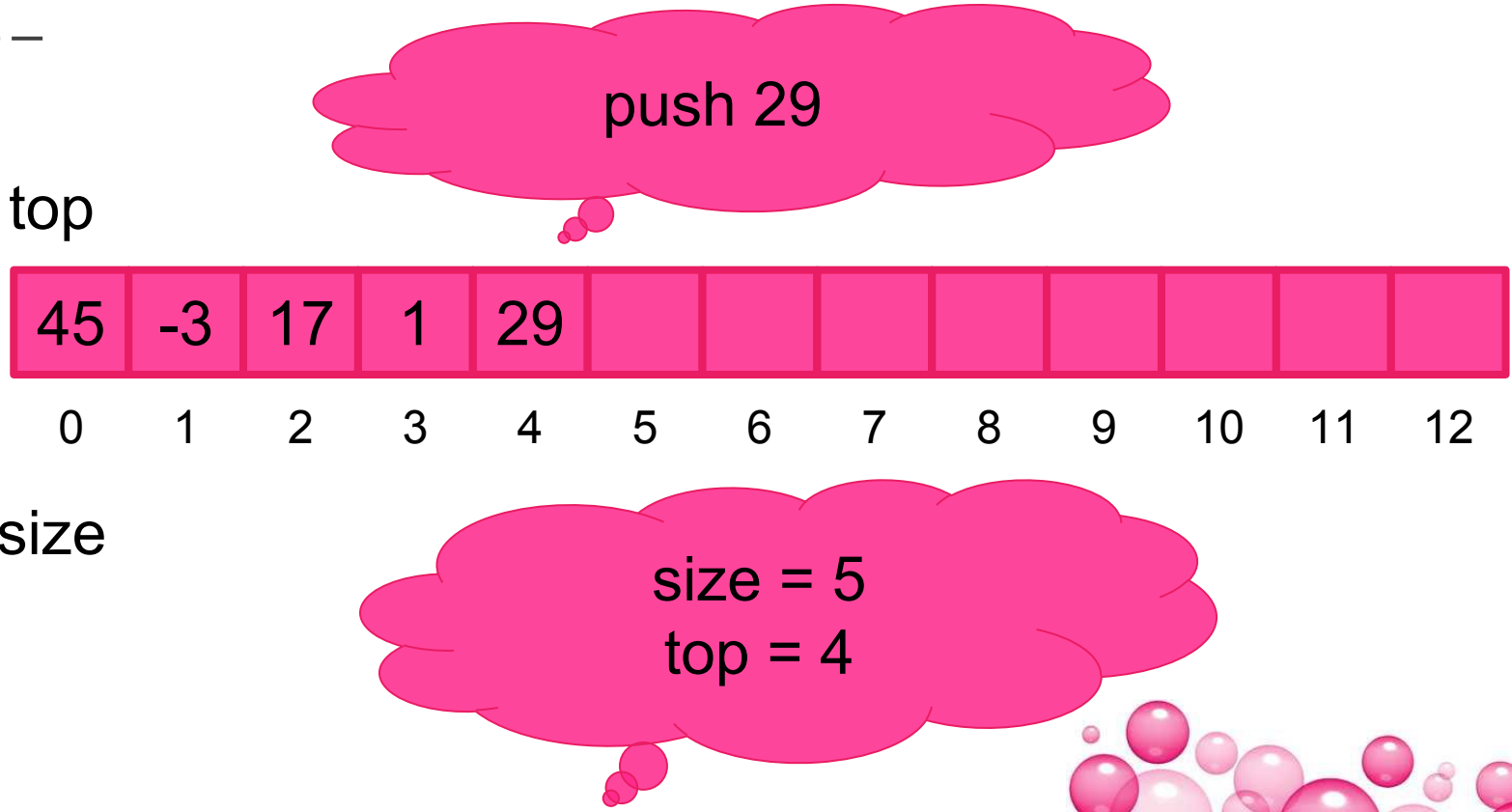
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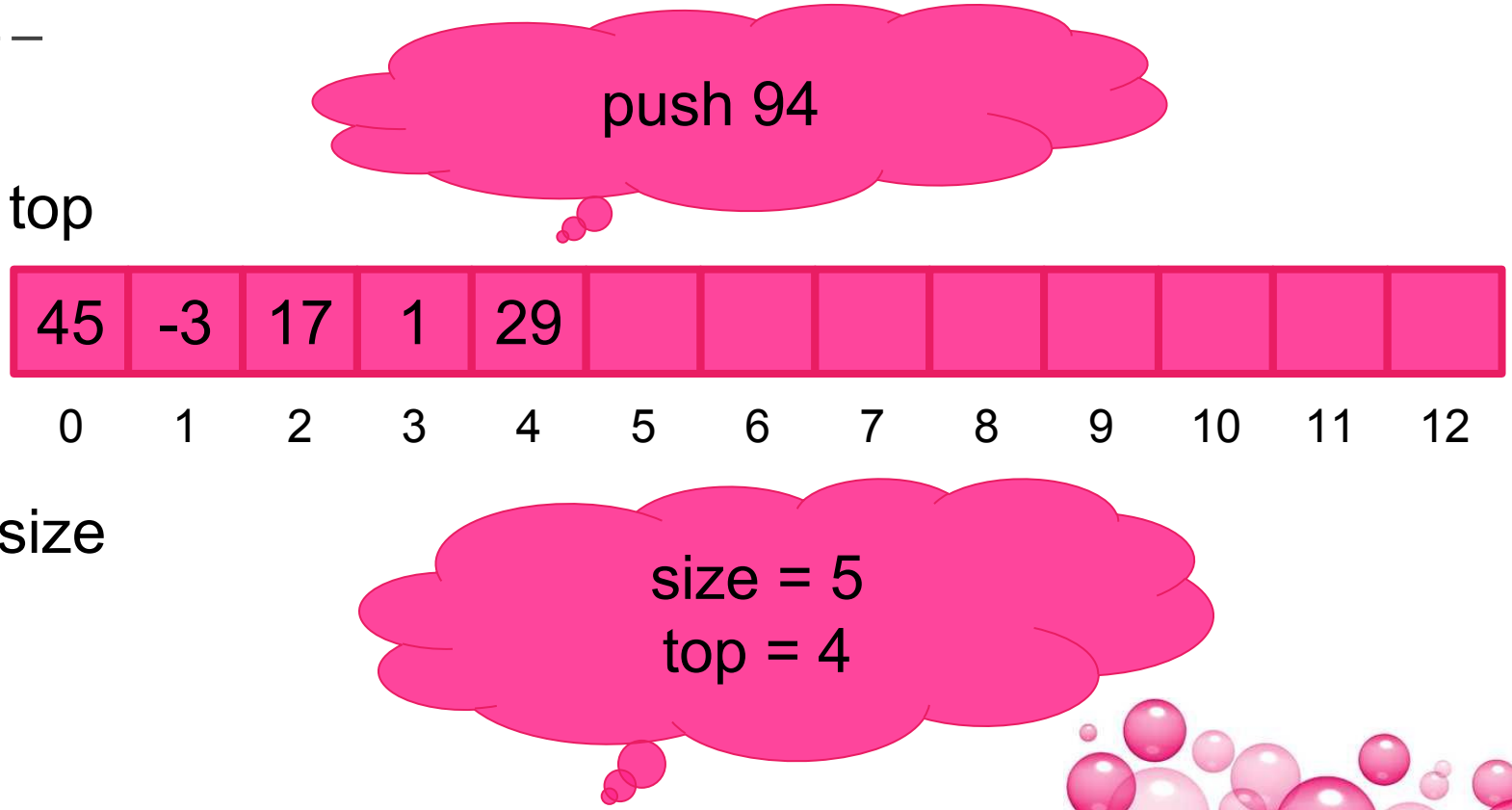
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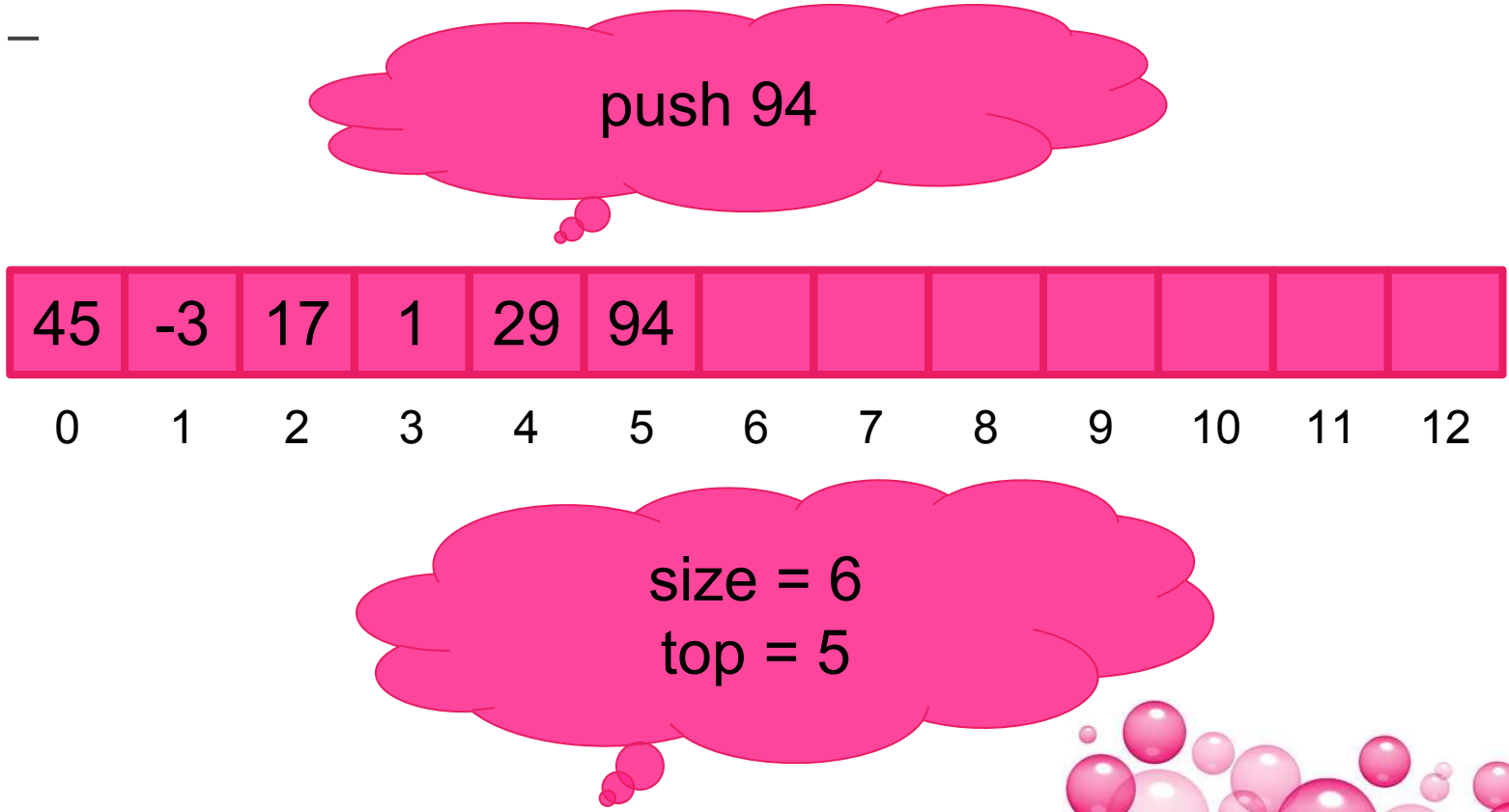
array implementation - push



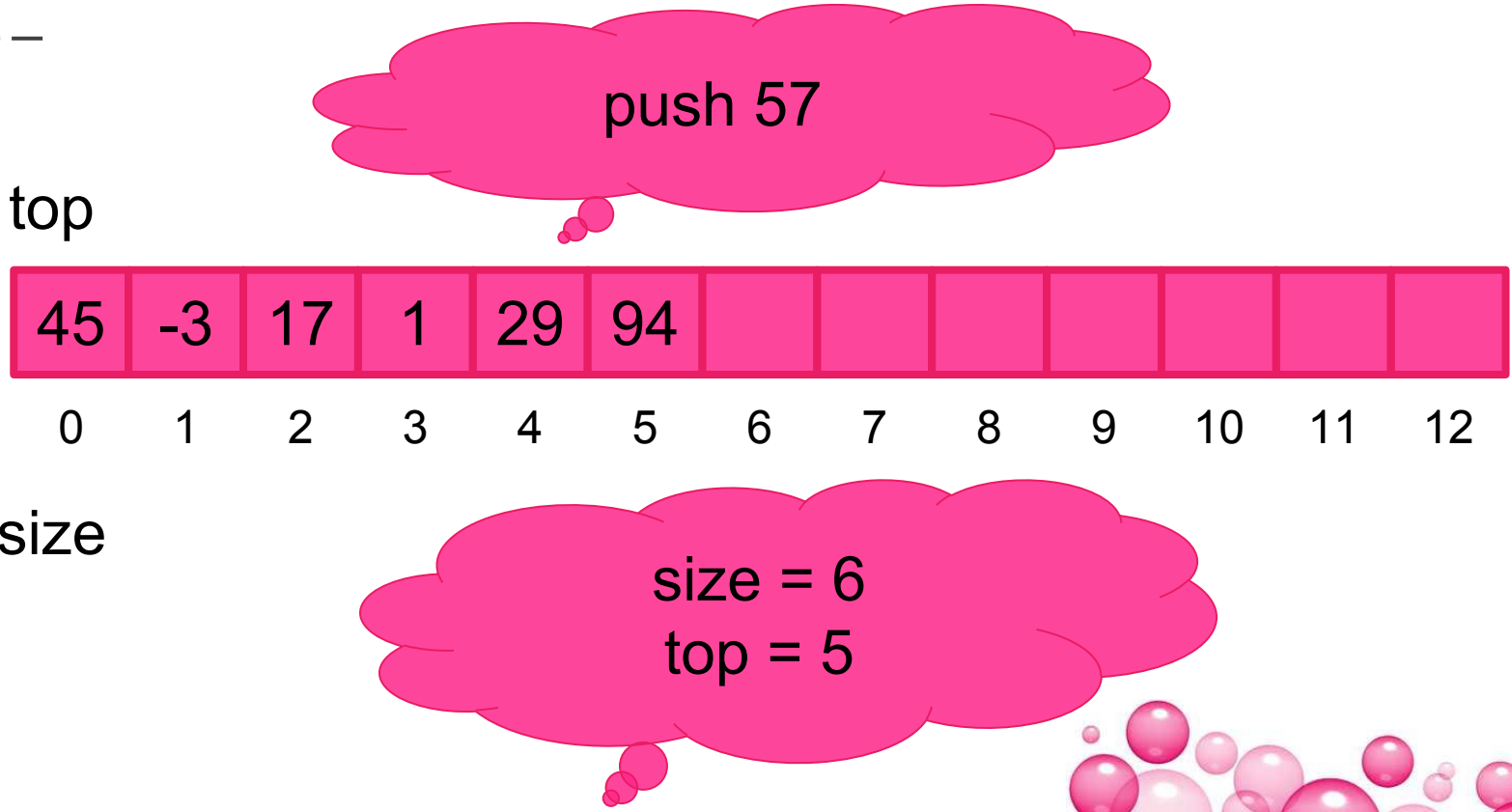
array implementation - push



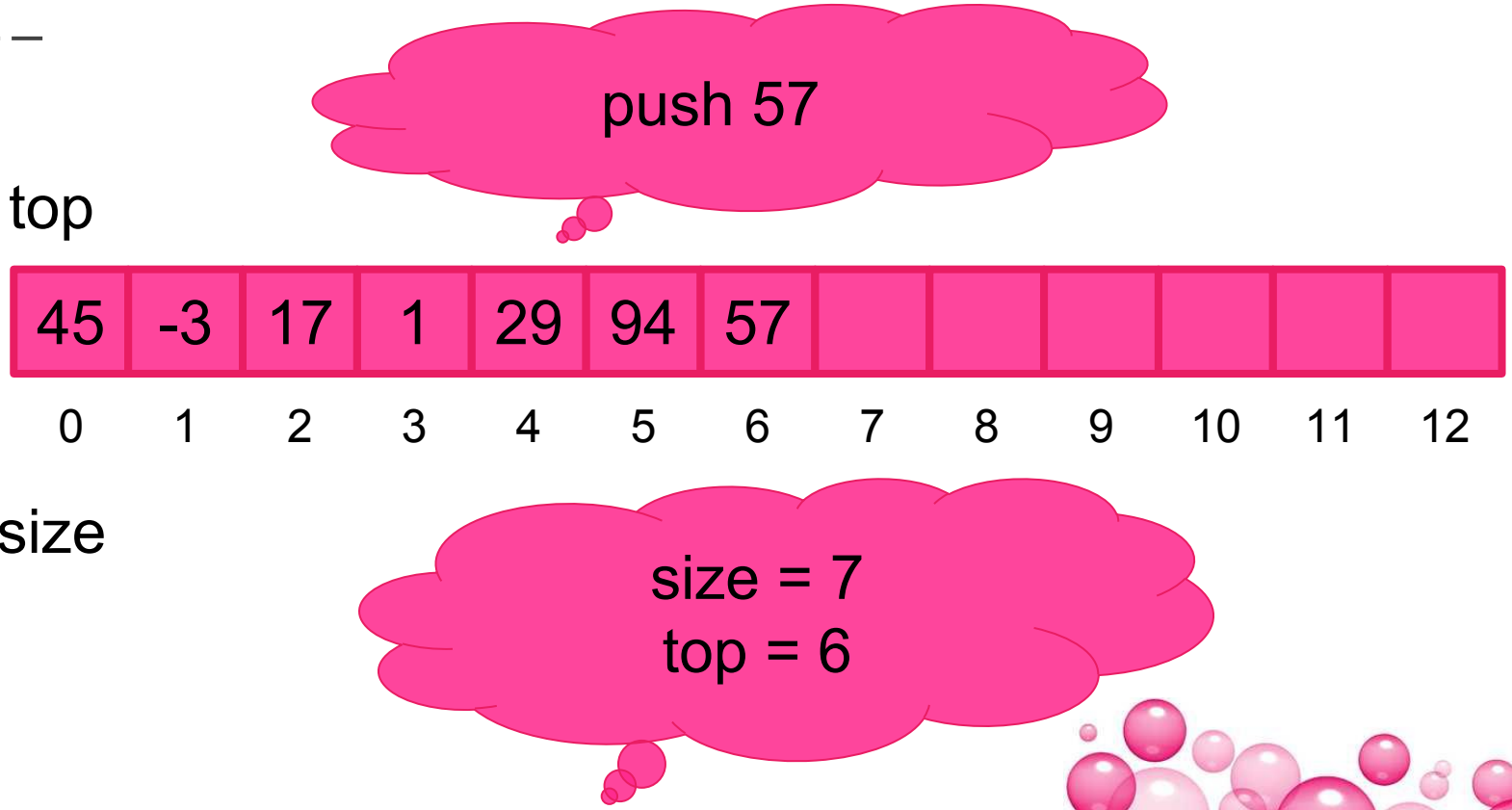
array implementation - push



array implementation - push



array implementation - push



array implementation - pop

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

size = 7
top = 6

array implementation - pop

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

size = 6
top = 5

array implementation - pop

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

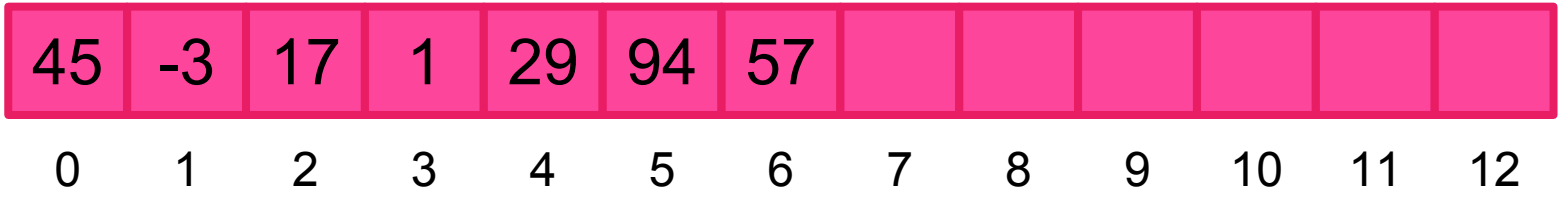
size = 5
top = 4

array implementation - pop

– – –

worrisome 94 and 57?

top



size

size = 5
top = 4

array implementation - push 35

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

size = 5
top = 4

array implementation - push 35

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

size = 6
top = 5

array implementation - push 68

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

size = 6
top = 5

array implementation - push 68

top



0 1 2 3 4 5 6 7 8 9 10 11 12

size

size = 7
top = 6

array implementation - data members



array implementation - data members

do we need top?

top



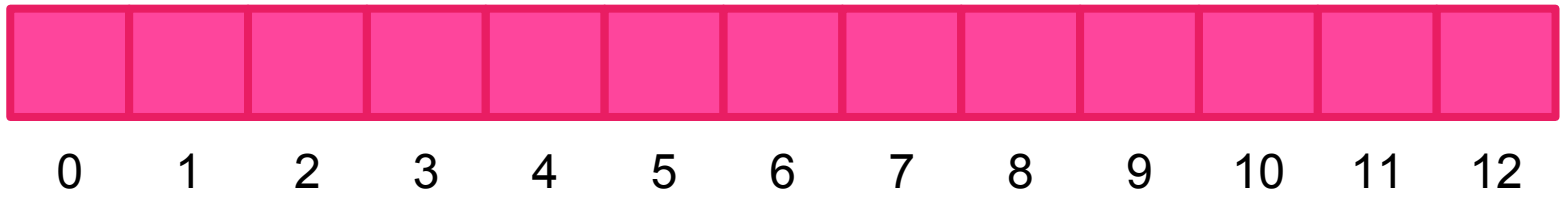
0 1 2 3 4 5 6 7 8 9 10 11 12

size

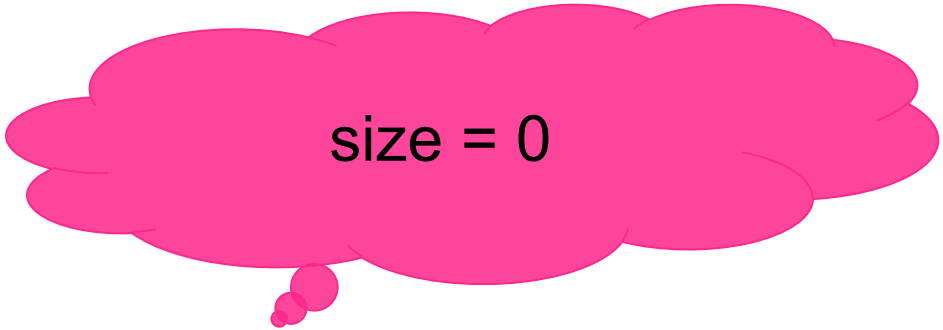
size = 7

~~top = 6~~

array implementation



size





array implementation

operation	array
push	O(1)
pop	O(1)
top	O(1)

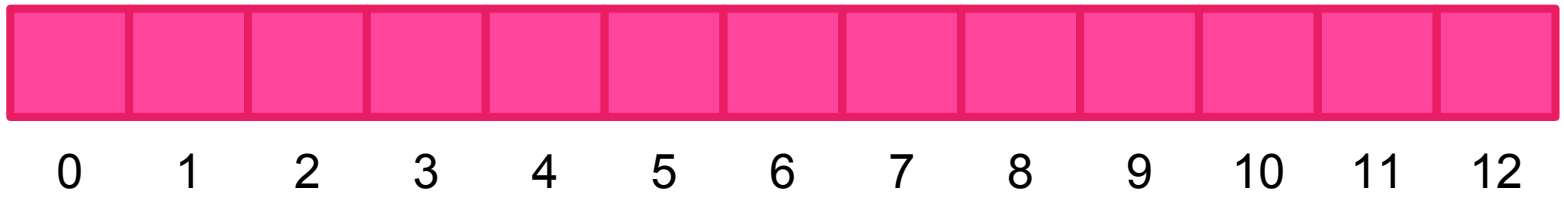


array implementation

- though all operations run in constant time, space is not dynamic
- dynamic memory schemes
  - vector
    - similar to array
    - but how to do the copying?
  - singly-linked list
  - doubly-linked list



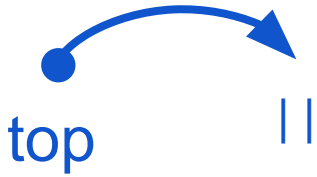
dynamic array implementation



size

size = 0

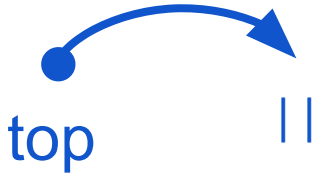
singly-linked list implementation

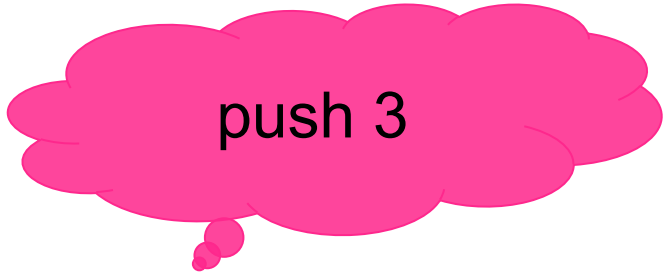


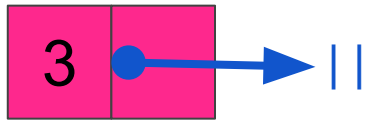




singly-linked list implementation

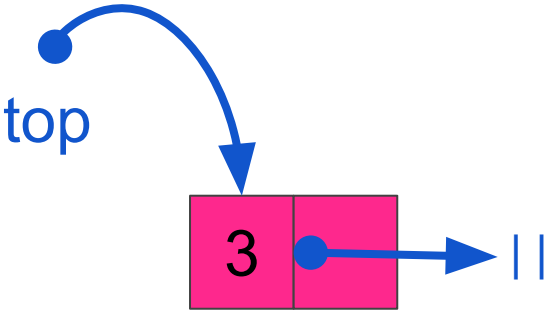


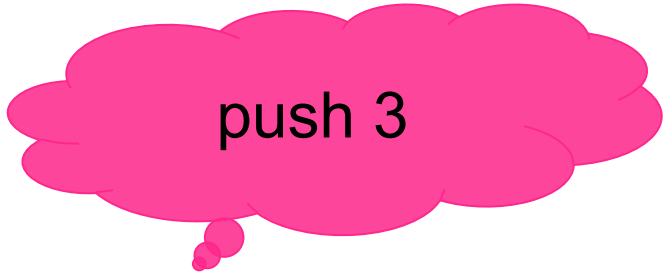




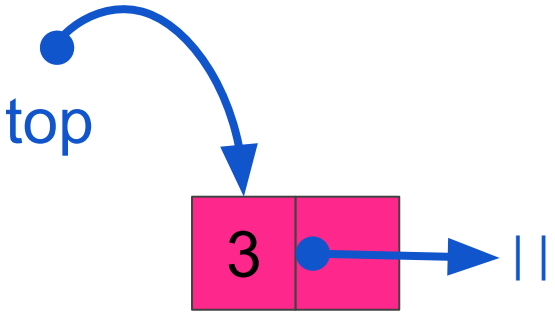


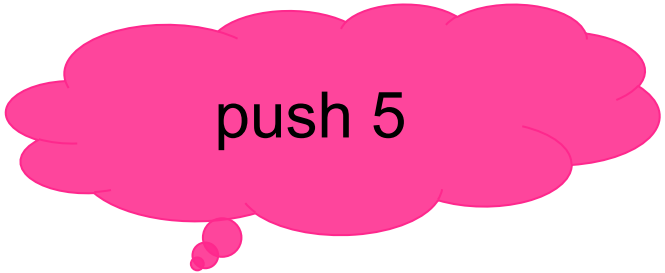
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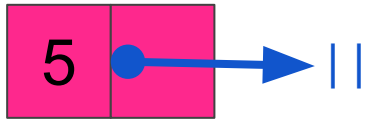




singly-linked list implementation

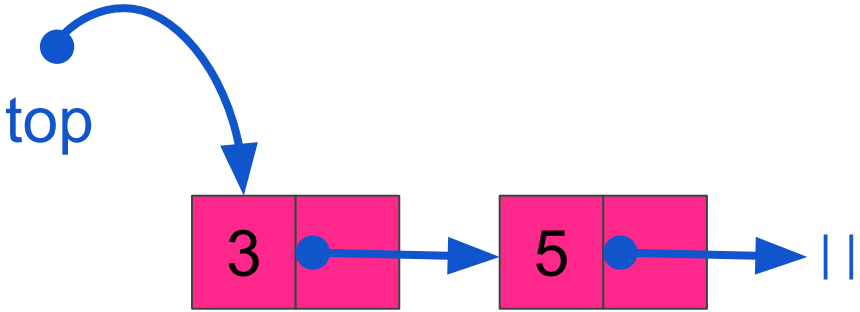


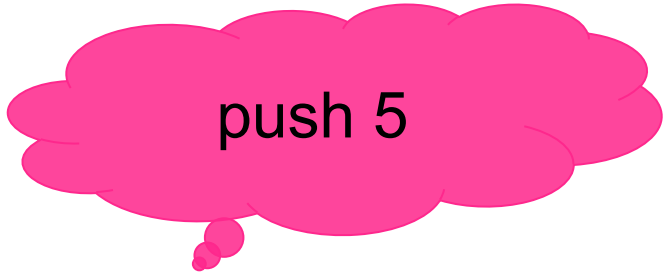




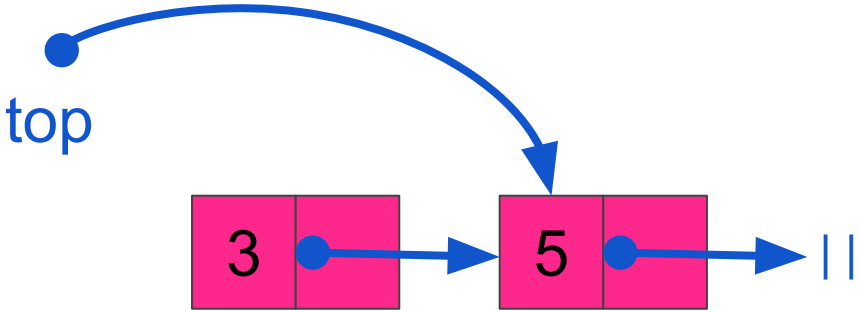


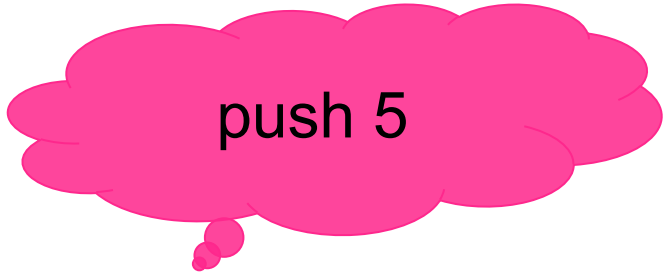
singly-linked list implementation



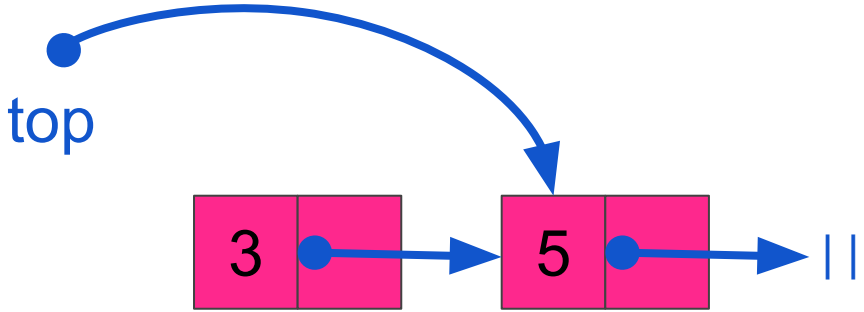


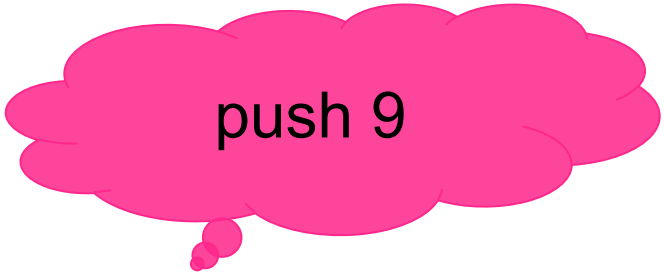
singly-linked list implementation

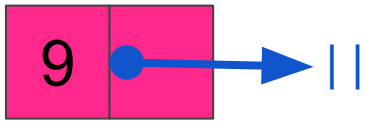




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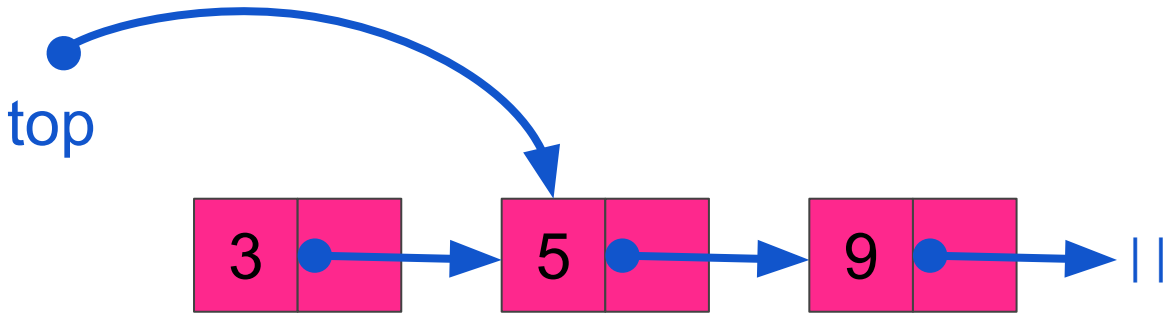


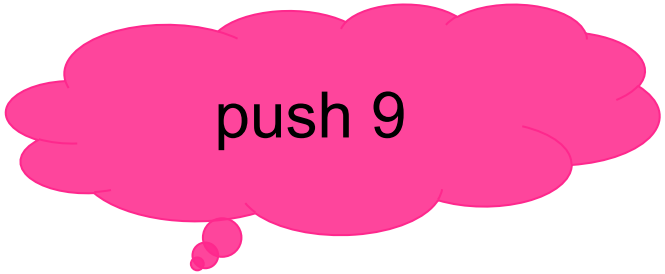




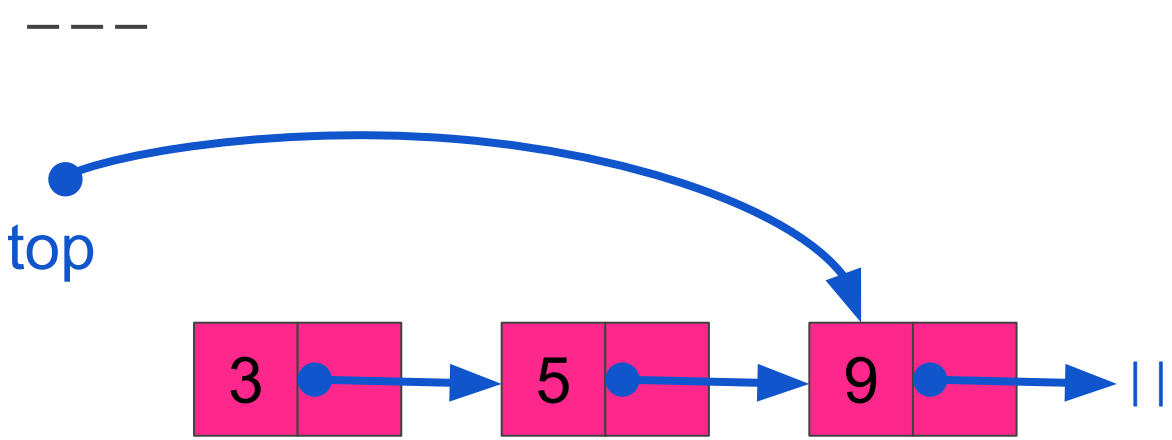


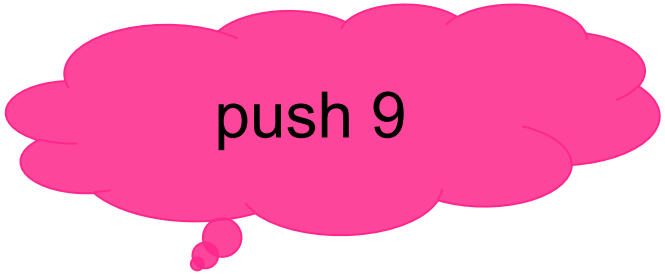
singly-linked list implementation



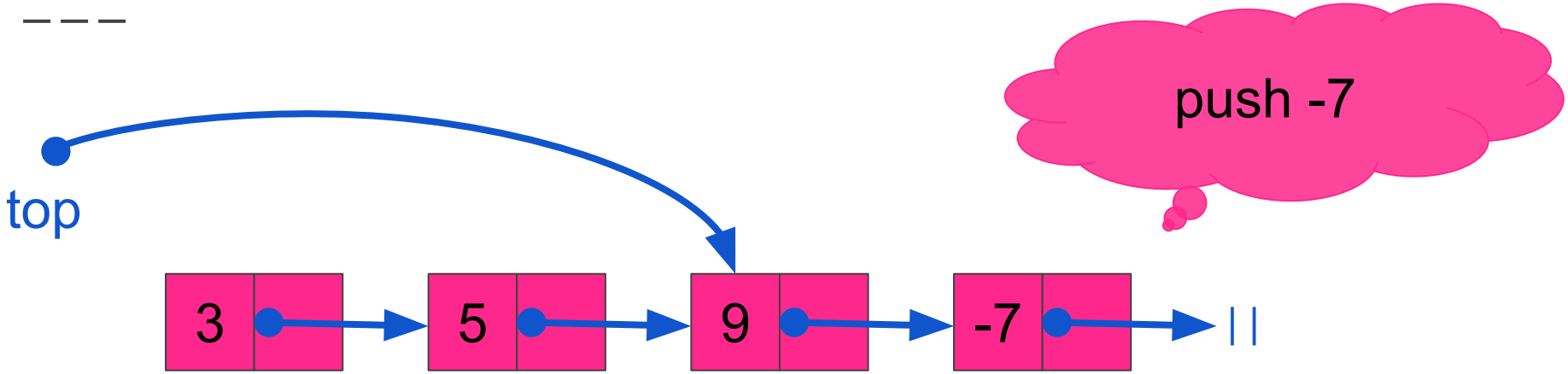


singly-linked list implementation

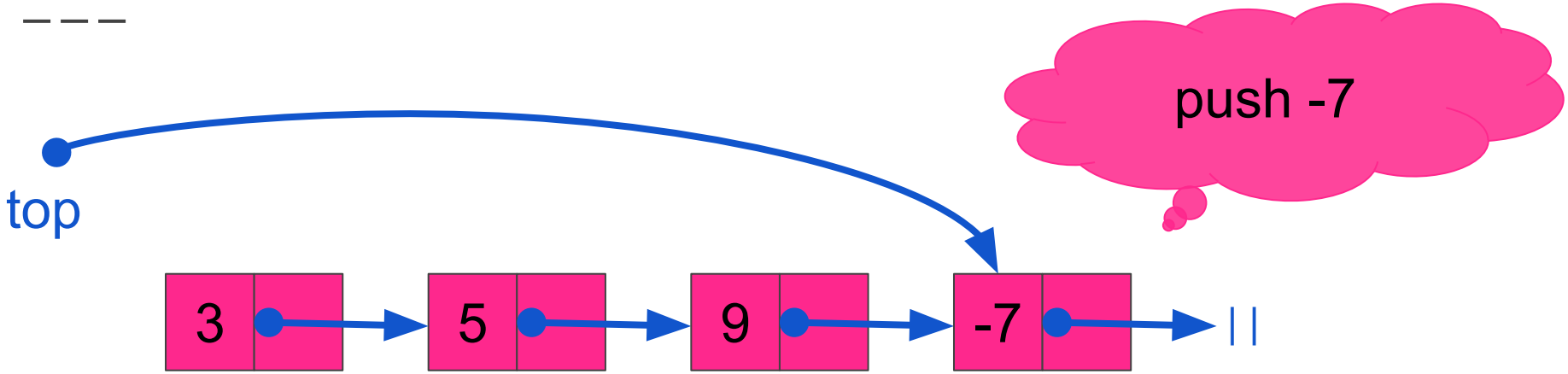




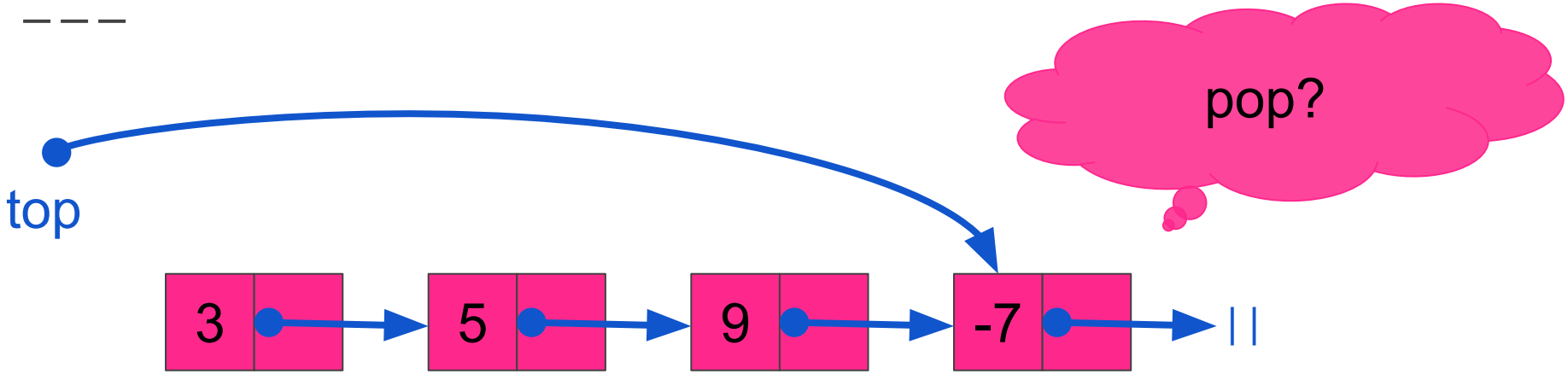
singly-linked list implementation



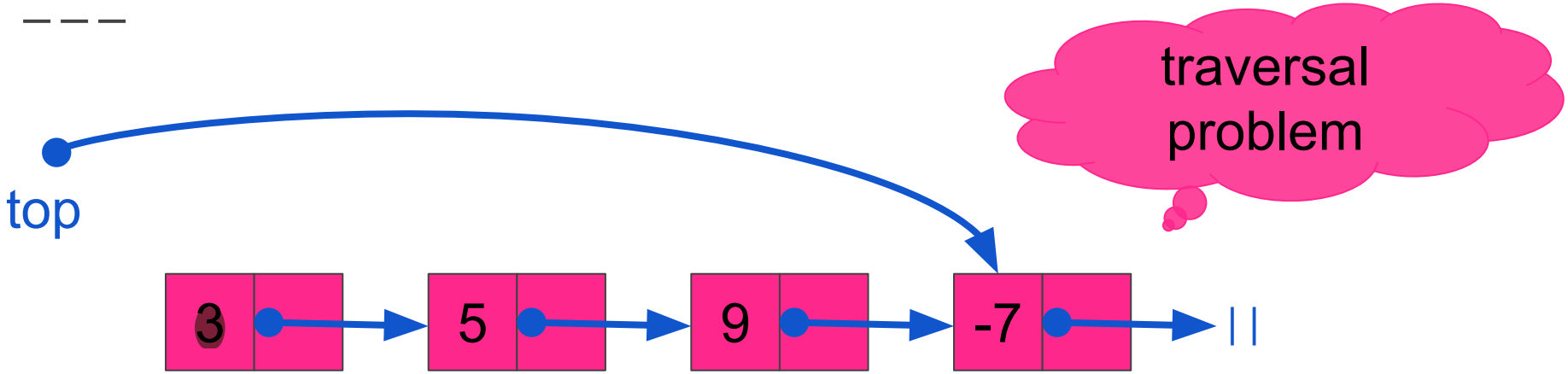
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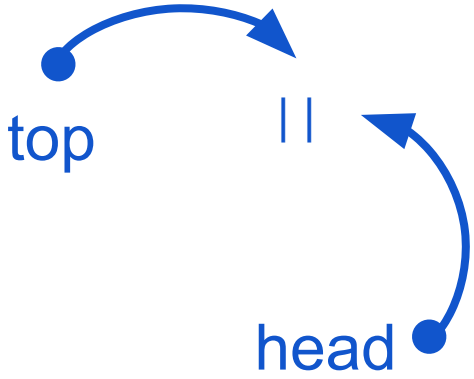
singly-linked list implementation



singly-linked list implementation



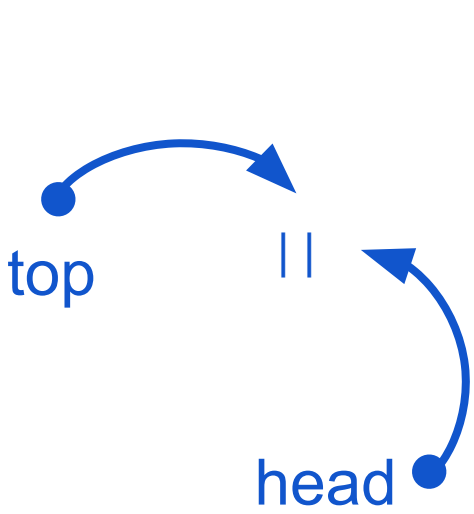
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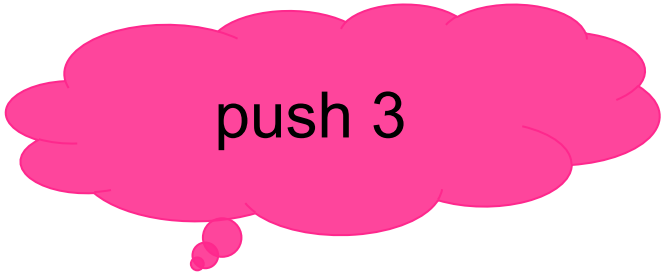


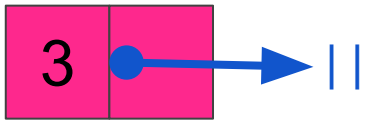




singly-linked list implementation

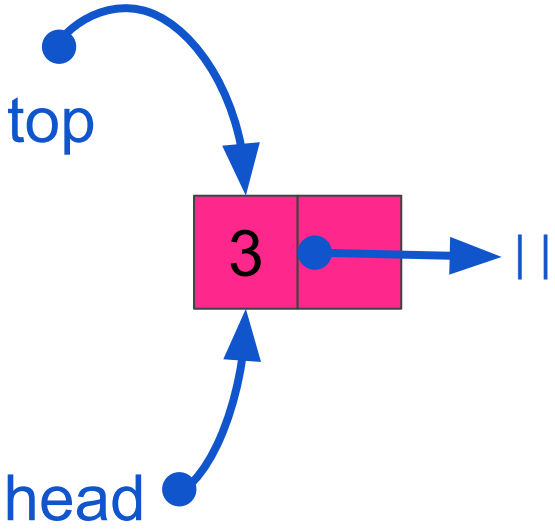


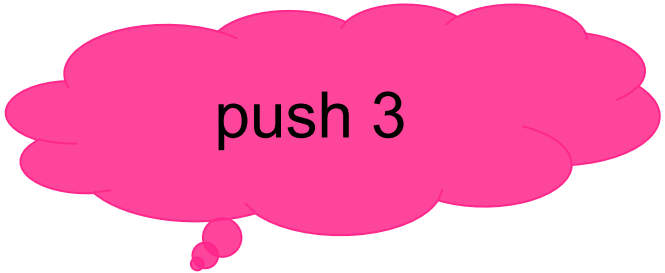






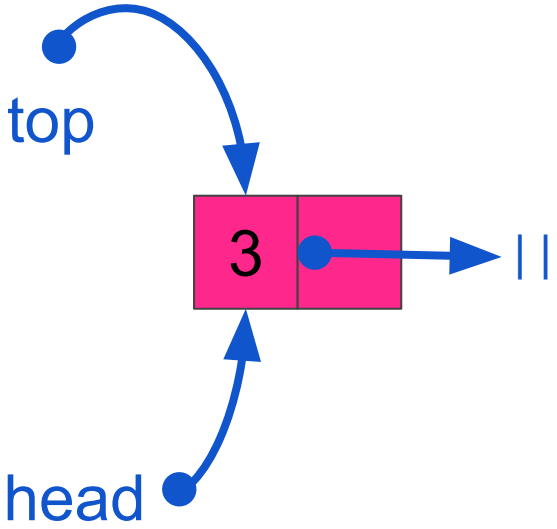
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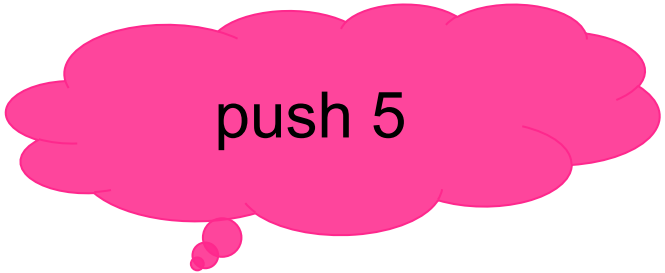


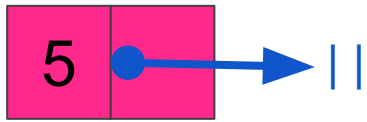




singly-linked list implementation

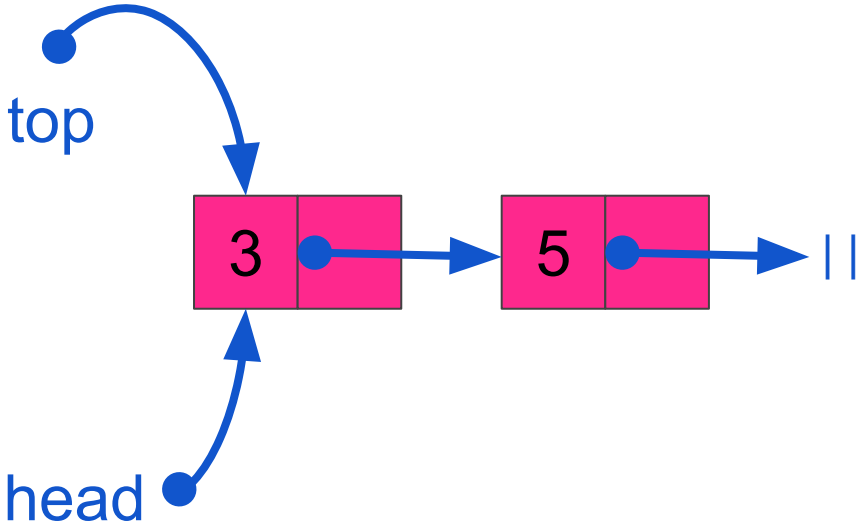


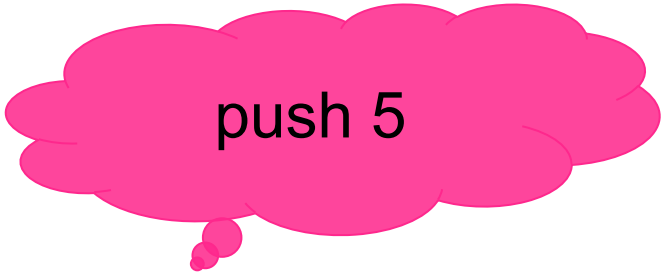




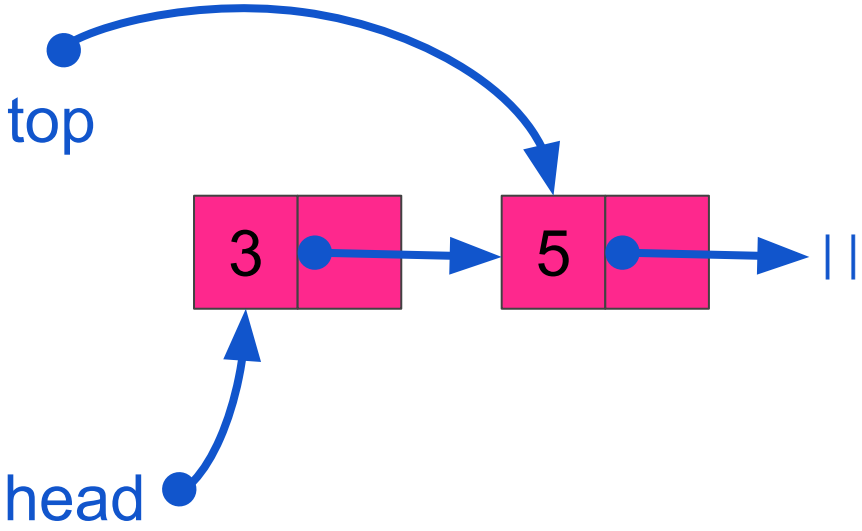


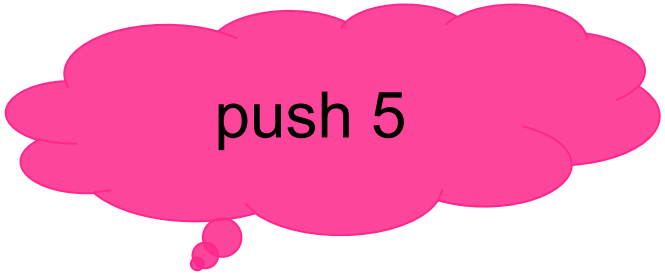
singly-linked list implementation



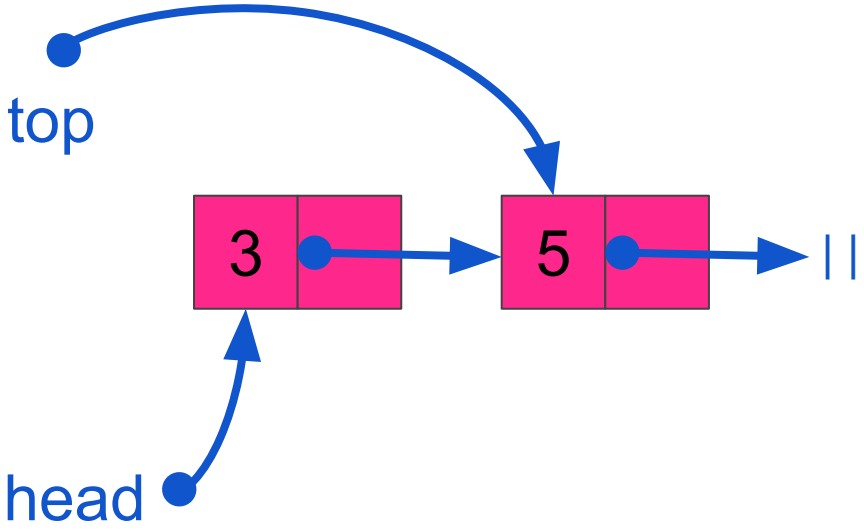


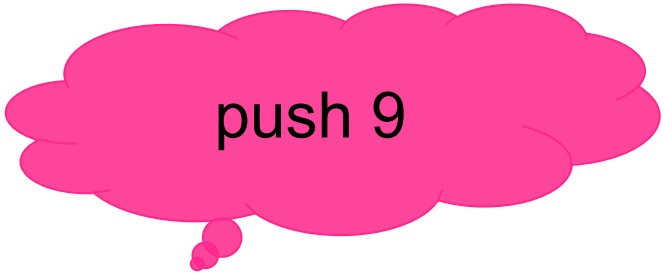
singly-linked list implementation

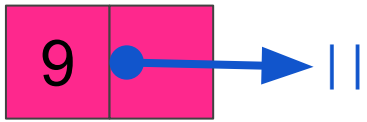




singly-linked list implementation

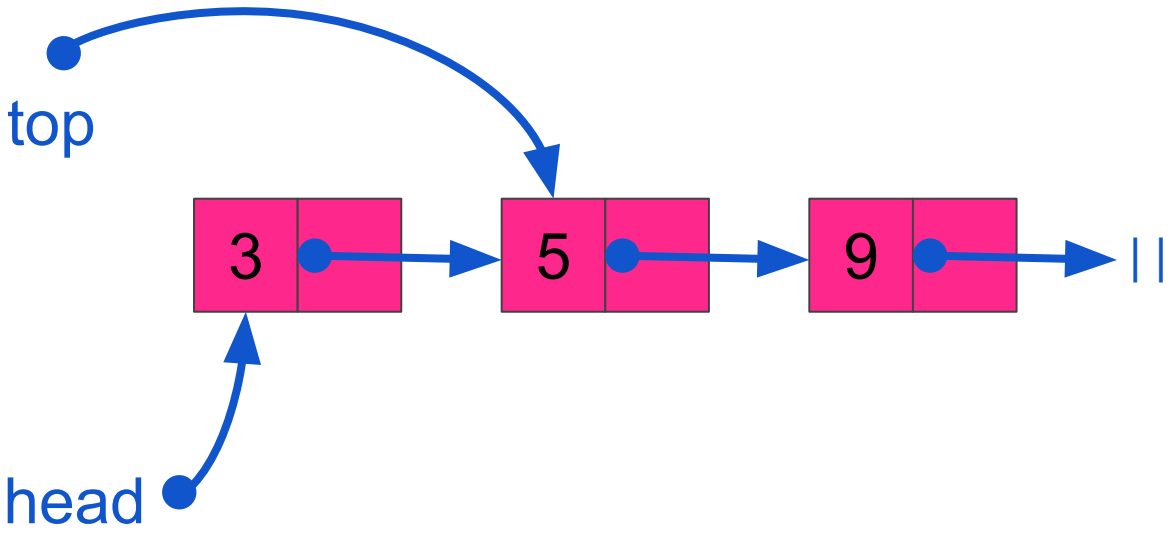


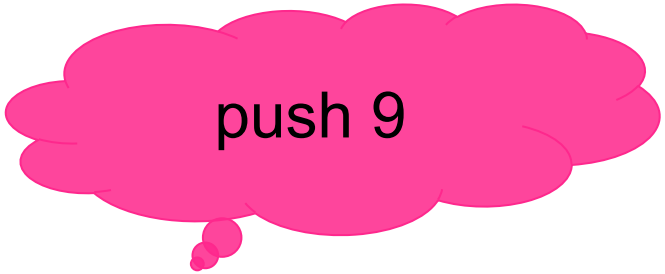




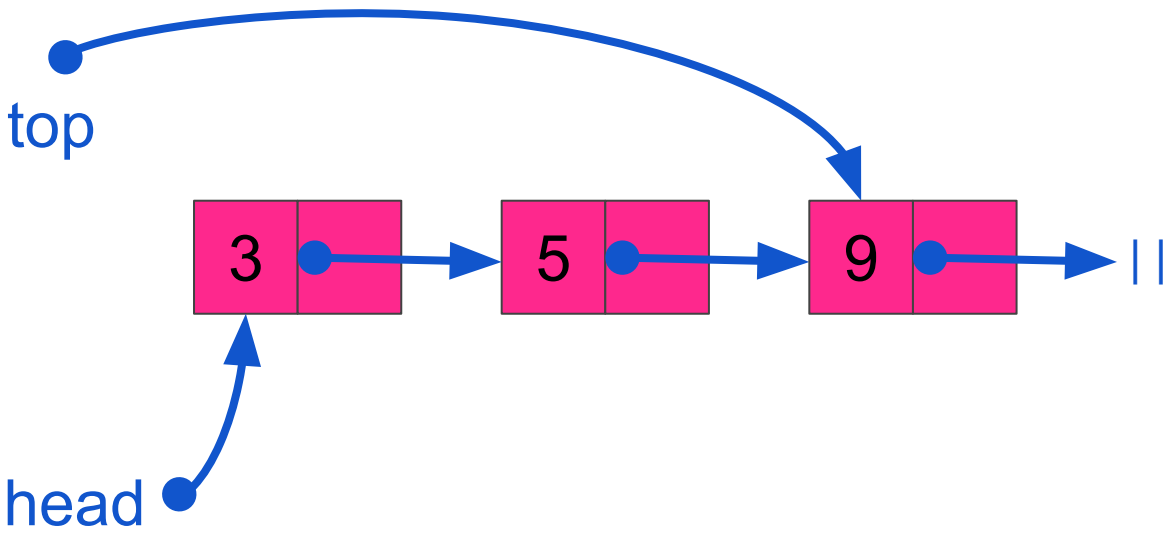


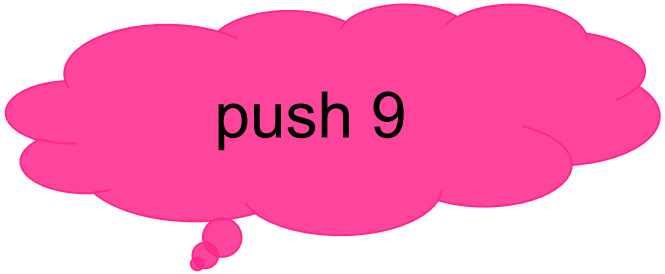
singly-linked list implementation



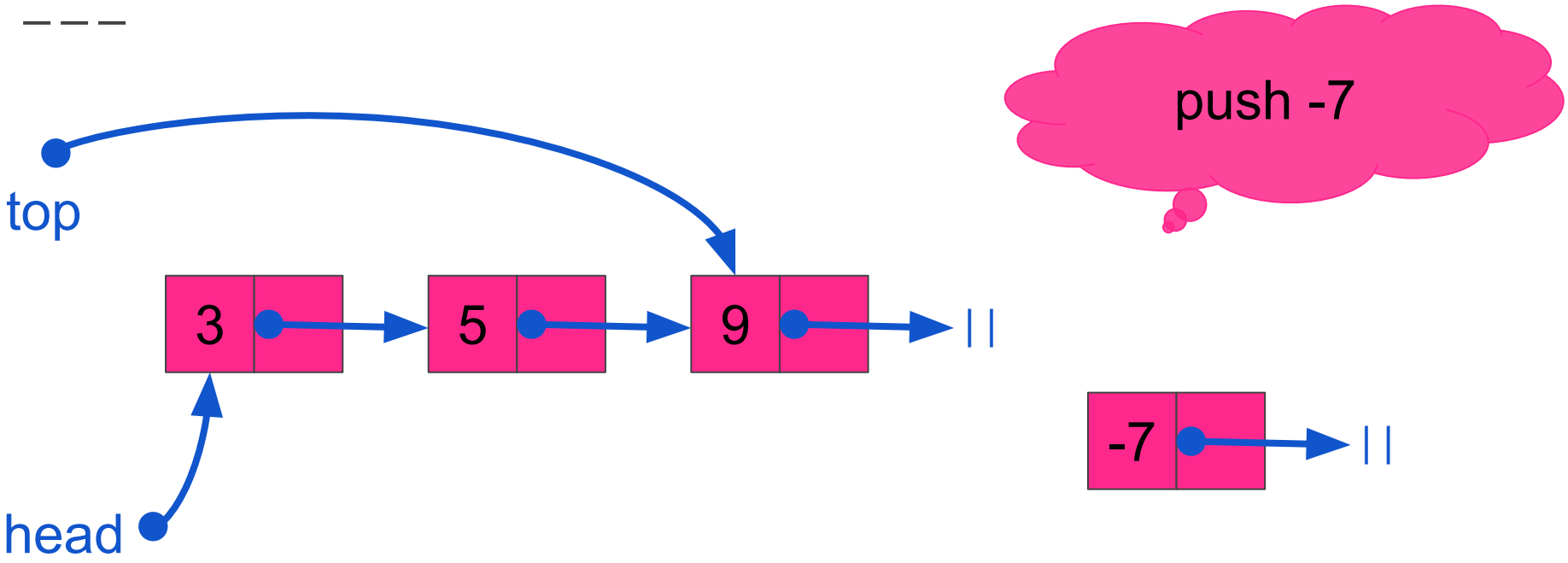


singly-linked list implementation

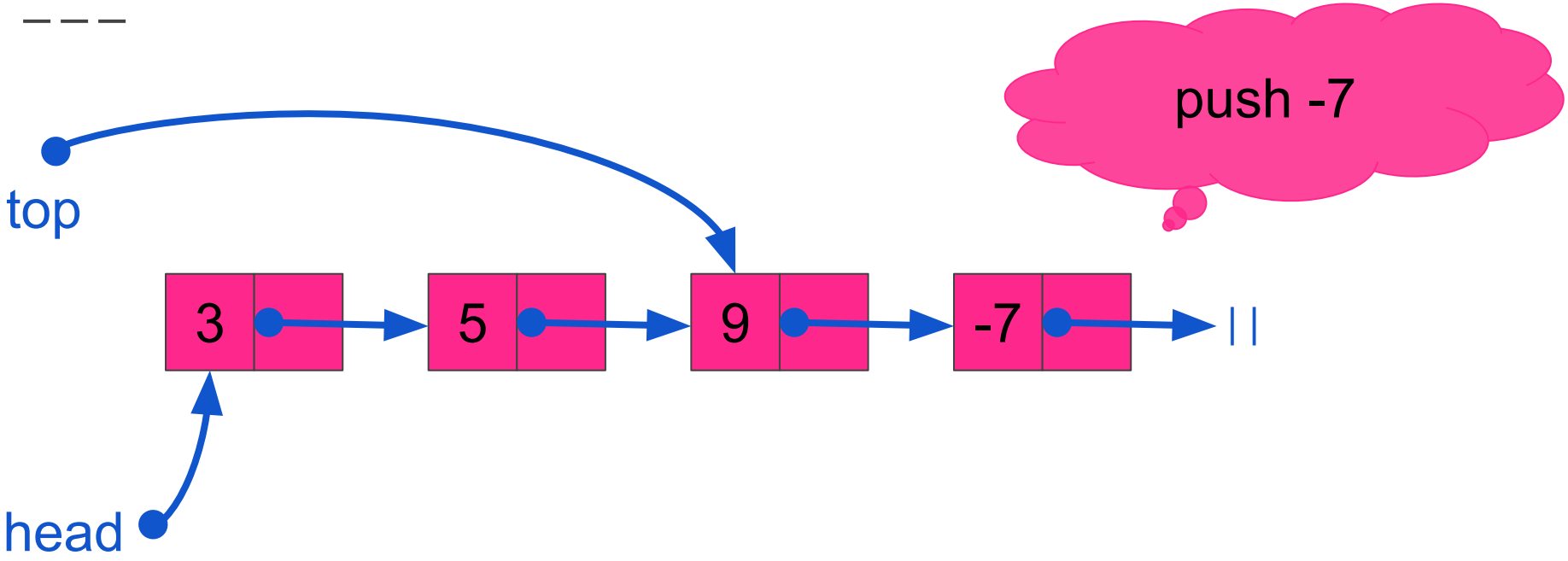




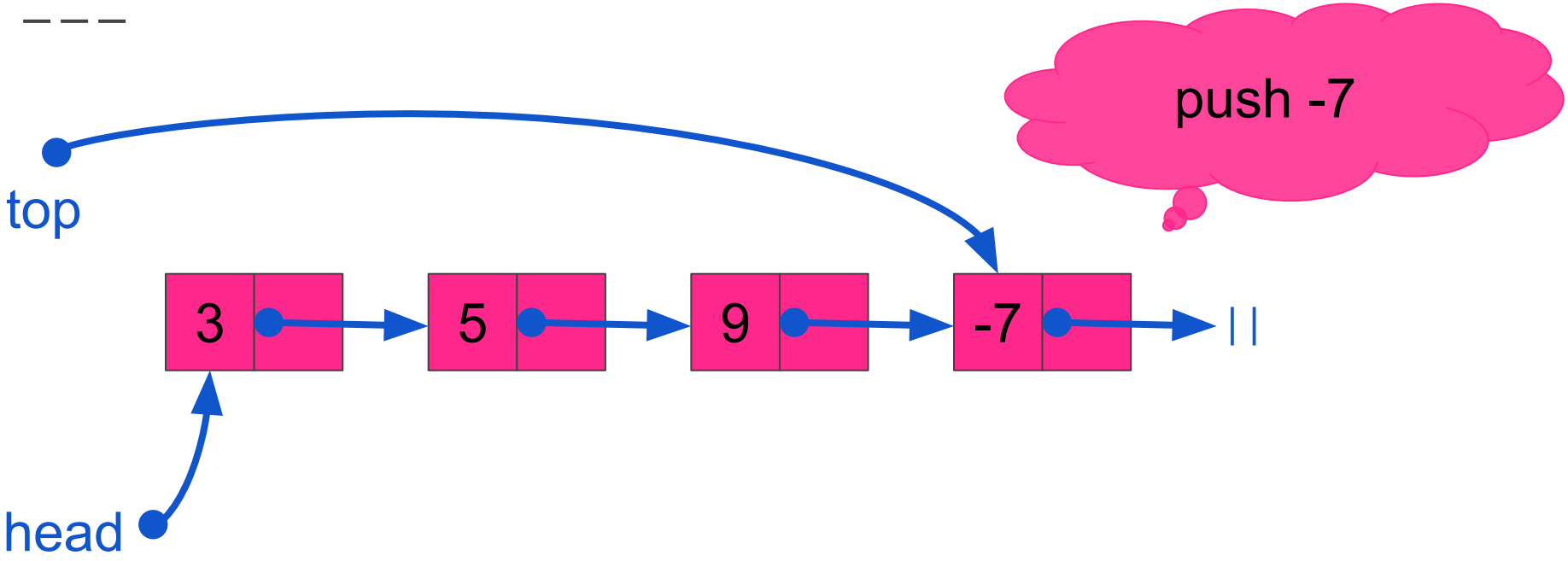
singly-linked list implementation



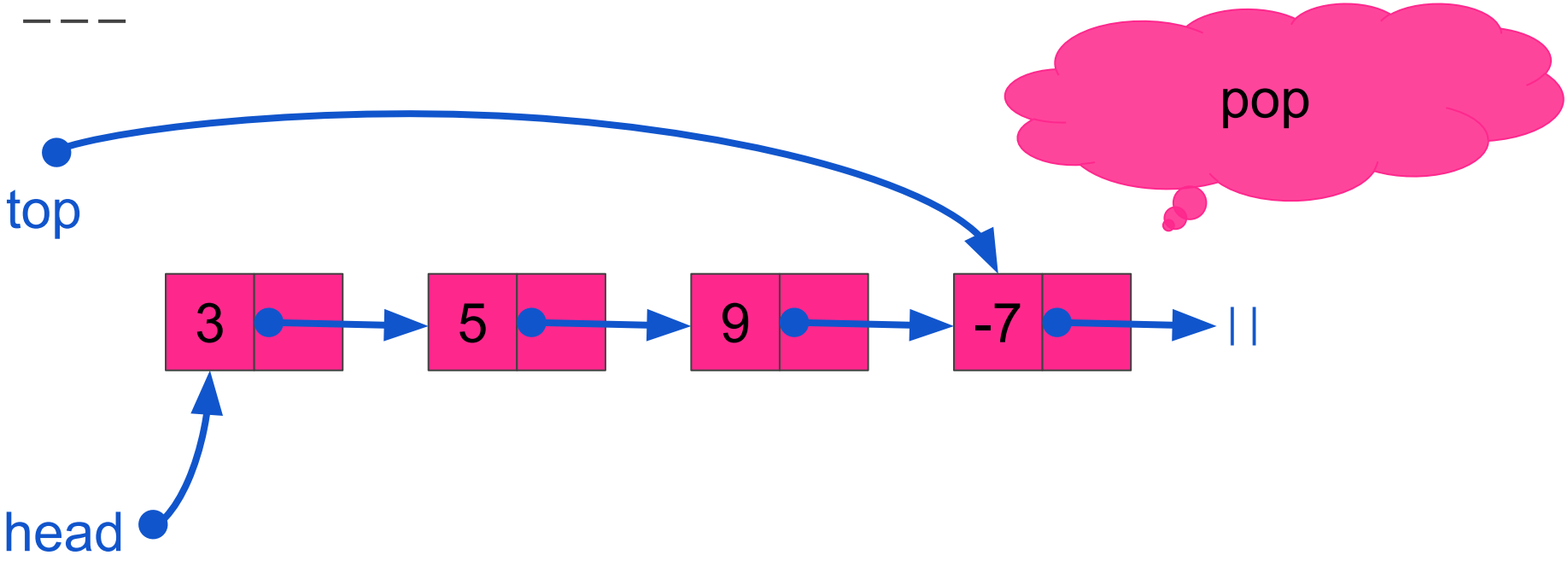
singly-linked list implementation



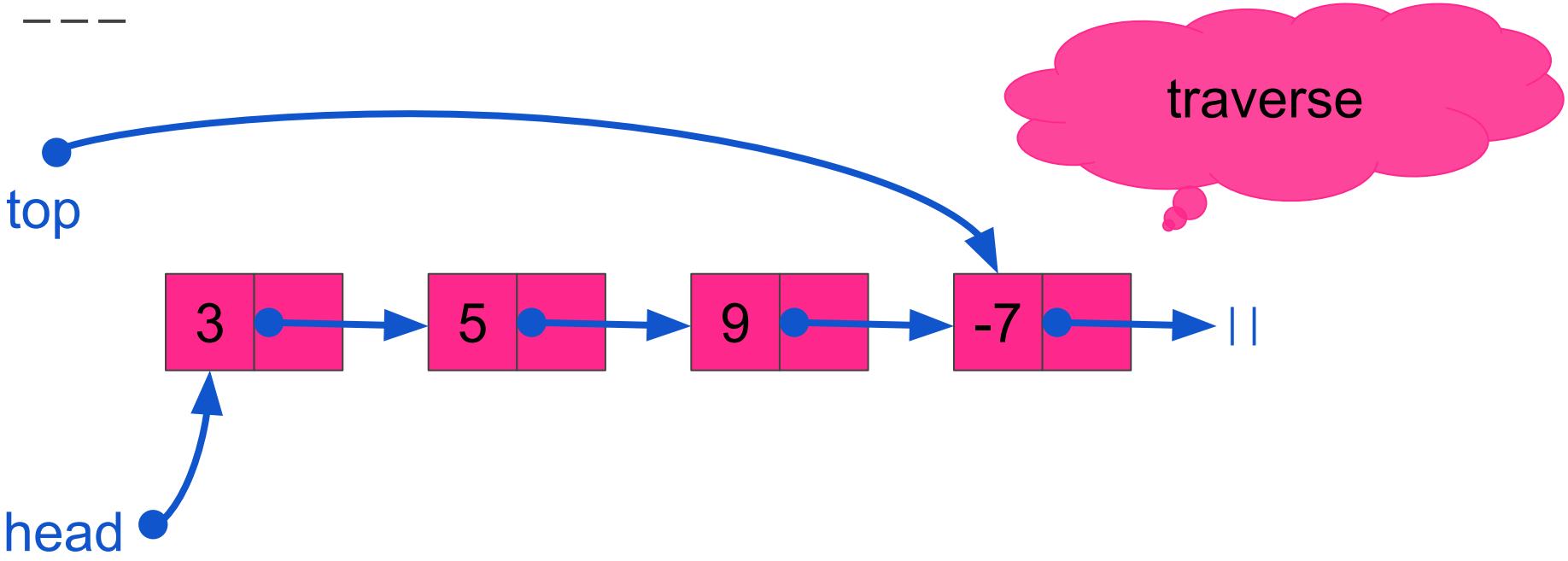
singly-linked list implementation



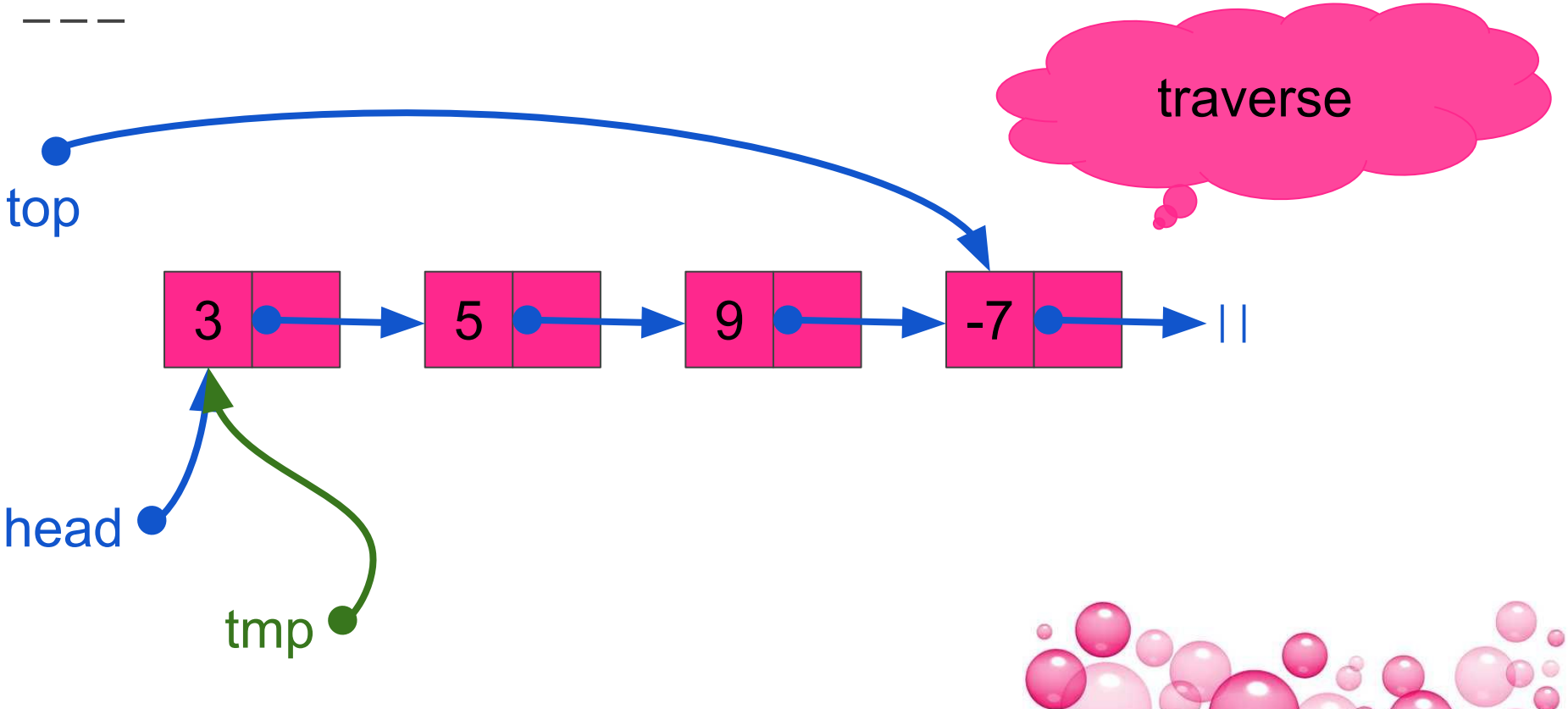
singly-linked list implementation



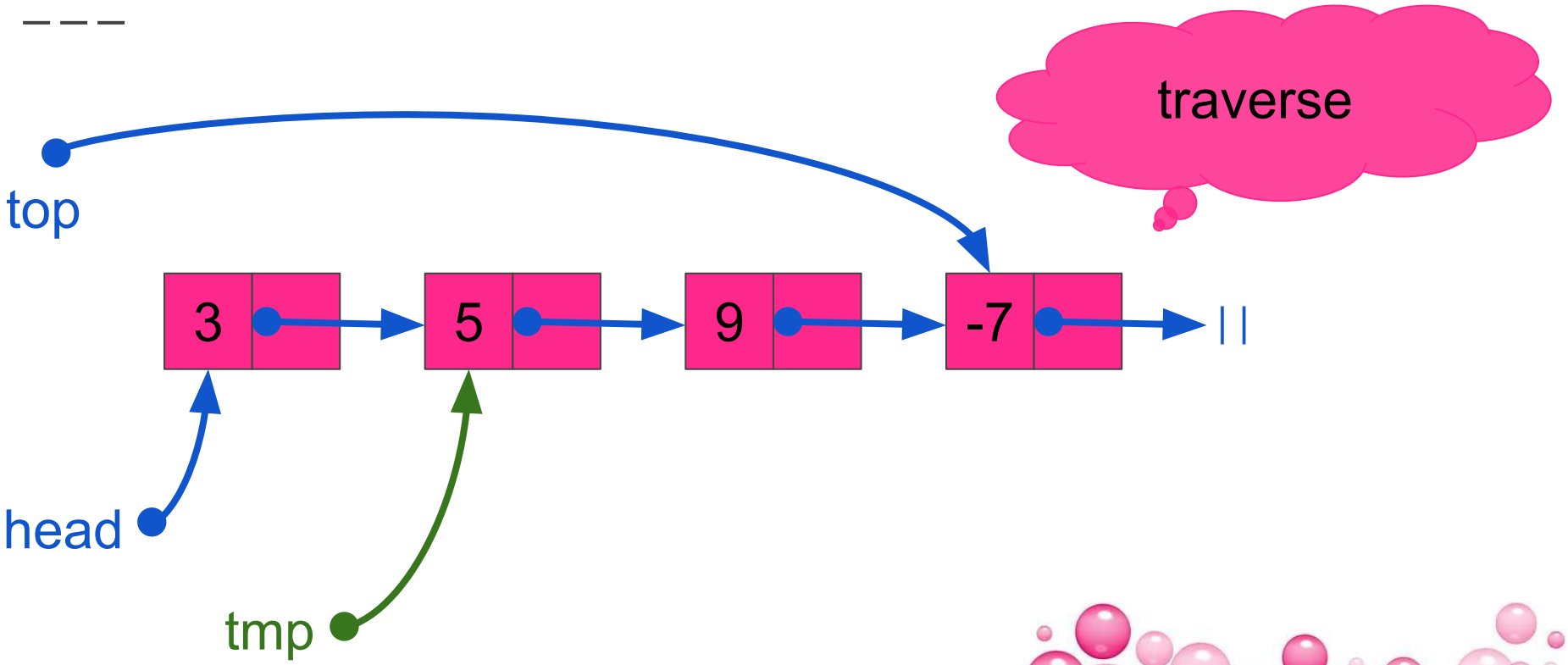
singly-linked list implementation - pop



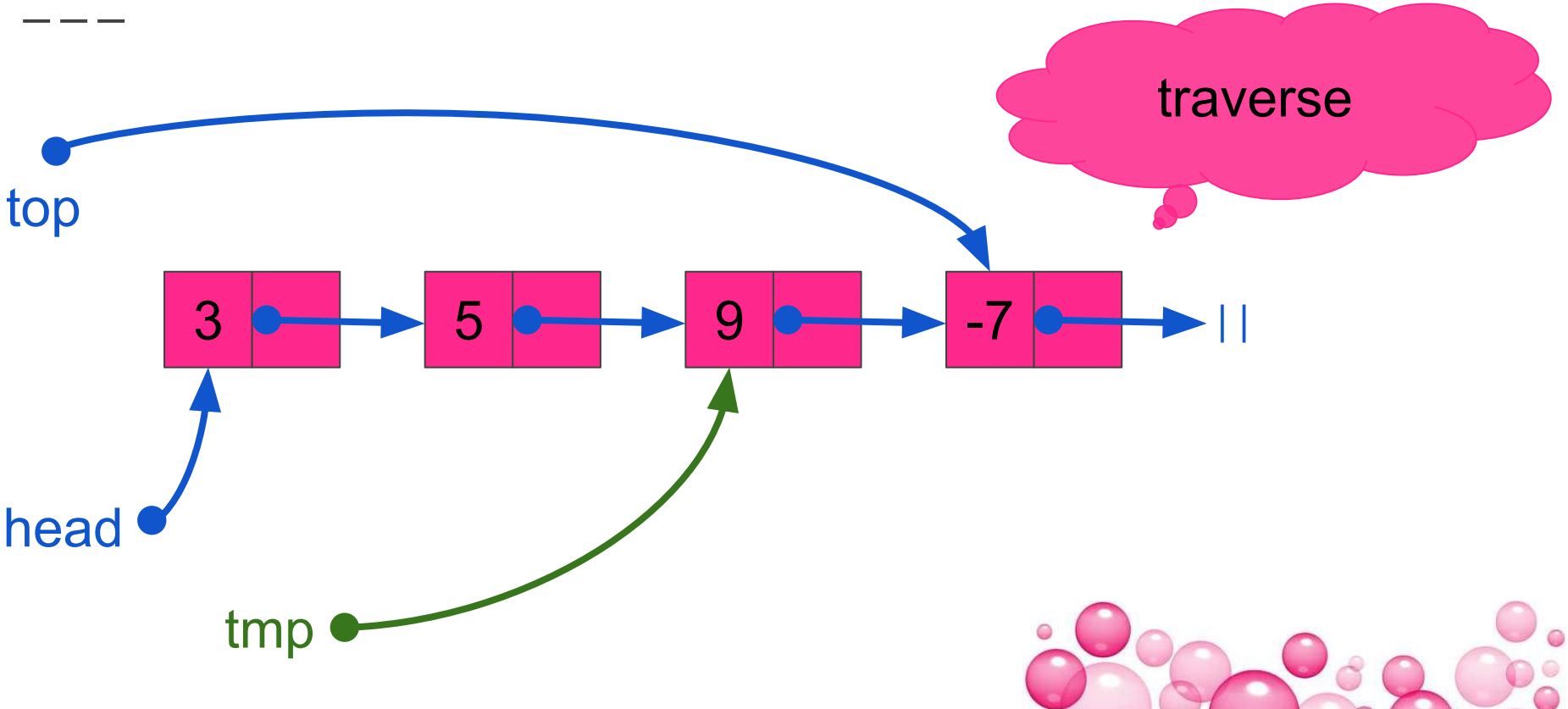
singly-linked list implementation - pop



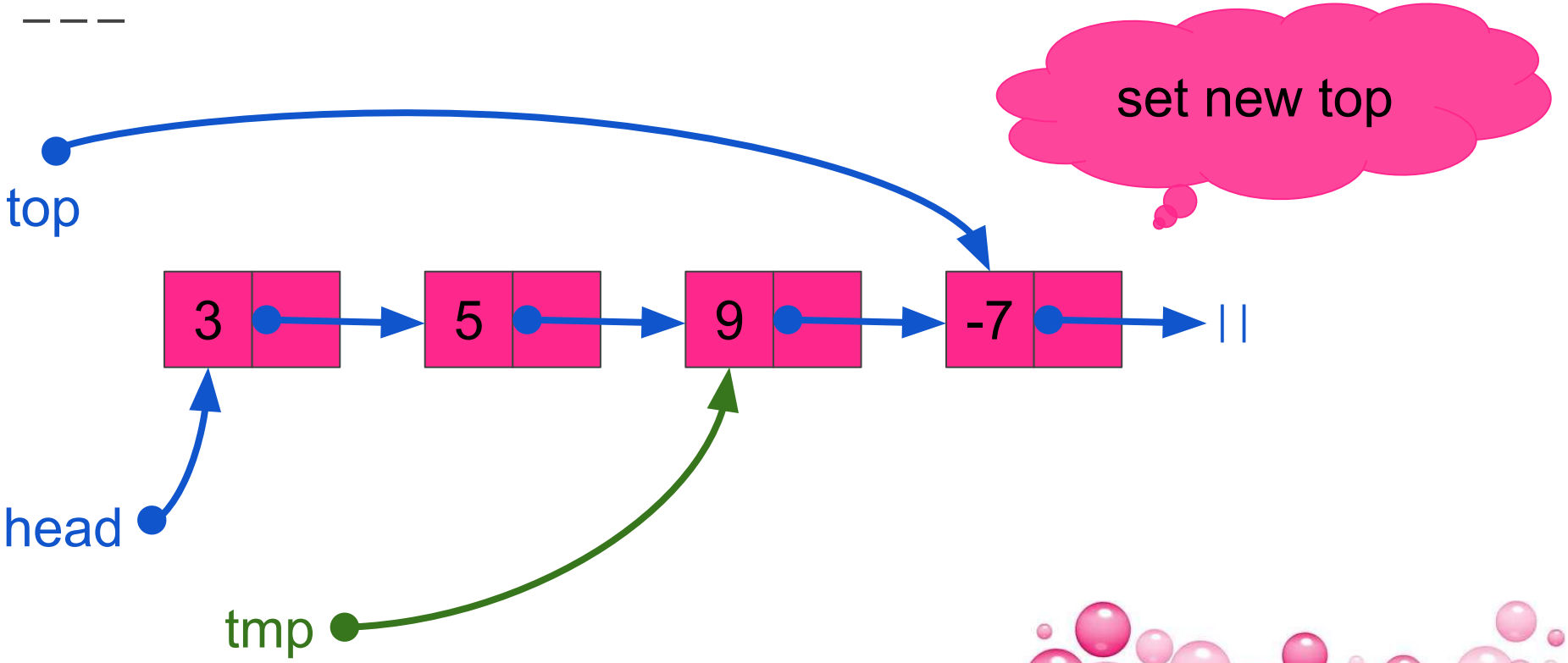
singly-linked list implementation - pop



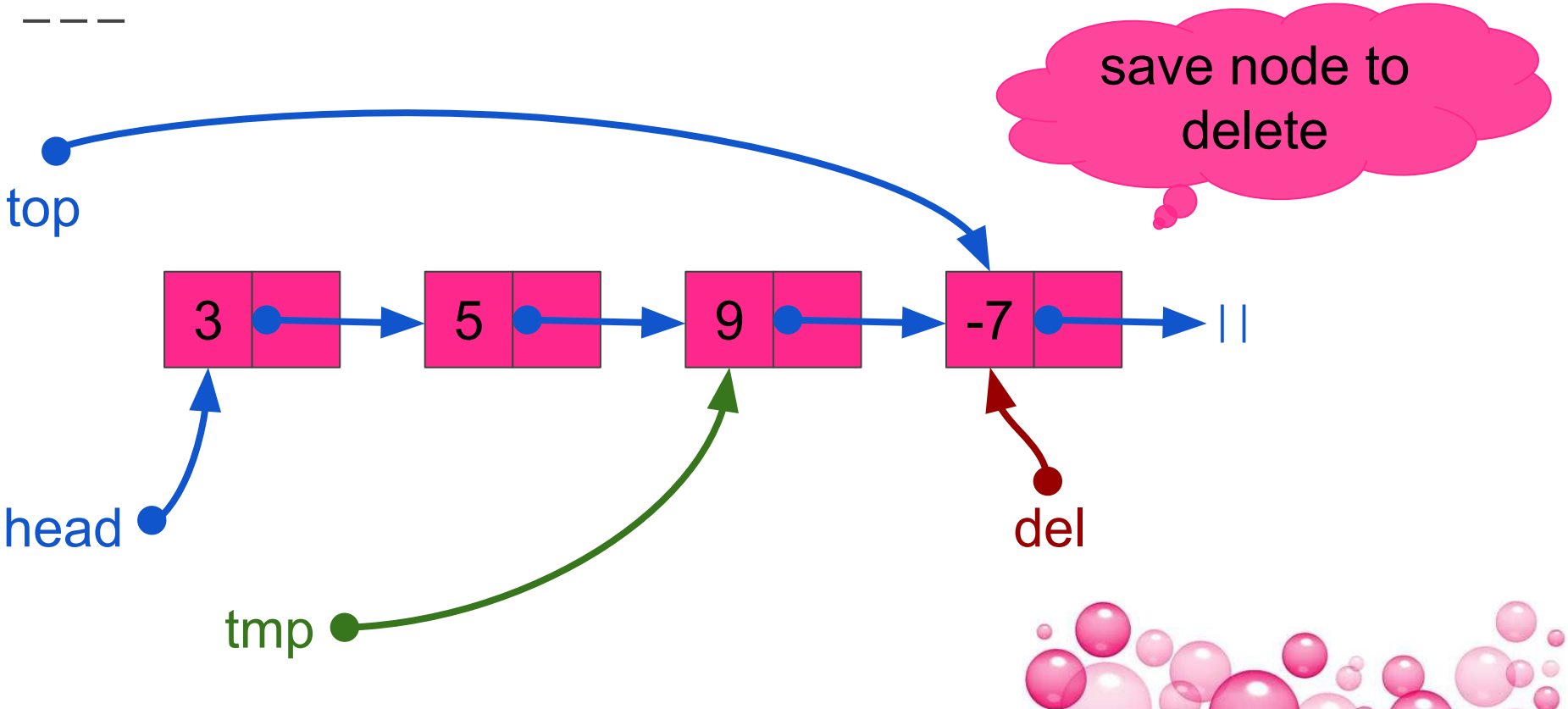
singly-linked list implementation - pop



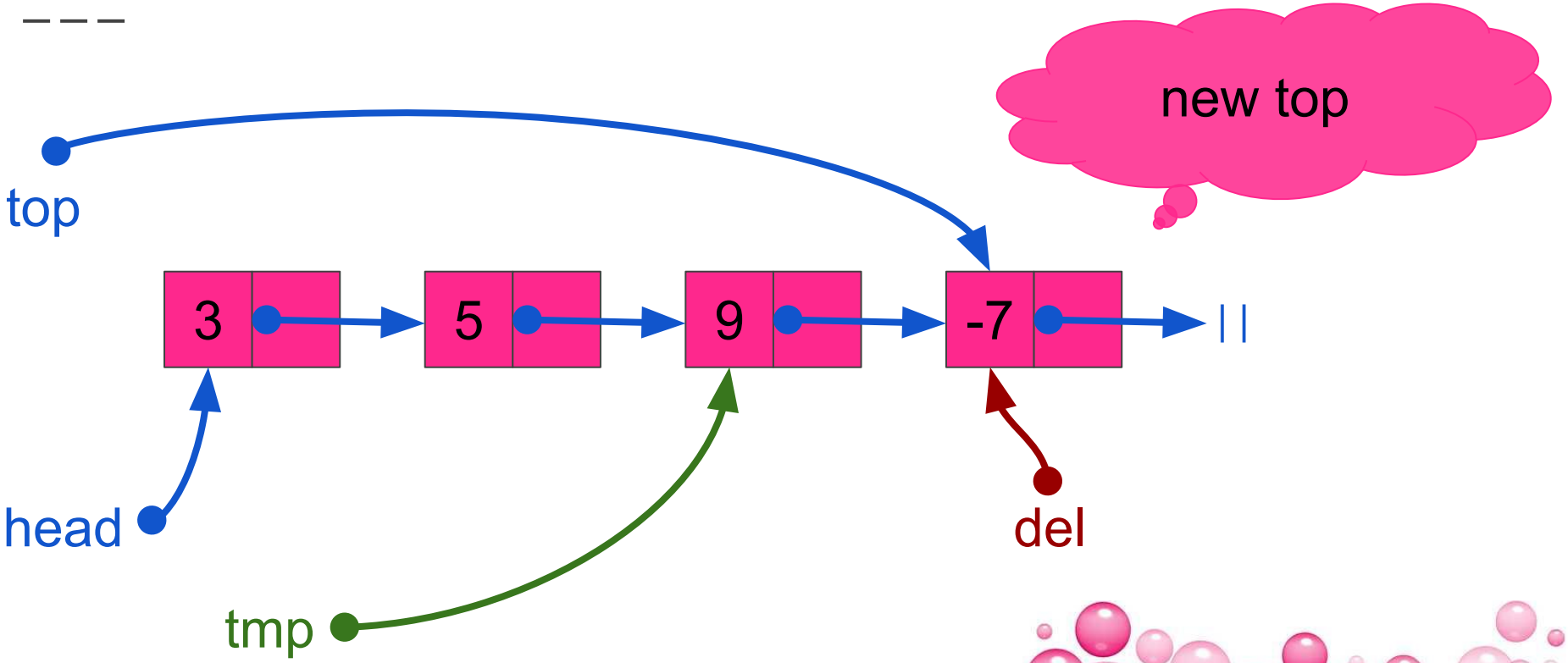
singly-linked list implementation - pop



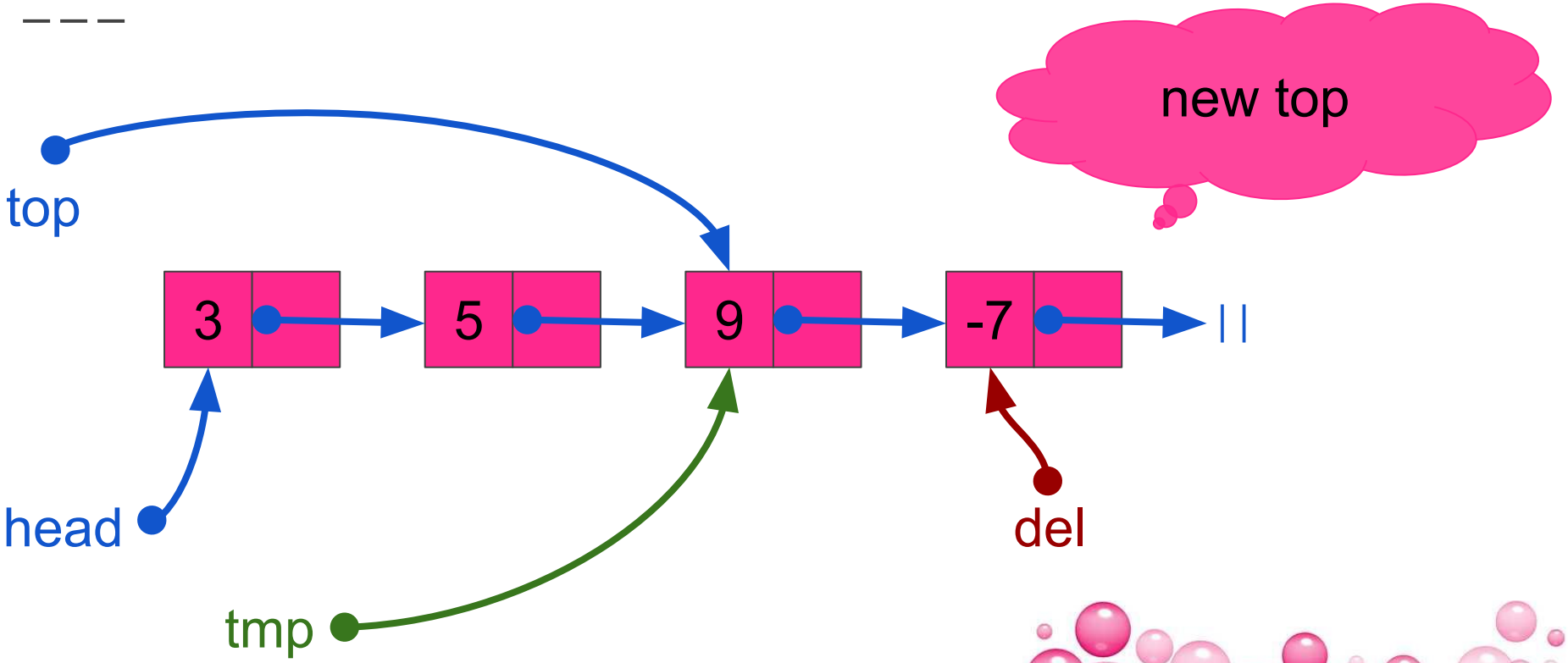
singly-linked list implementation - pop



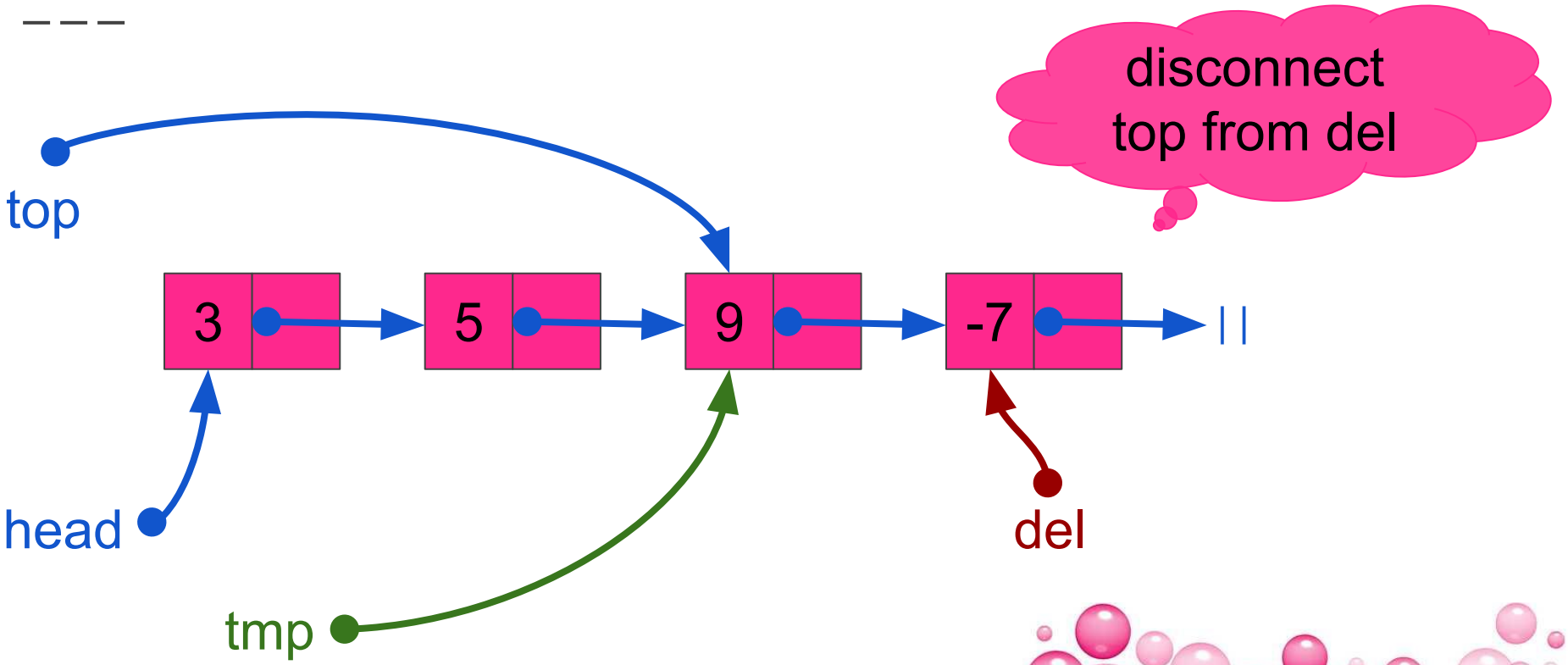
singly-linked list implementation - pop



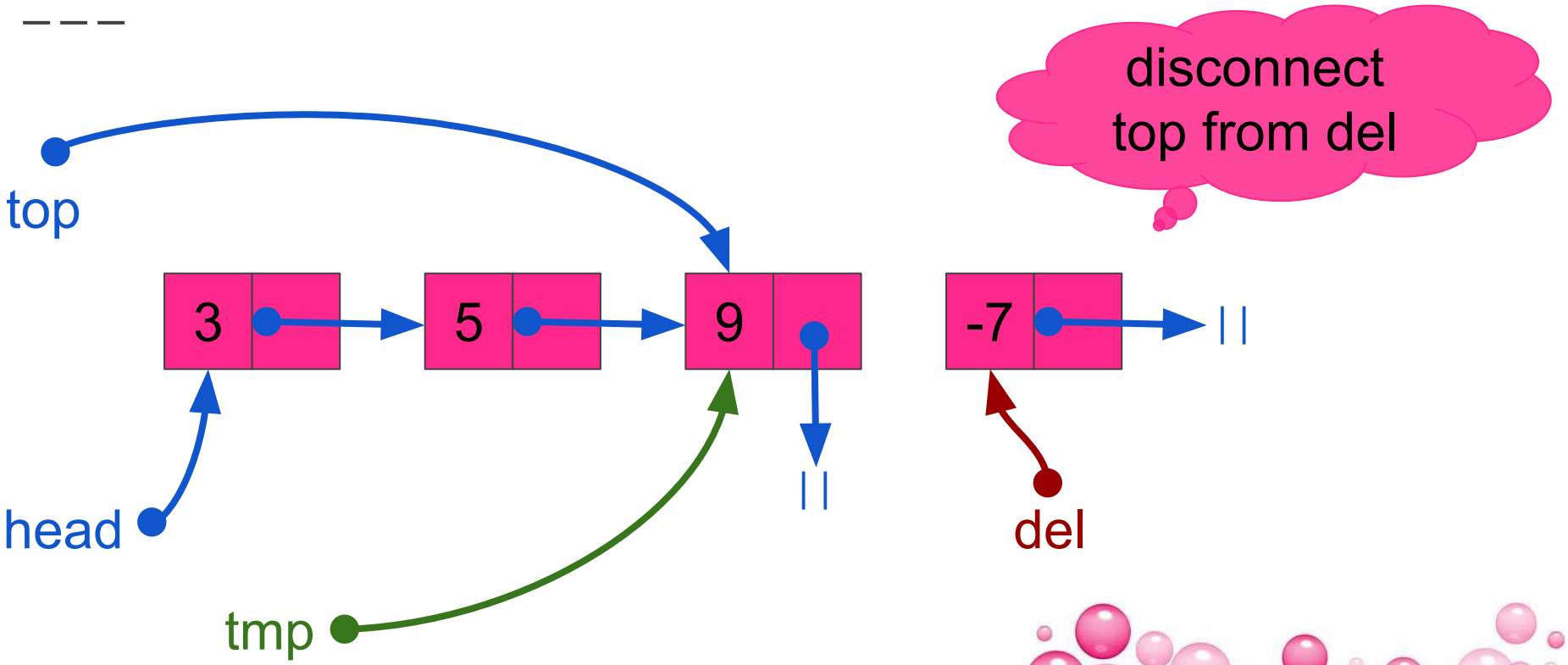
singly-linked list implementation - pop



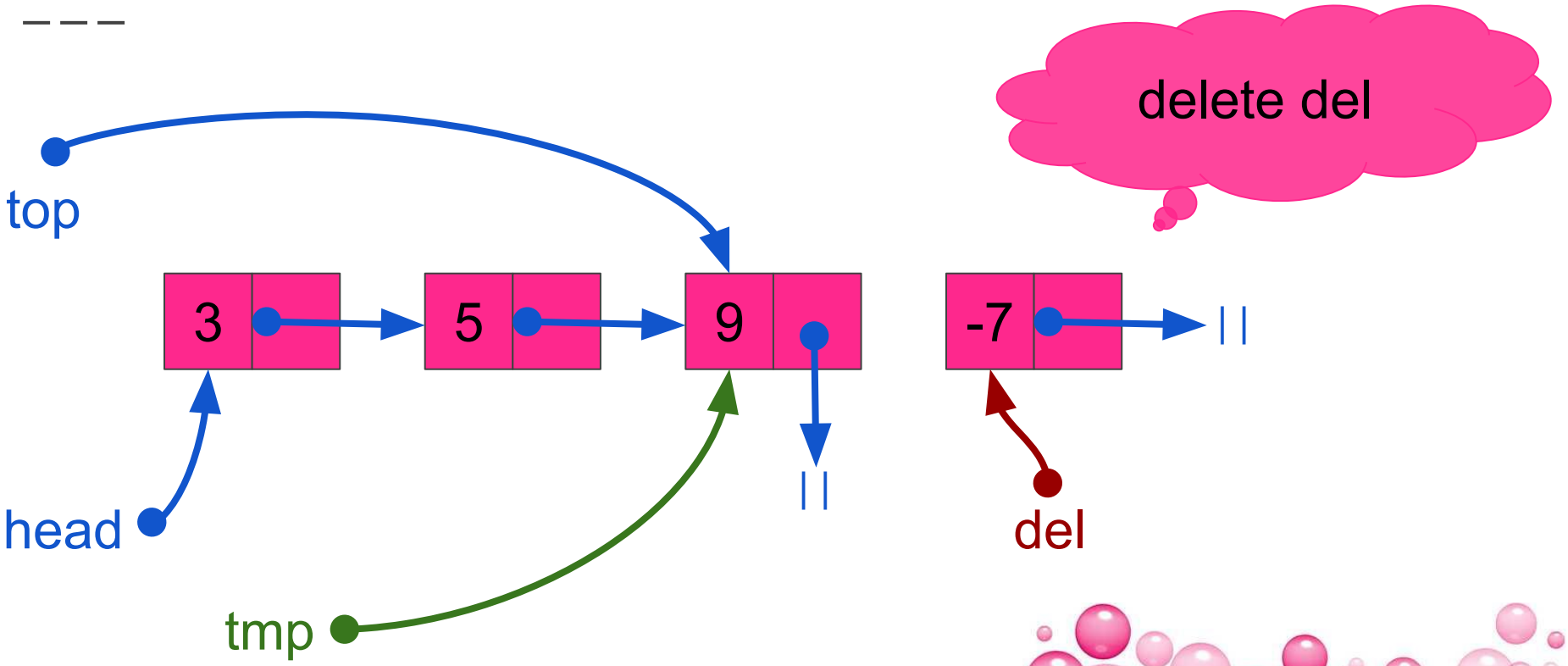
singly-linked list implementation - pop



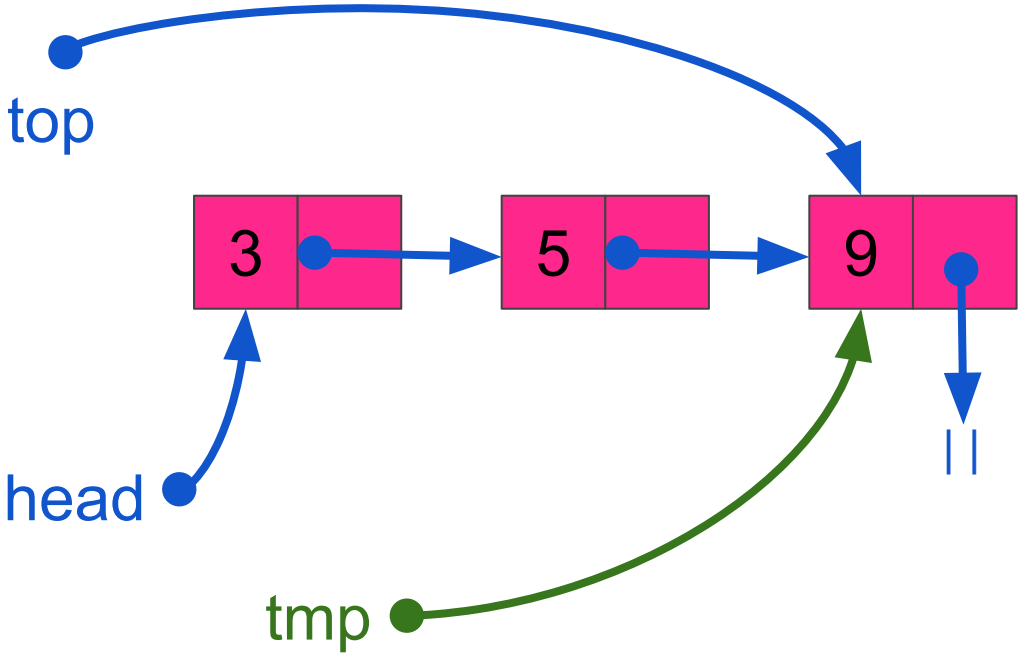
singly-linked list implementation - pop



singly-linked list implementation - pop

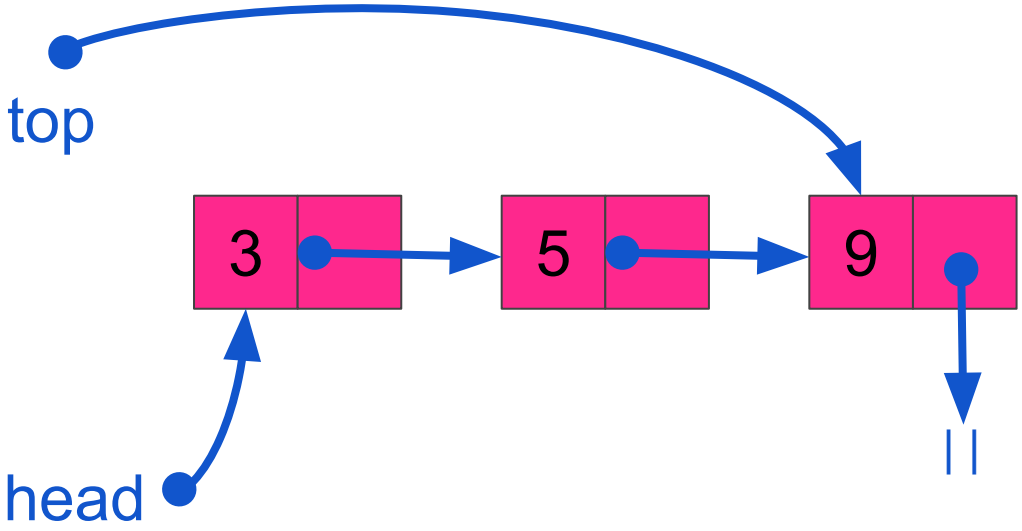


singly-linked list implementation - pop



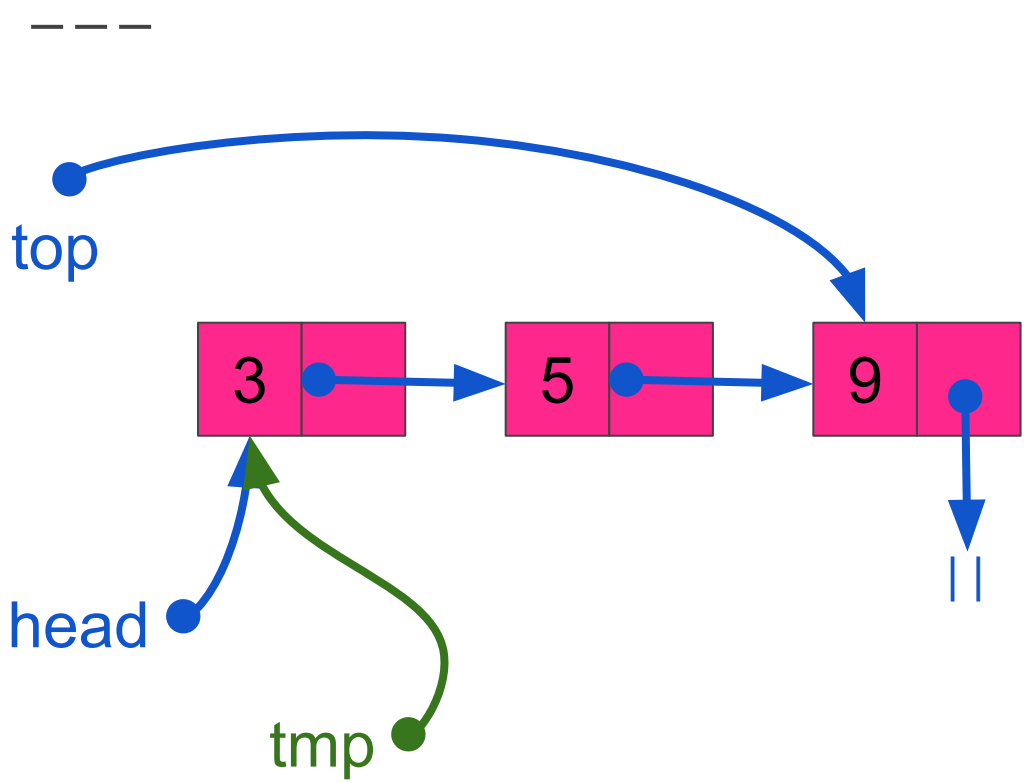


singly-linked list implementation - pop



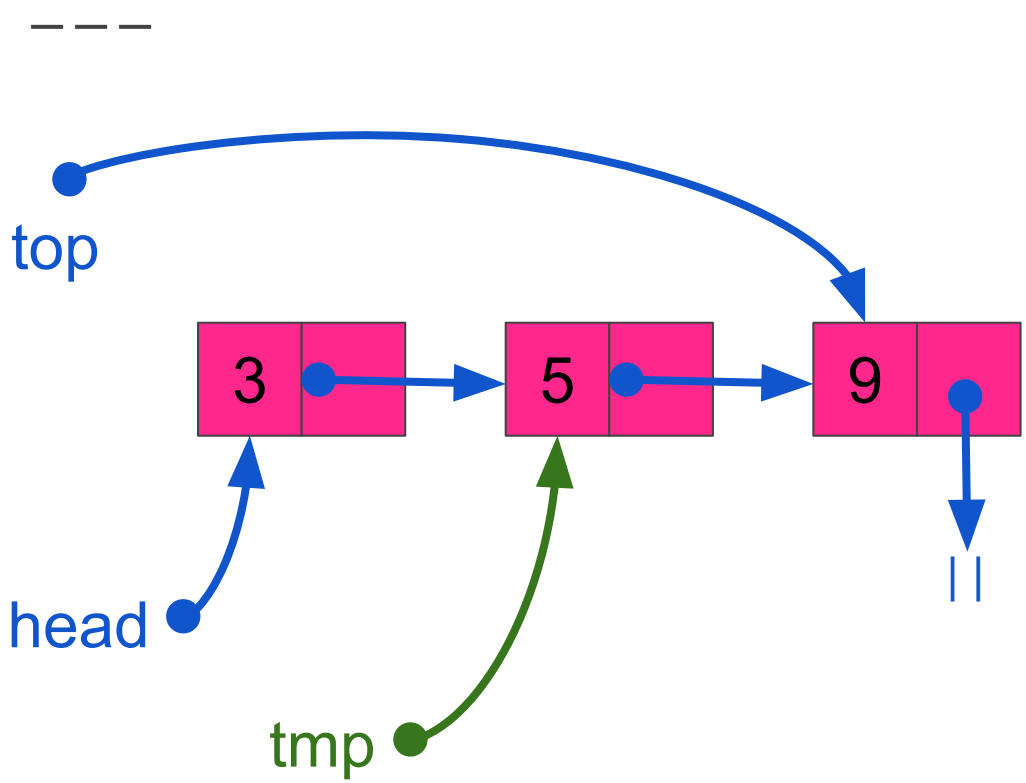


singly-linked list implementation - pop



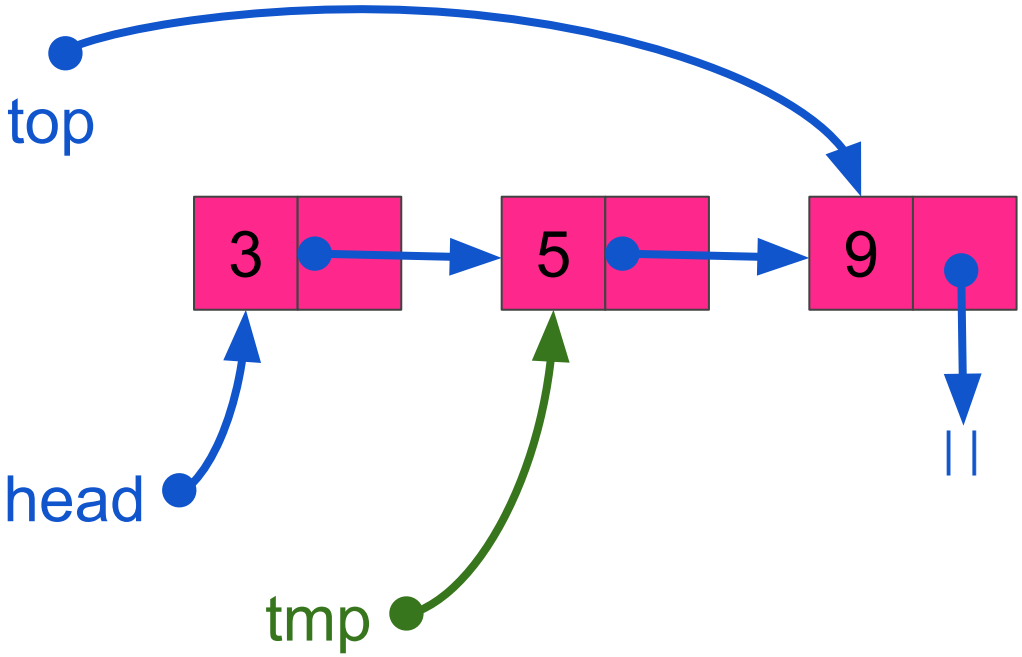
traverse

singly-linked list implementation - pop



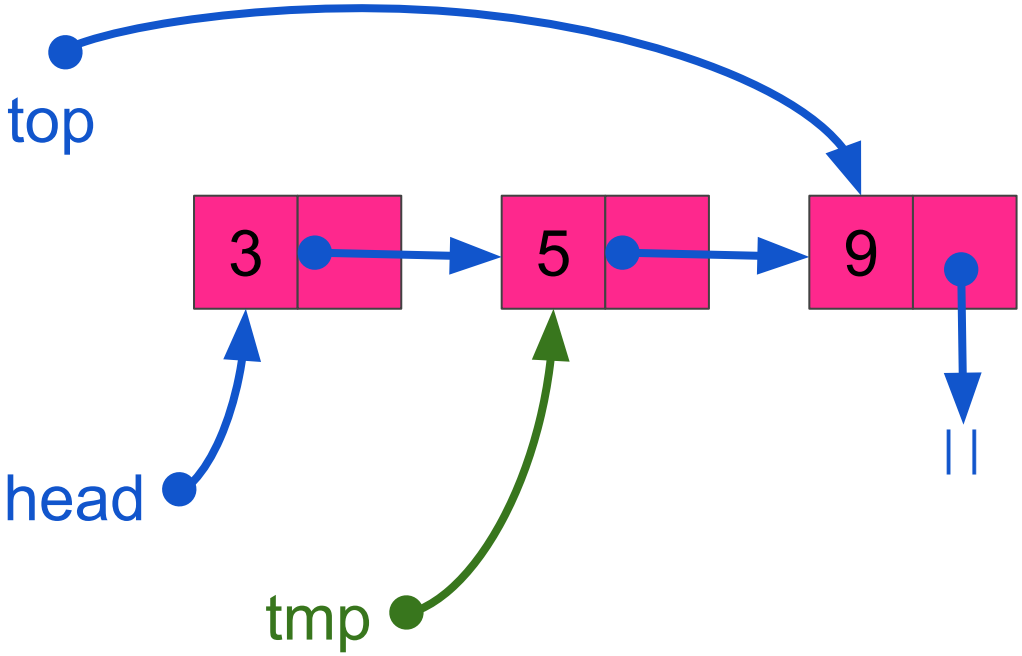
traverse

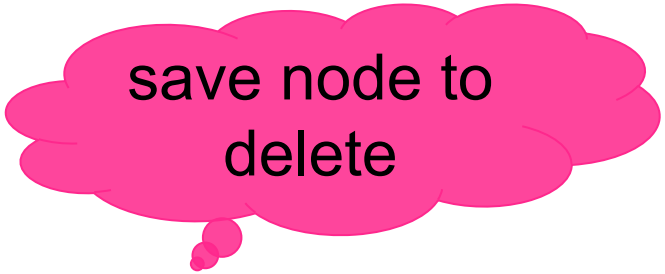
singly-linked list implementation - pop



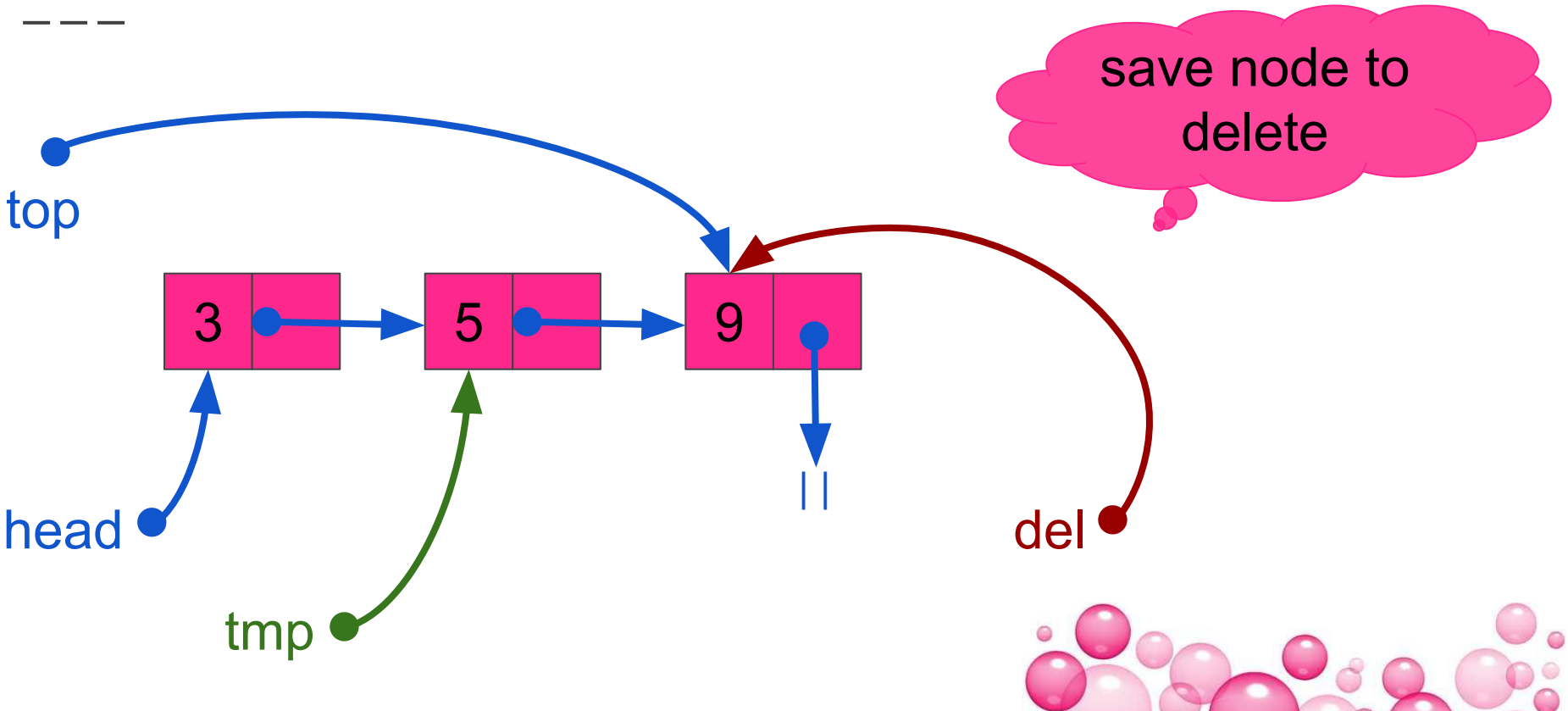
set new top

singly-linked list implementation - pop

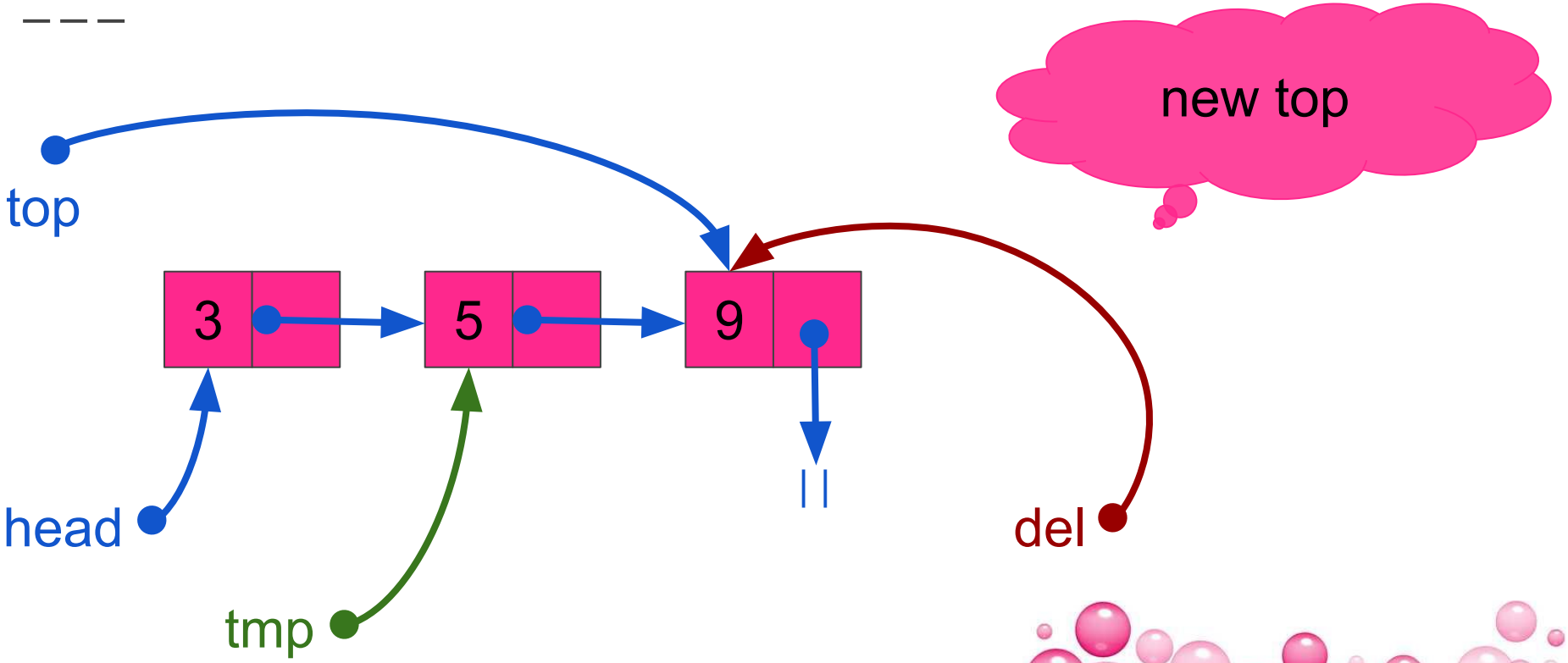




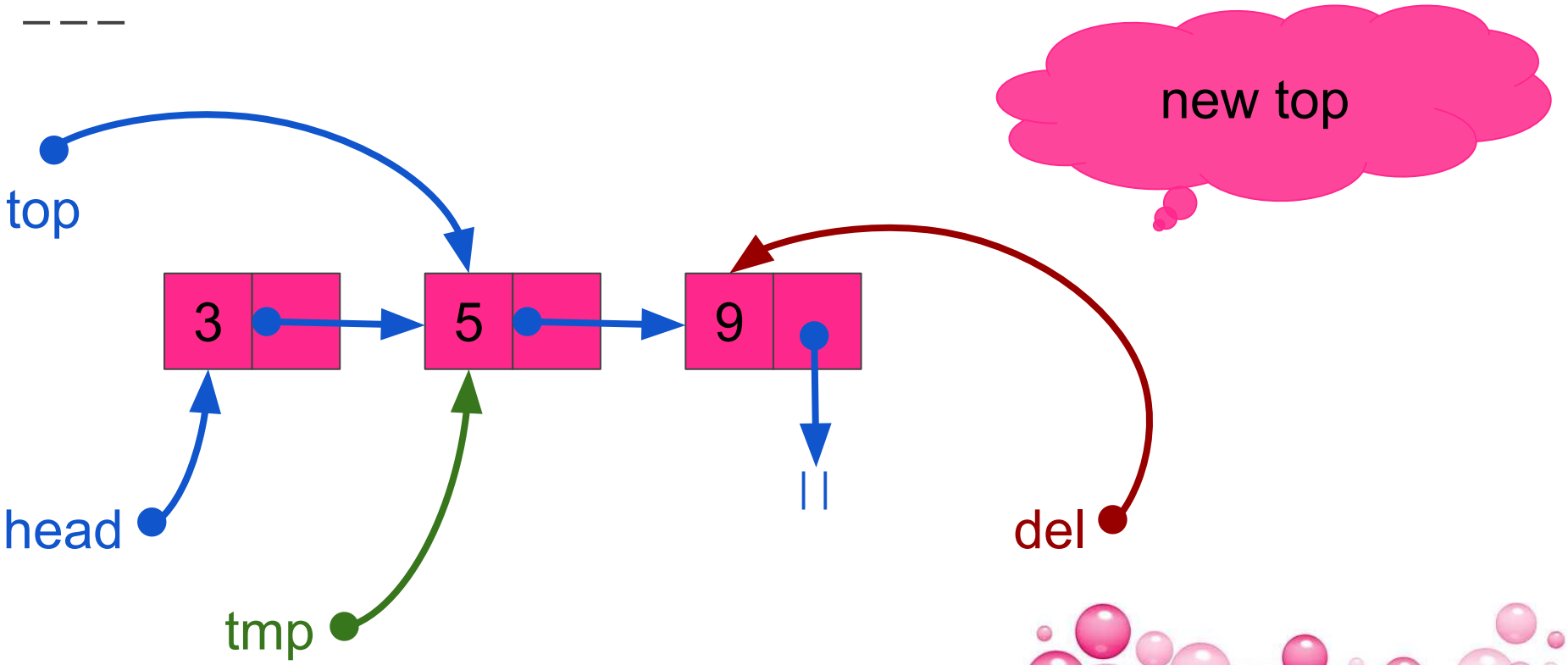
singly-linked list implementation - pop



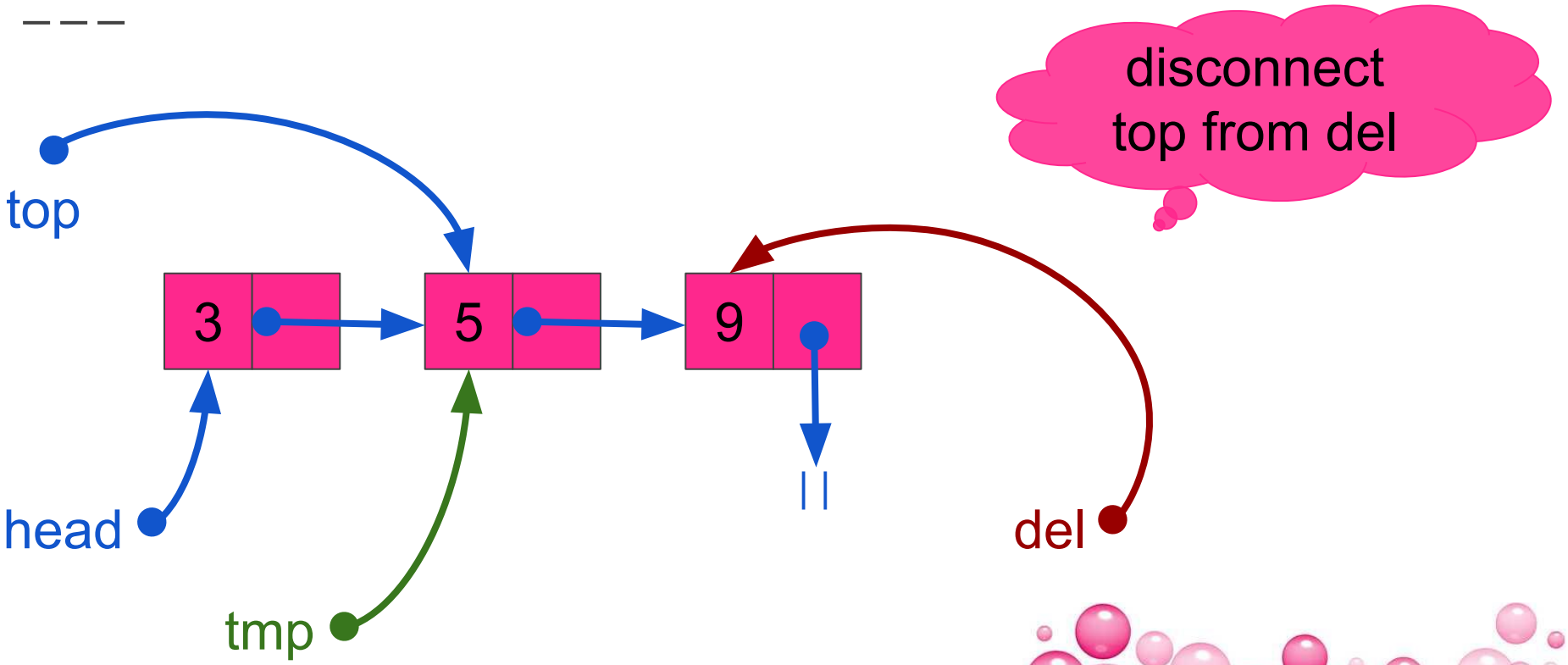
singly-linked list implementation - pop



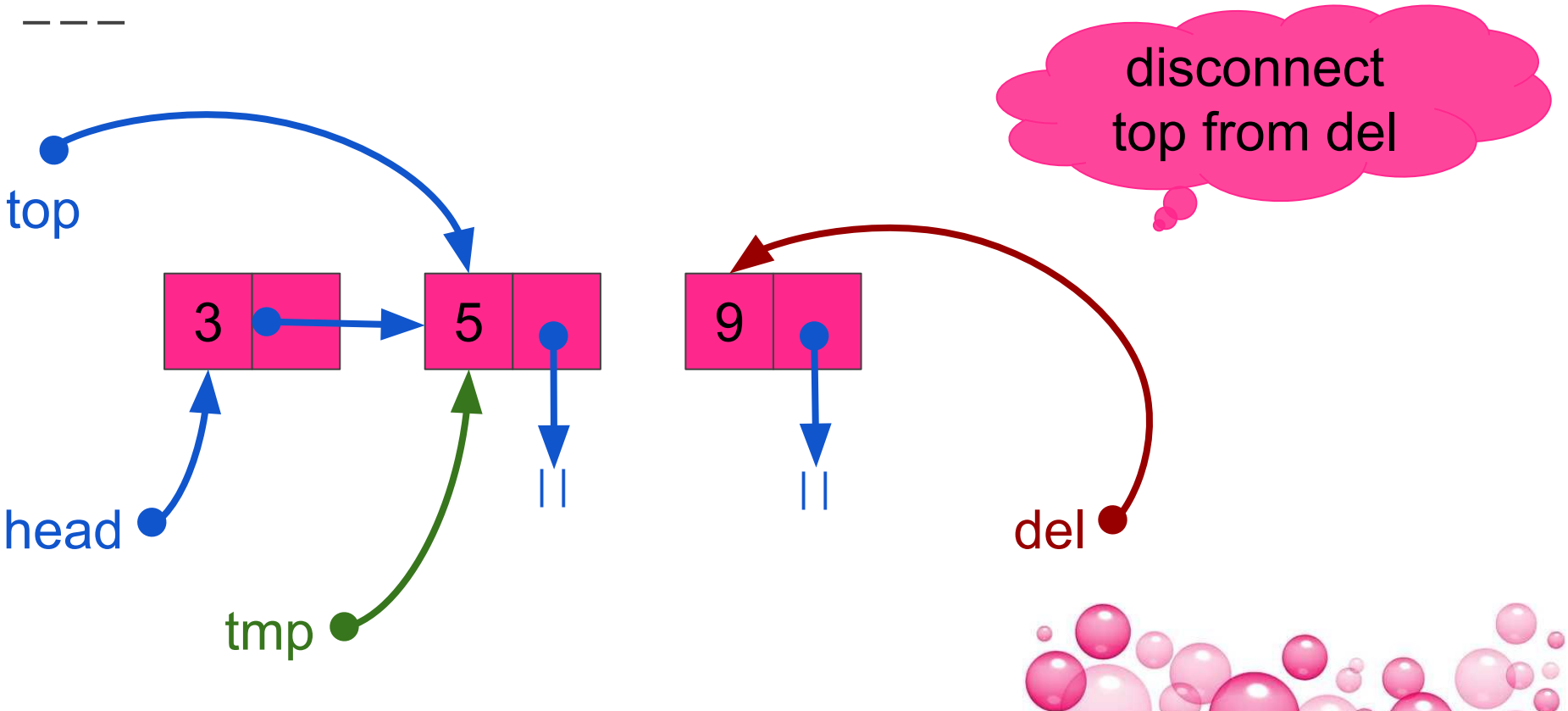
singly-linked list implementation - pop



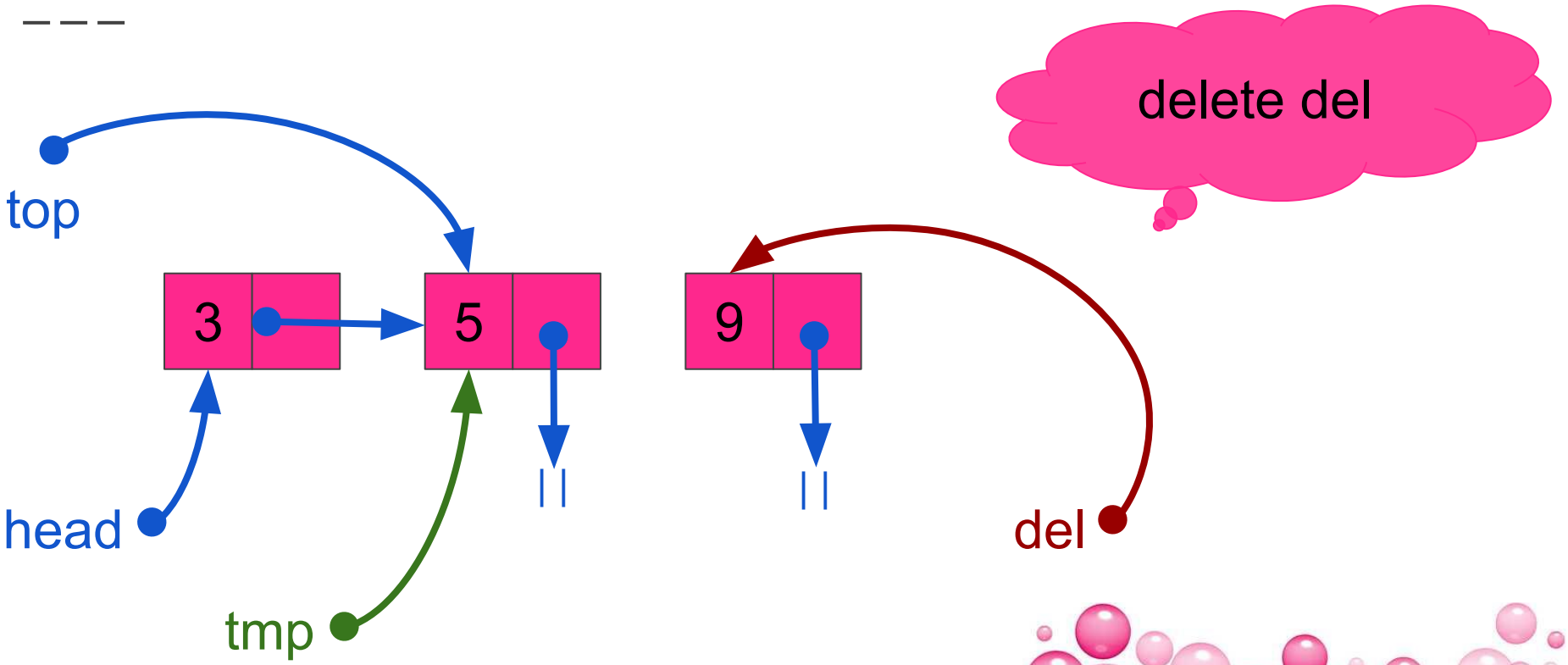
singly-linked list implementation - pop



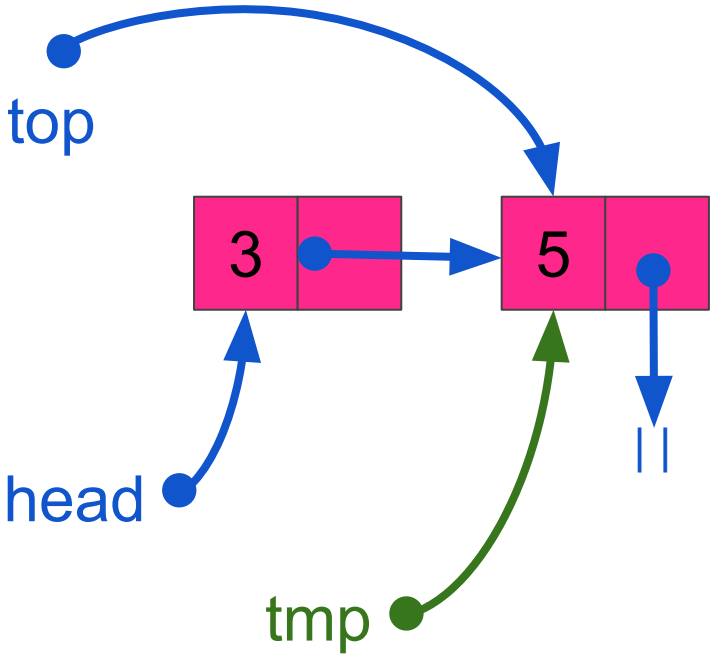
singly-linked list implementation - pop



singly-linked list implementation - pop



singly-linked list implementation - pop

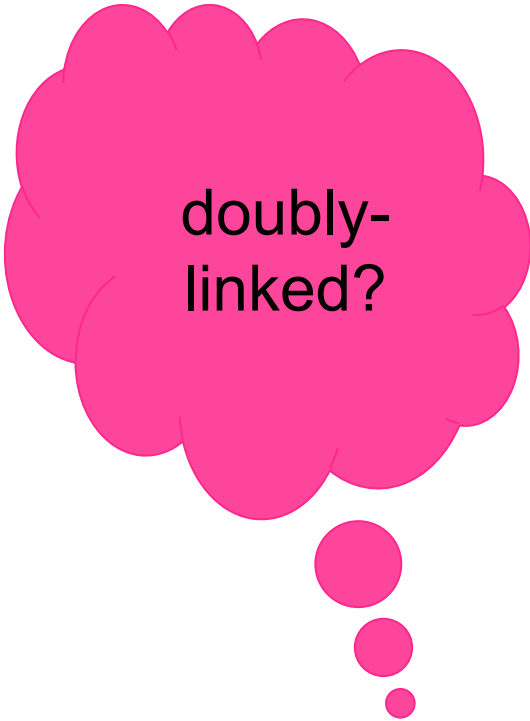






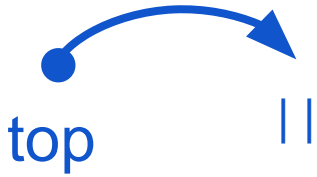
singly-linked list implementation

operation	array	singly-linked
push	O(1)	O(1)
pop	O(1)	O(n)
top	O(1)	O(1)





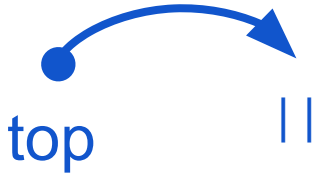
doubly-linked list implementation

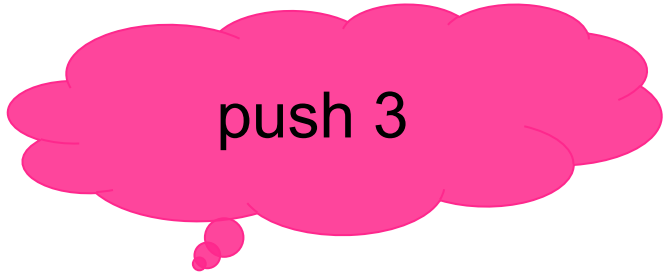


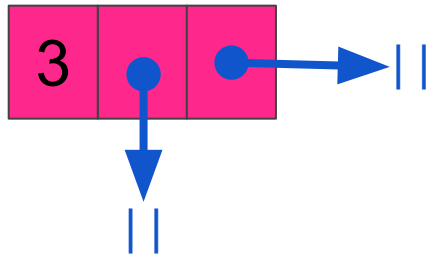




doubly-linked list implementation

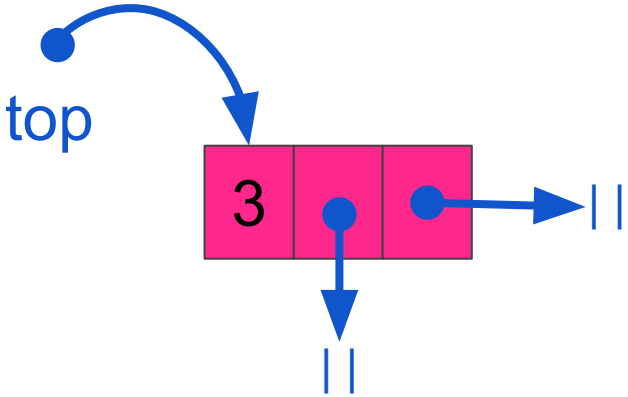


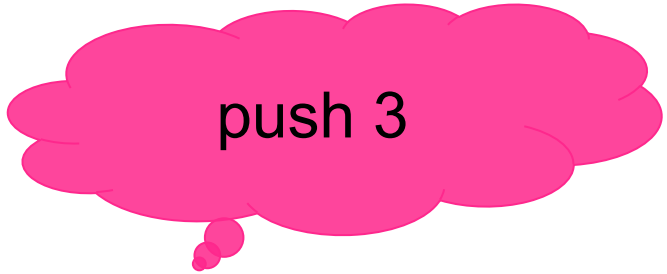




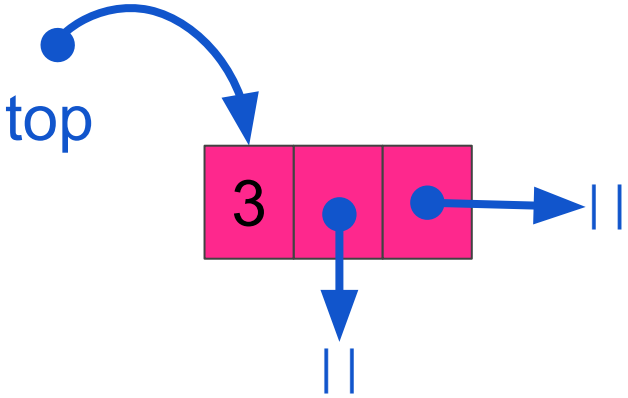


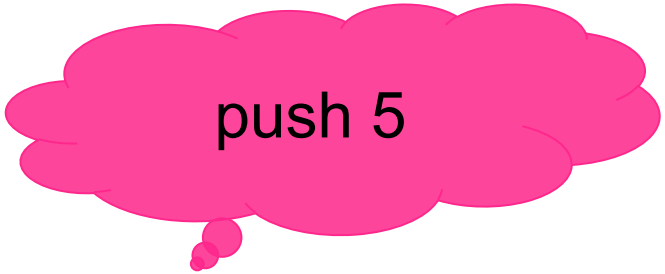
doubly-linked list implementation

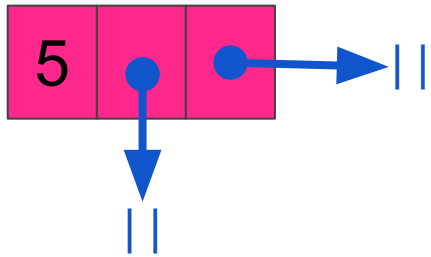




doubly-linked list implementation

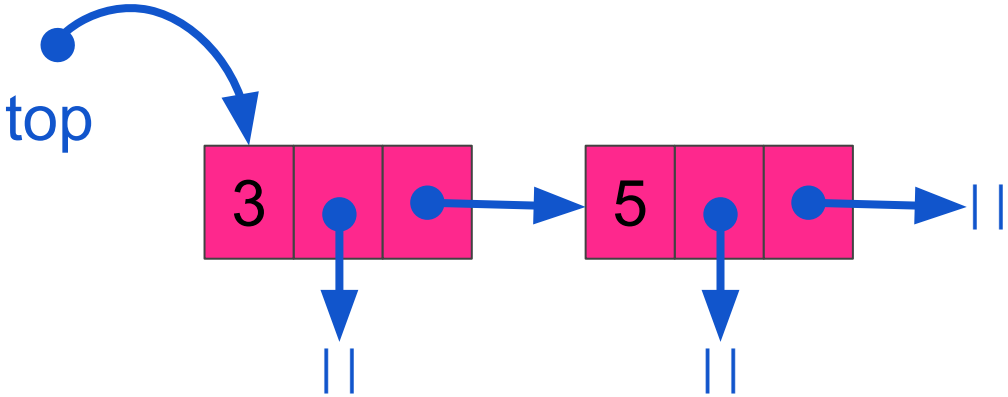


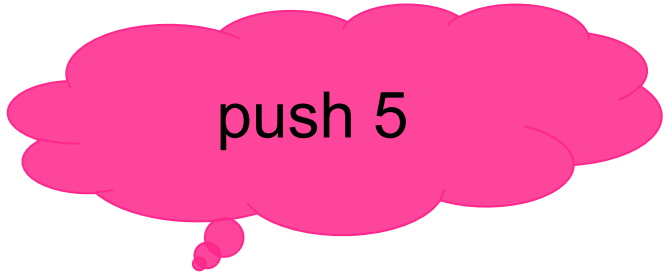




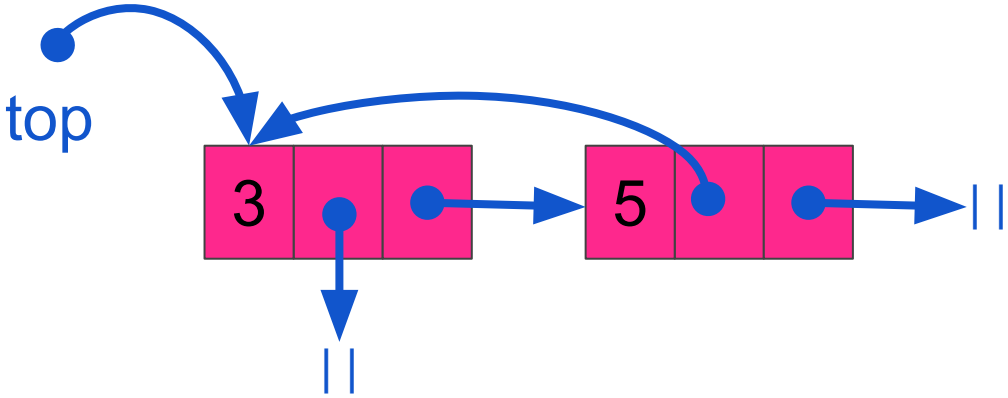


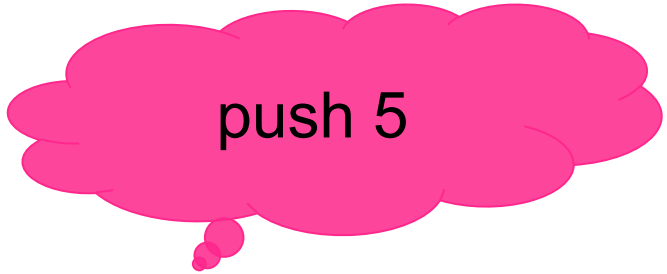
doubly-linked list implementation



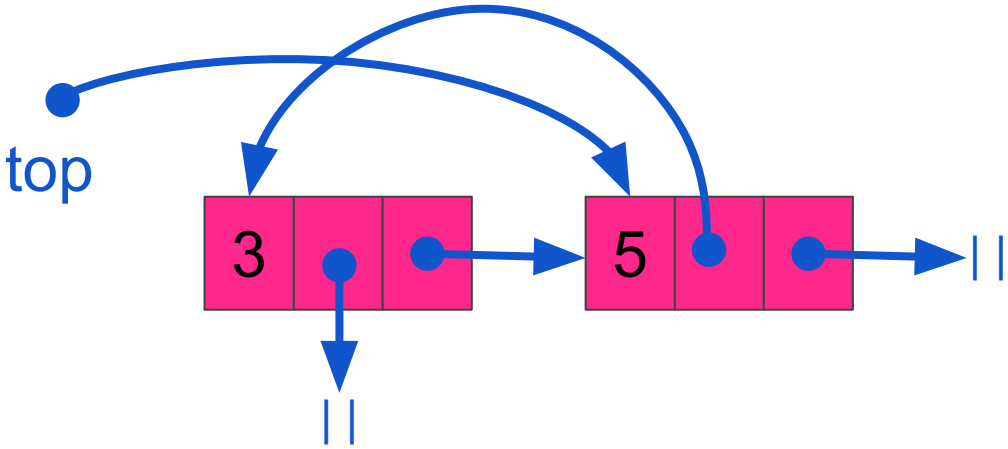


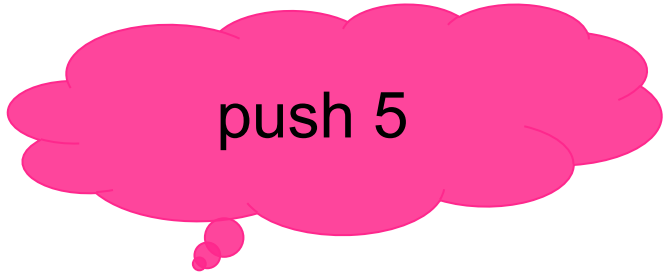
doubly-linked list implementation



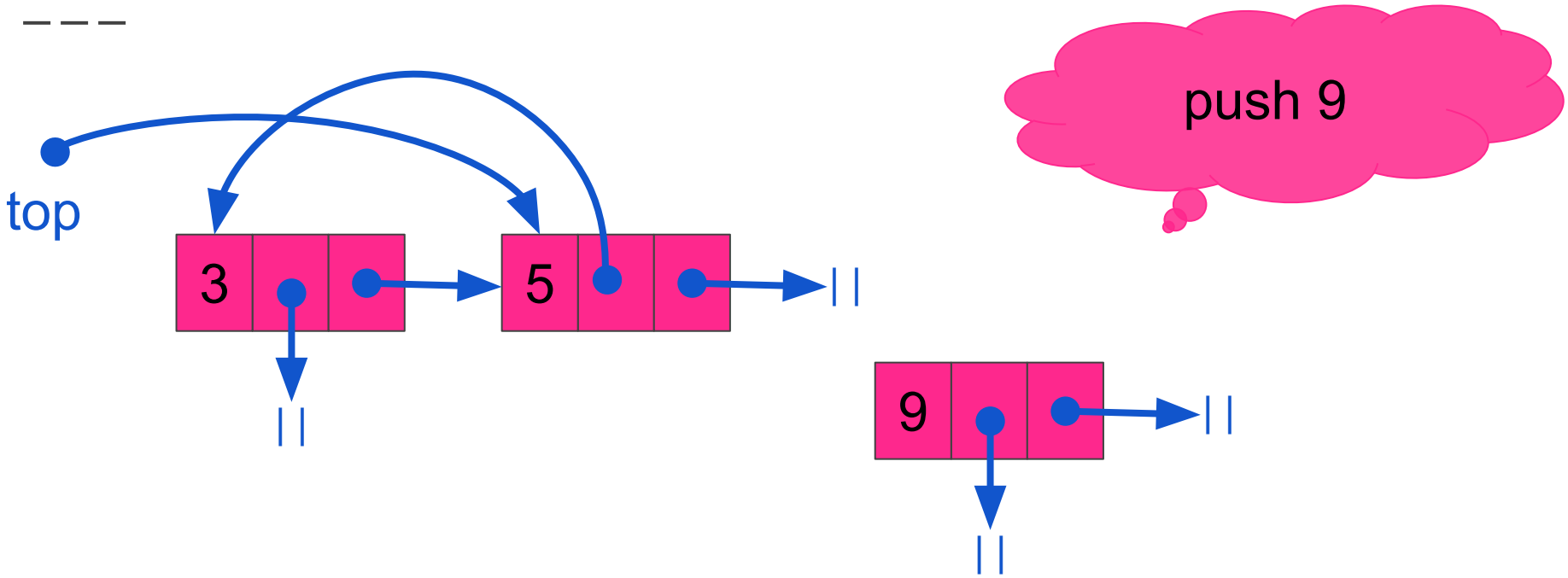


doubly-linked list implementation

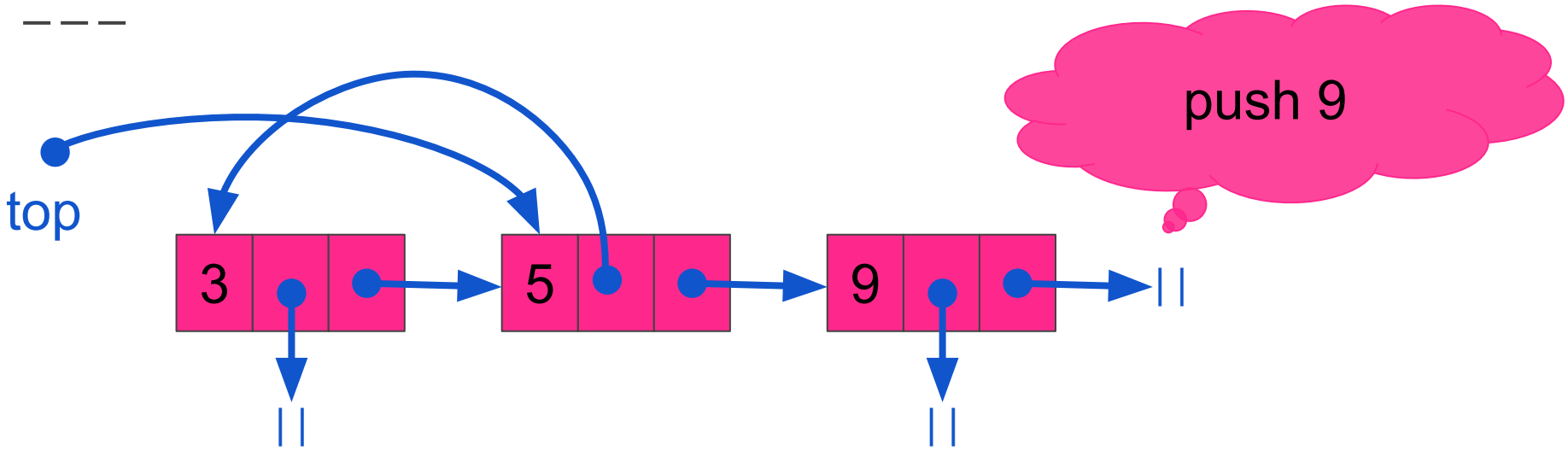




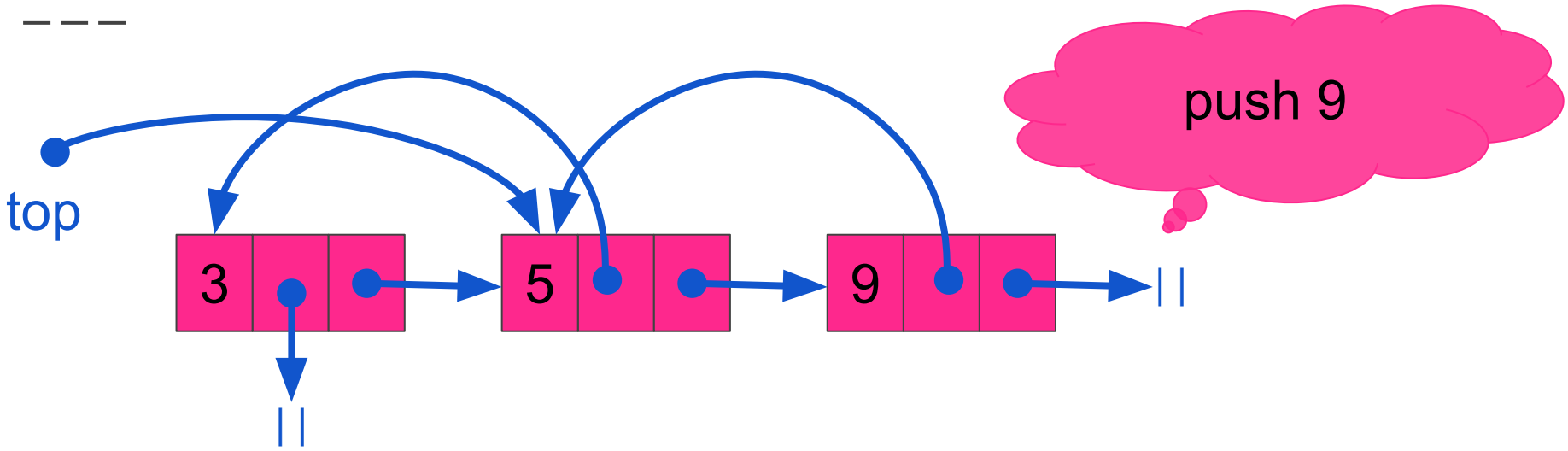
doubly-linked list implementation



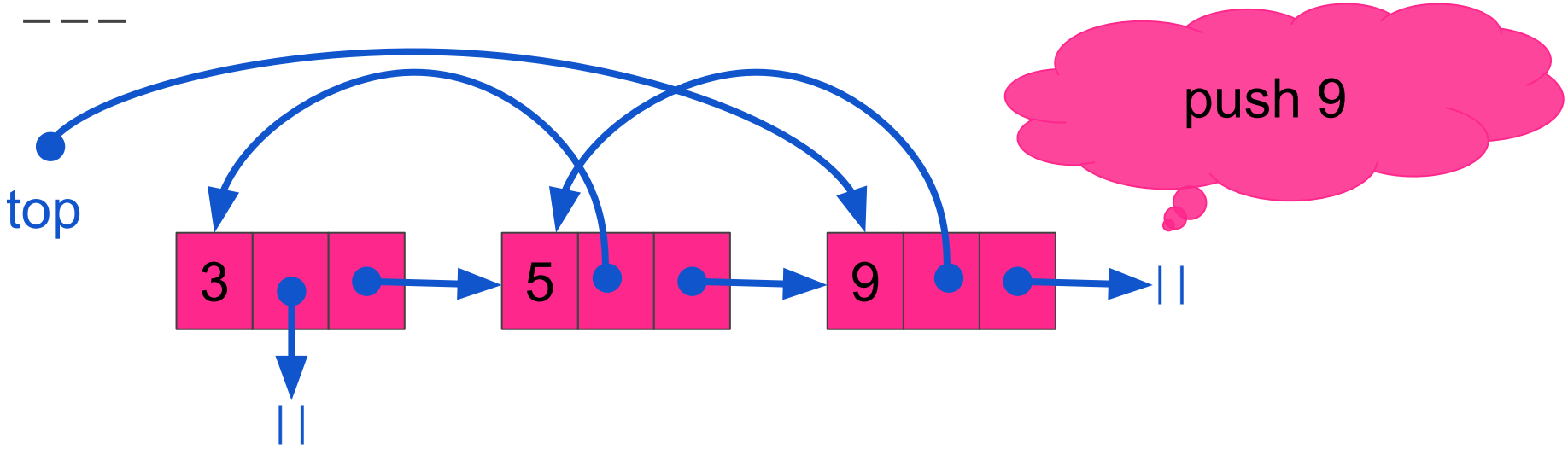
doubly-linked list implementation



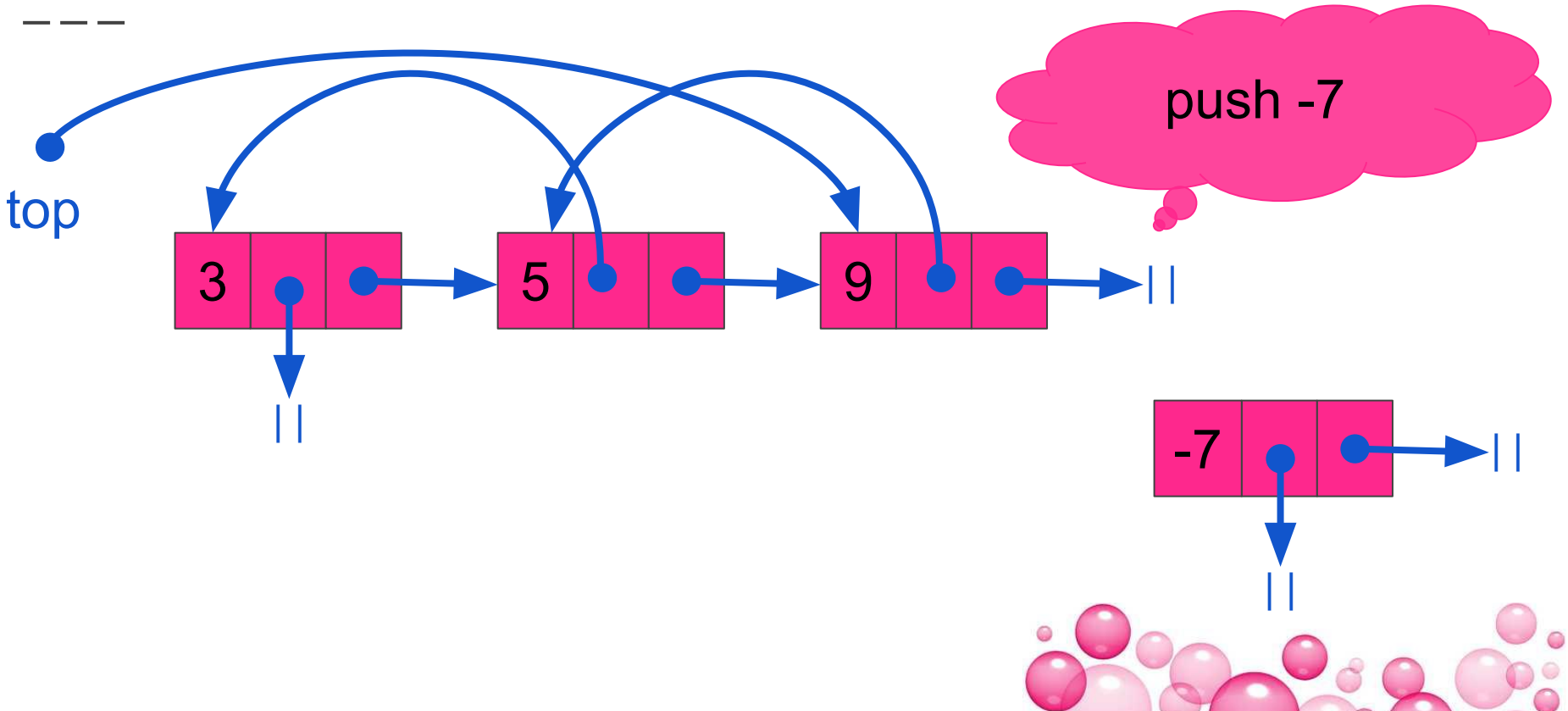
doubly-linked list implementation



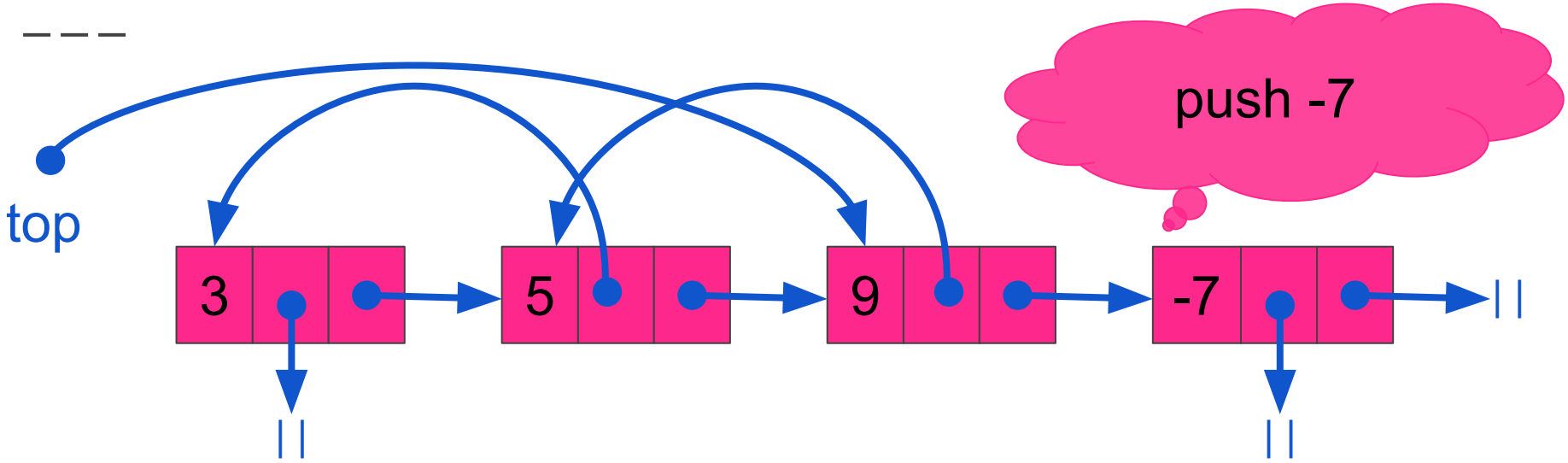
doubly-linked list implementation



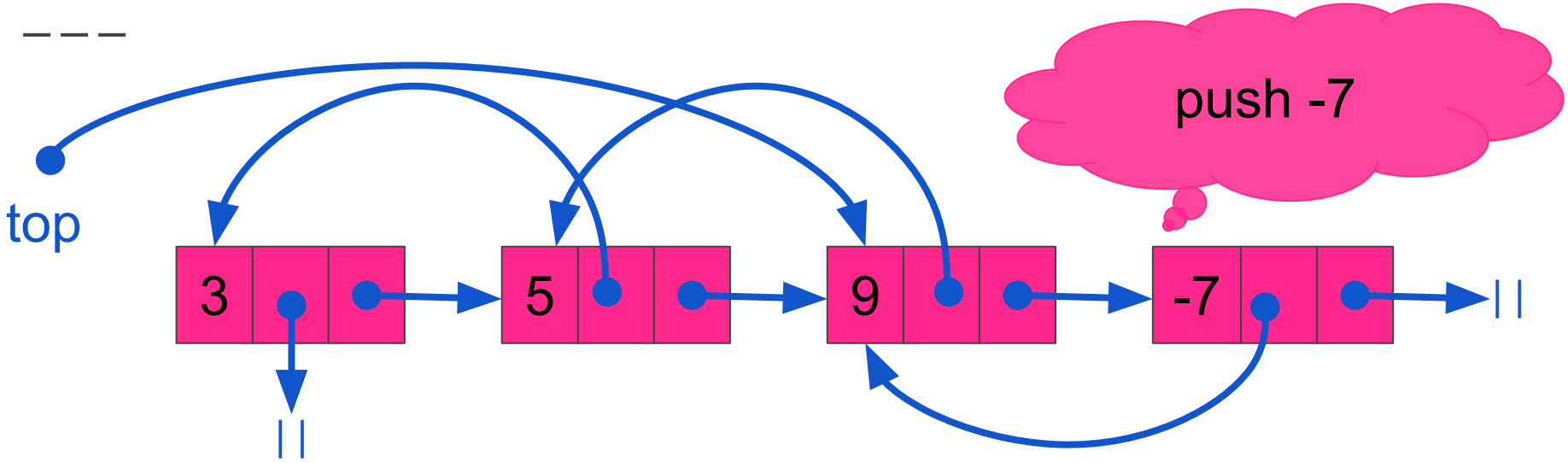
doubly-linked list implementation



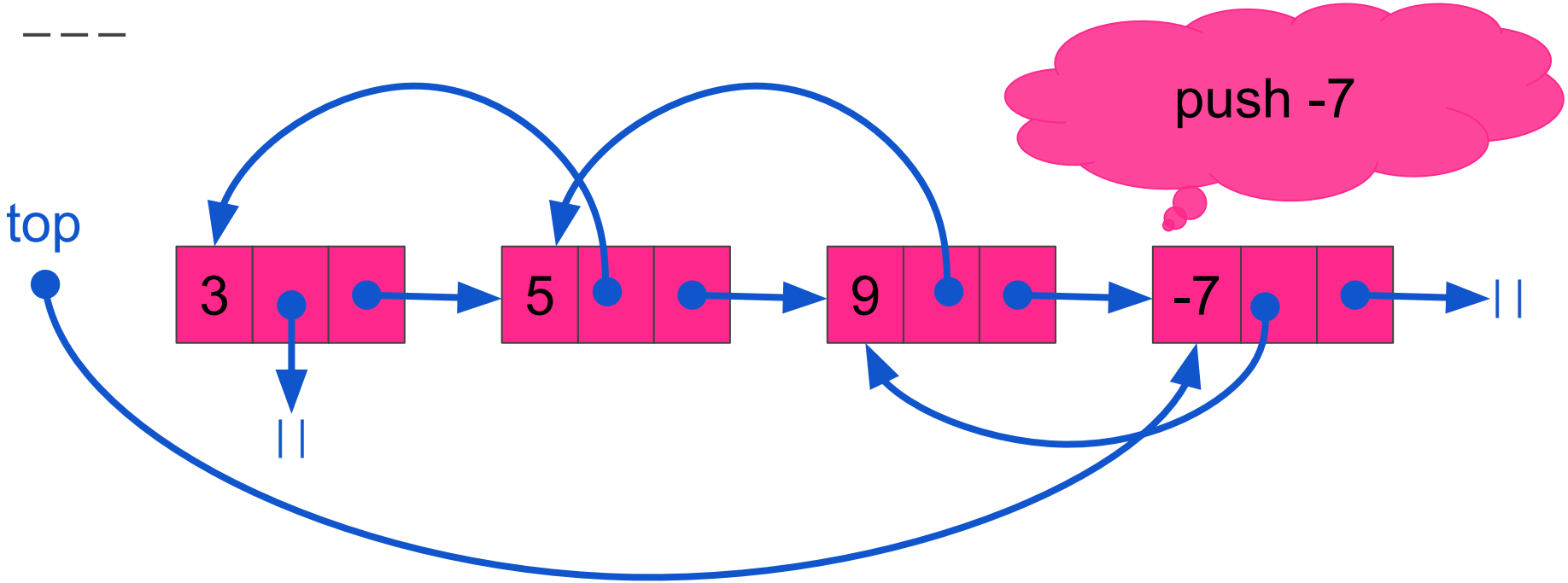
doubly-linked list implementation



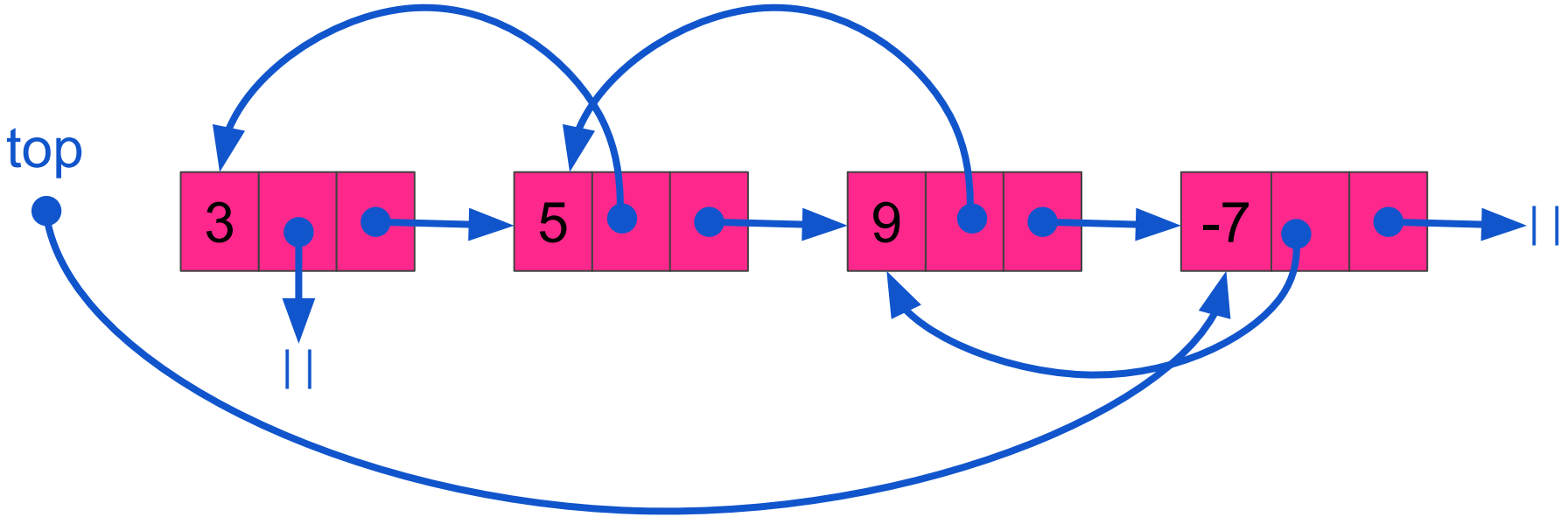
doubly-linked list implementation



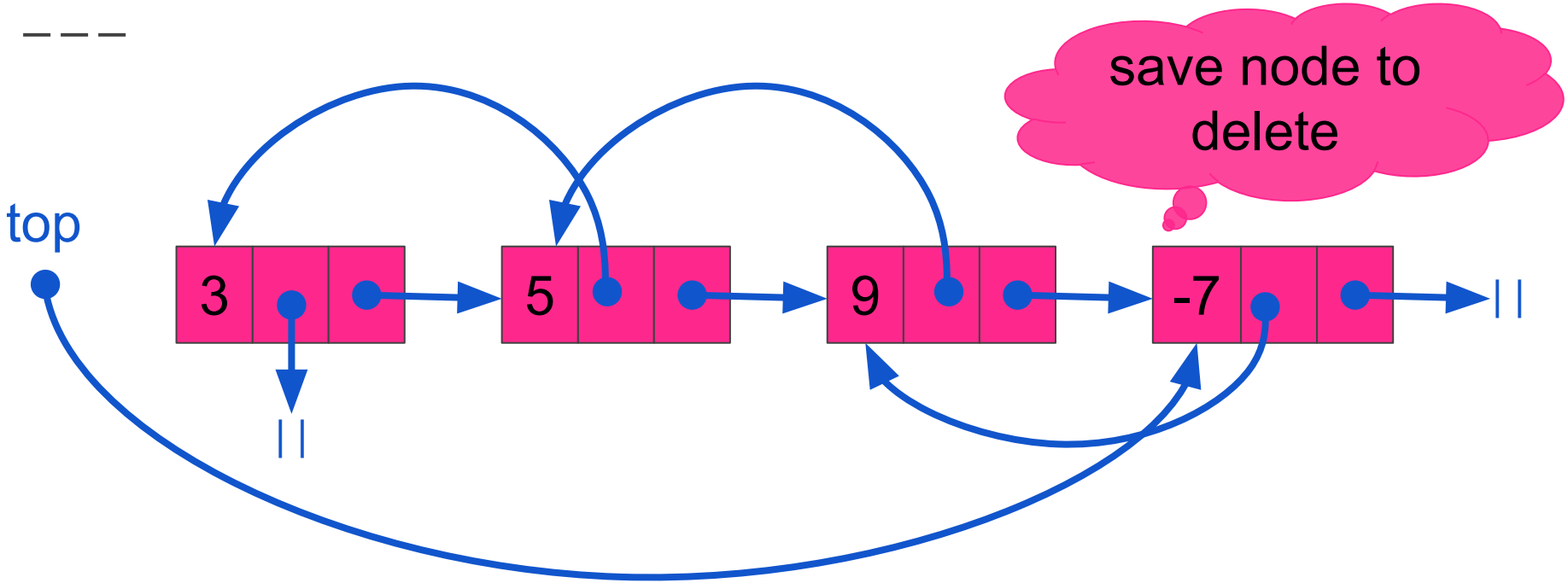
doubly-linked list implementation



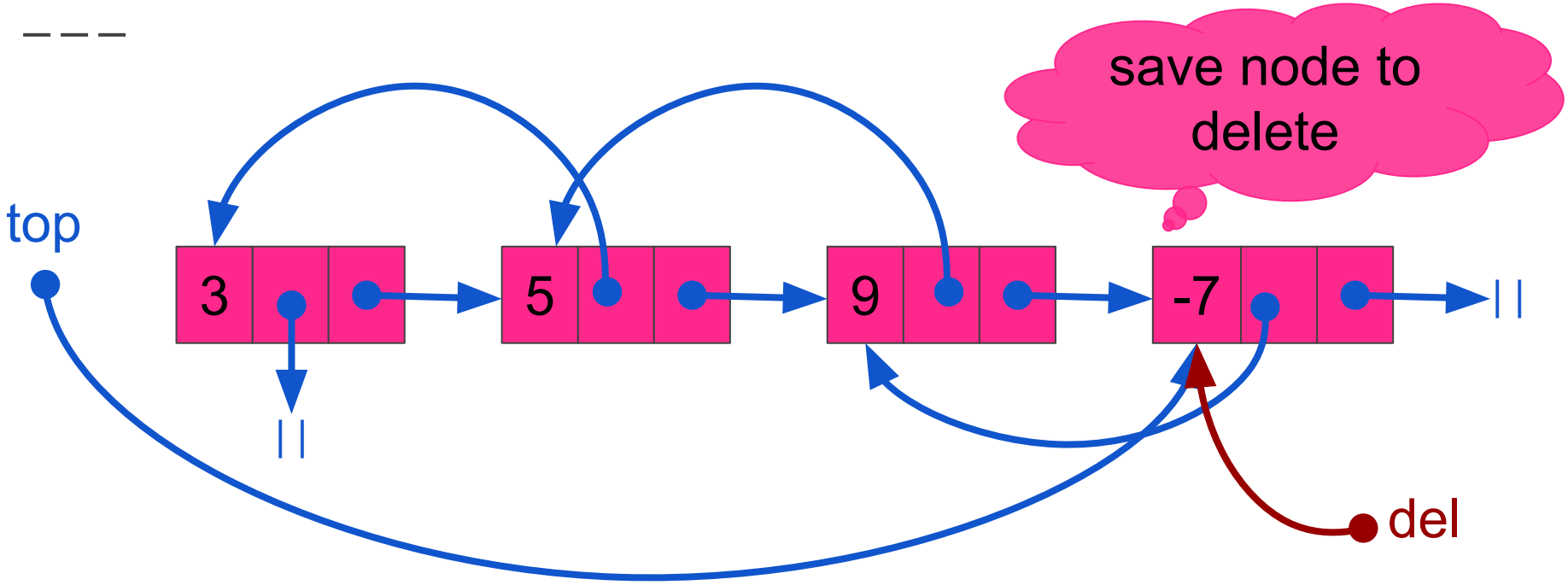
doubly-linked list implementation - pop



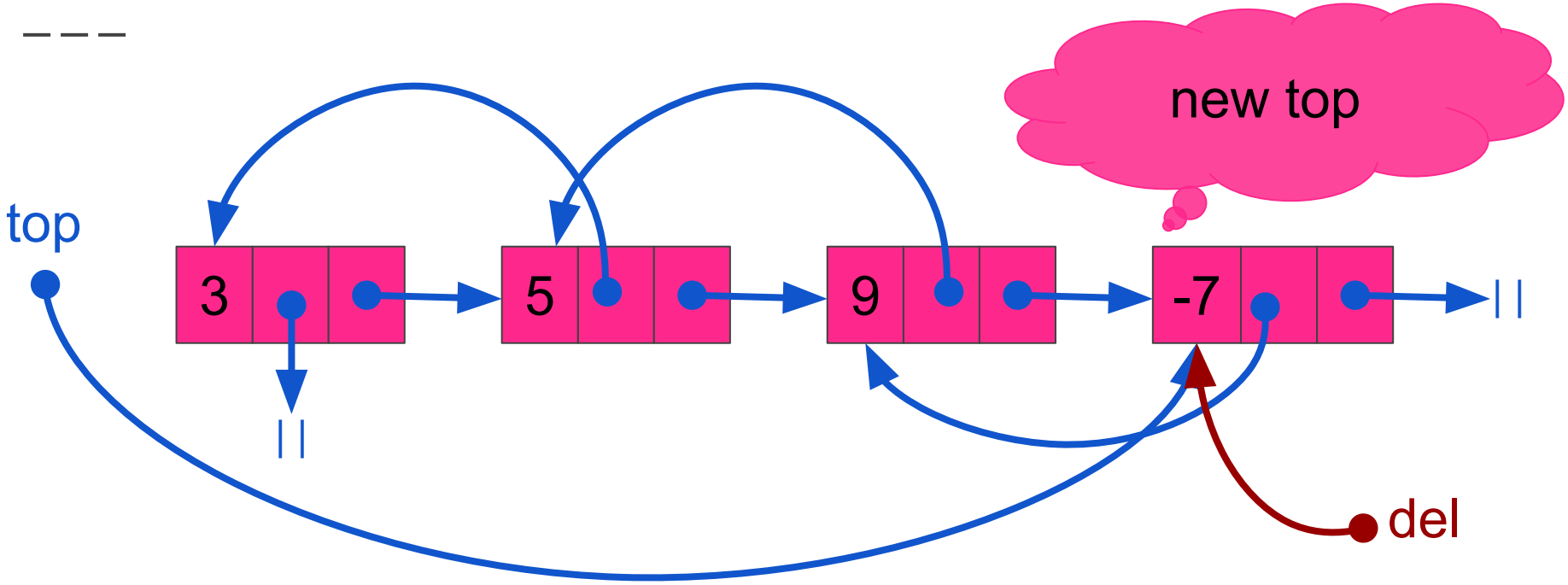
doubly-linked list implementation



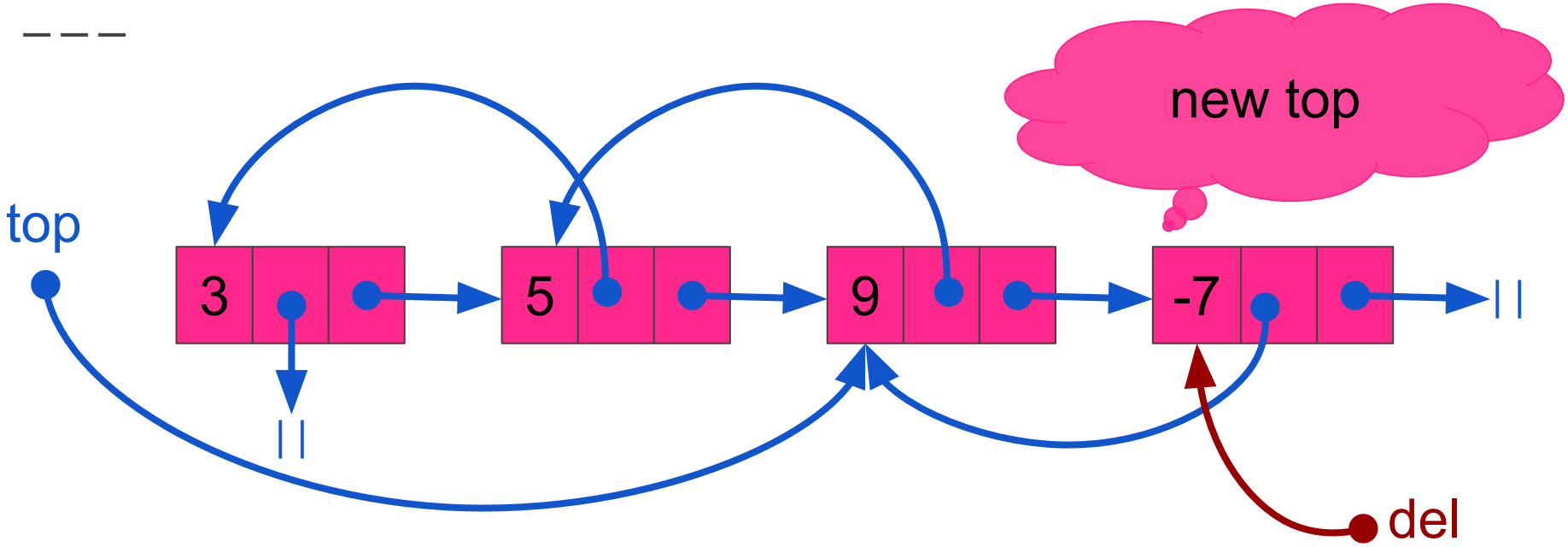
doubly-linked list implementation



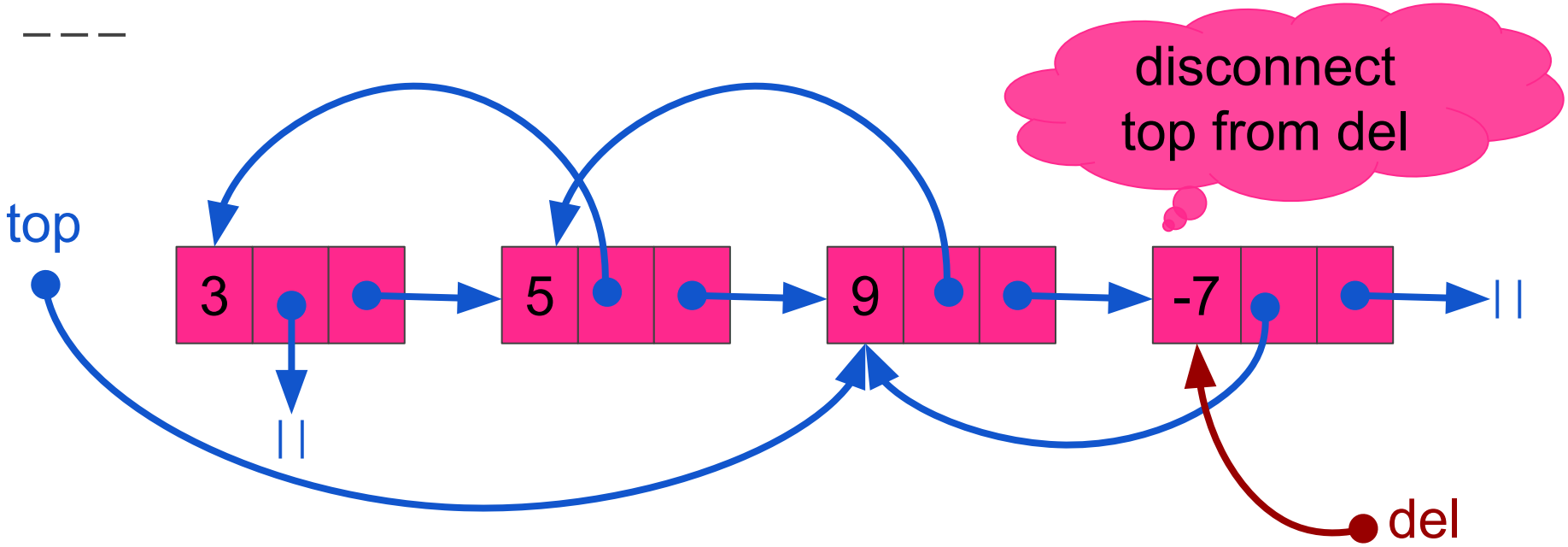
doubly-linked list implementation



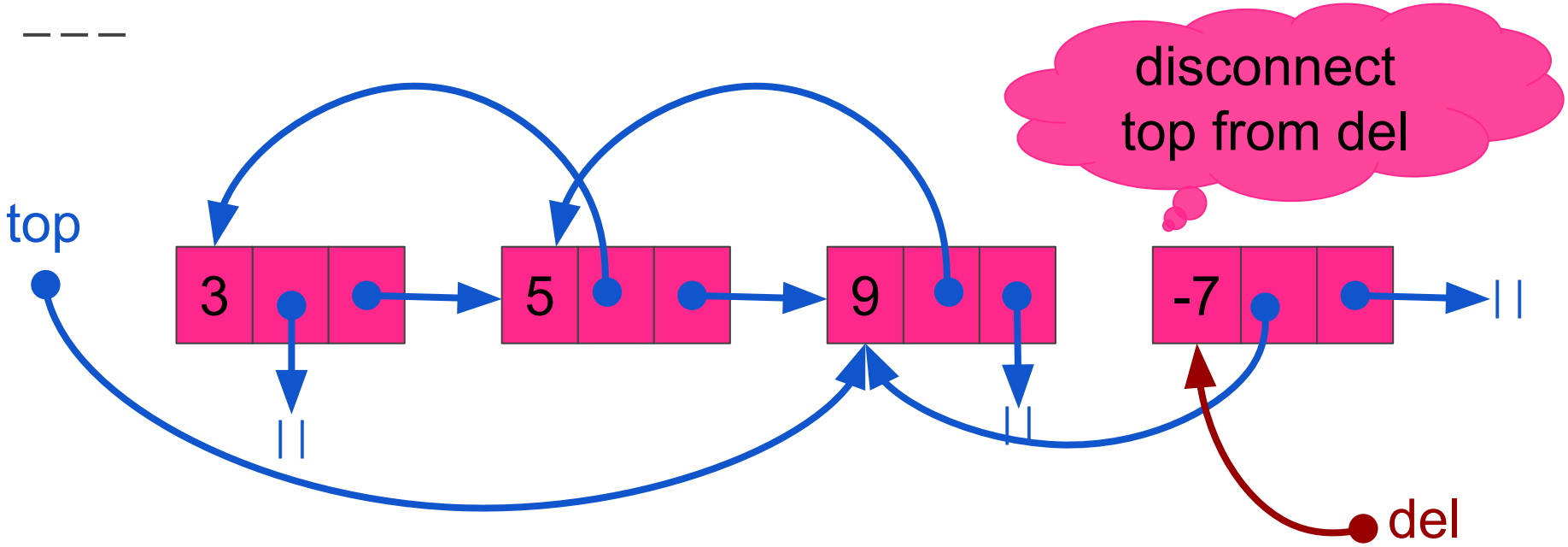
doubly-linked list implementation



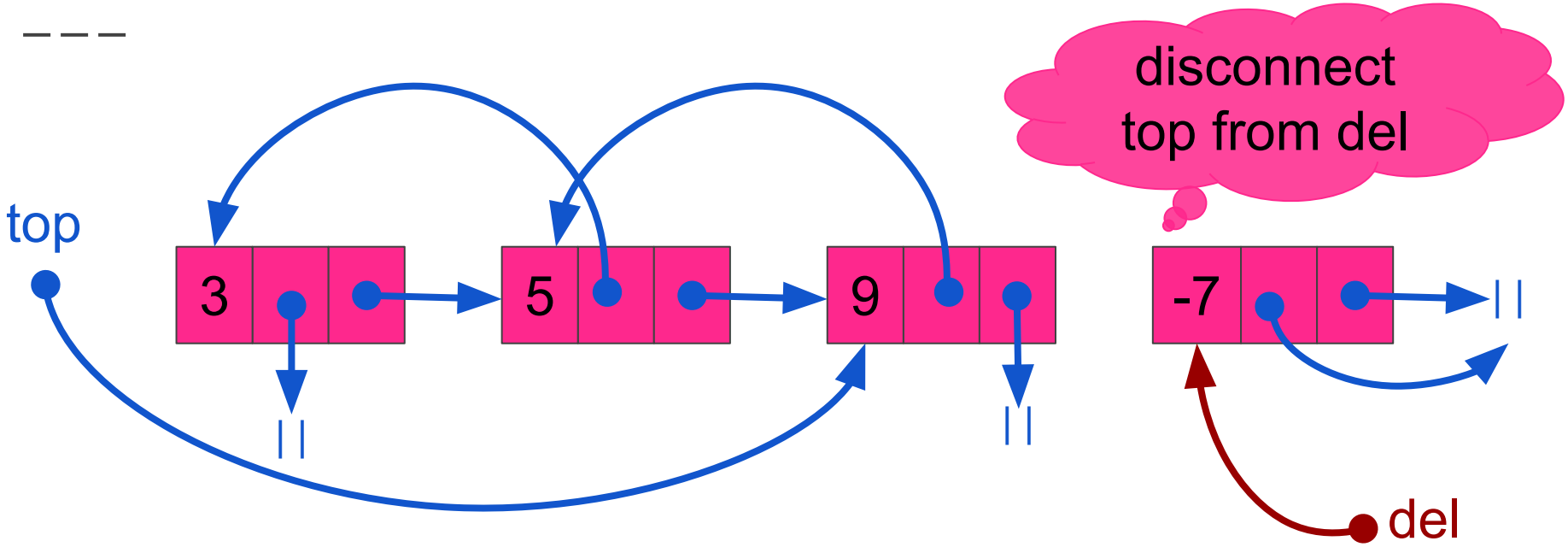
doubly-linked list implementation



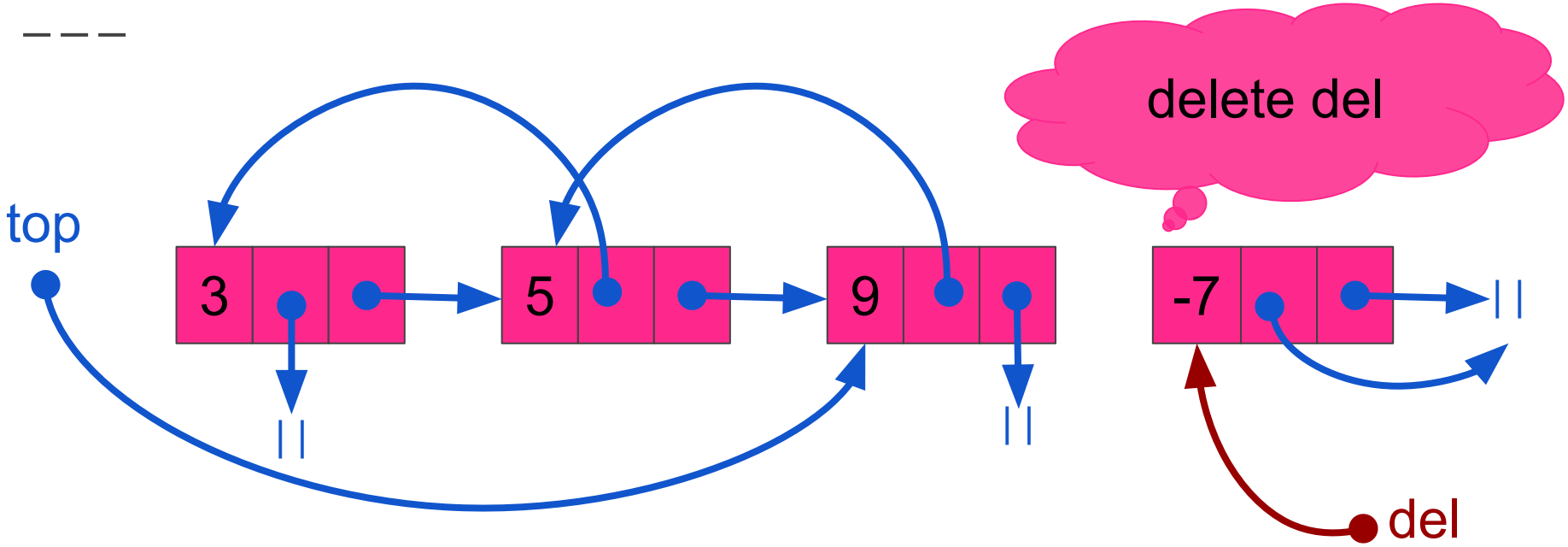
doubly-linked list implementation



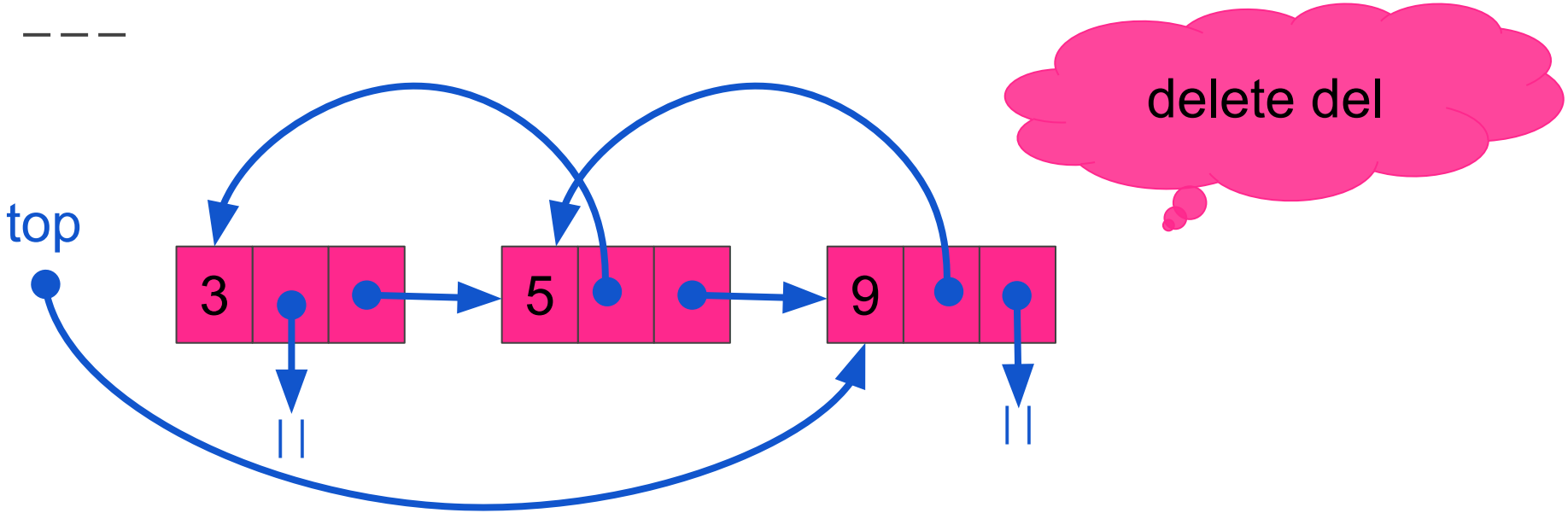
doubly-linked list implementation



doubly-linked list implementation



doubly-linked list implementation



doubly-linked list implementation

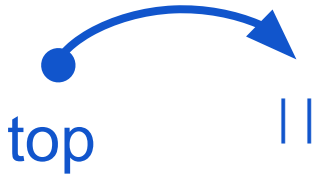
– – –



operation	array	singly-linked	doubly-linked
push	O(1)	O(1)	O(1)
pop	O(1)	O(n)	O(1)
top	O(1)	O(1)	O(1)



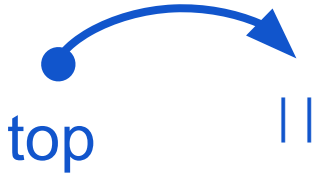
singly-linked list implementation

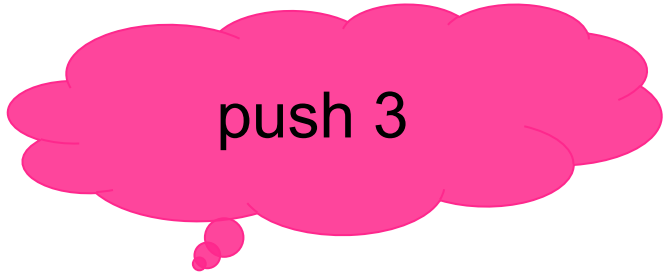


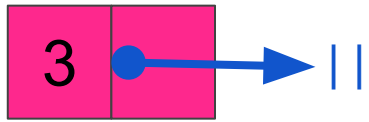




singly-linked list implementation

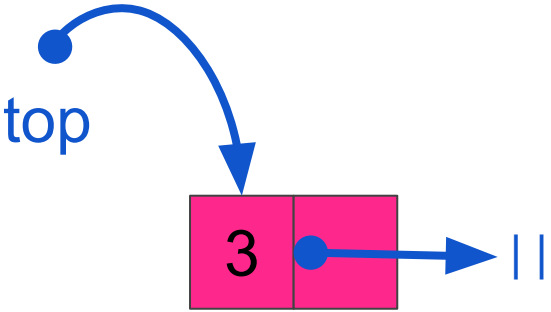


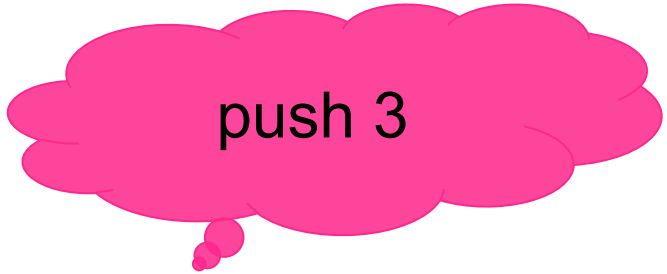




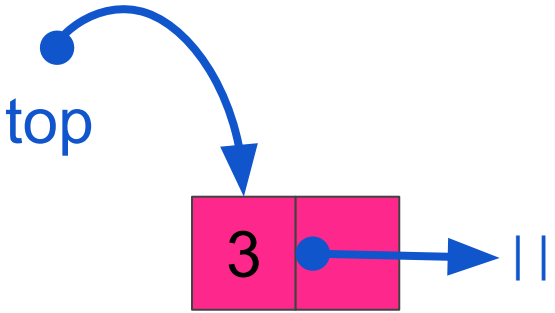


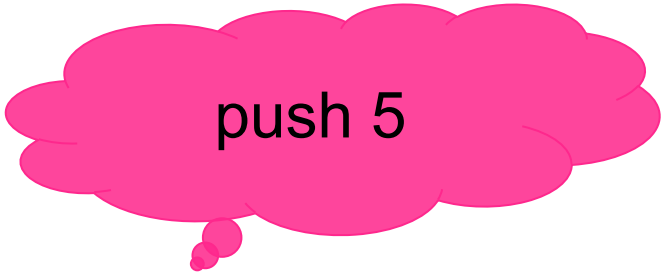
singly-linked list implementation

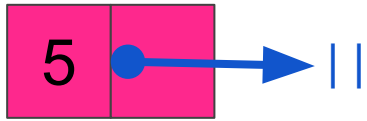




singly-linked list implementation

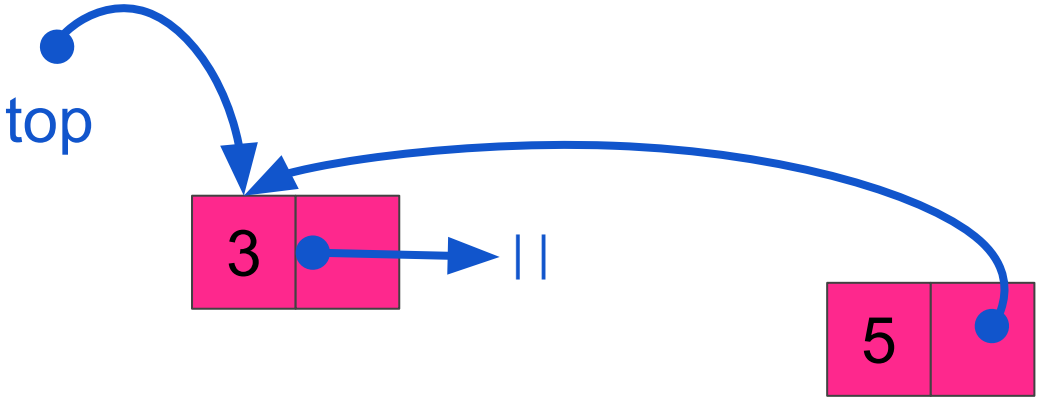


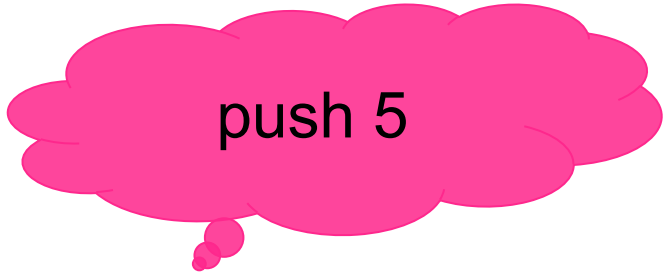




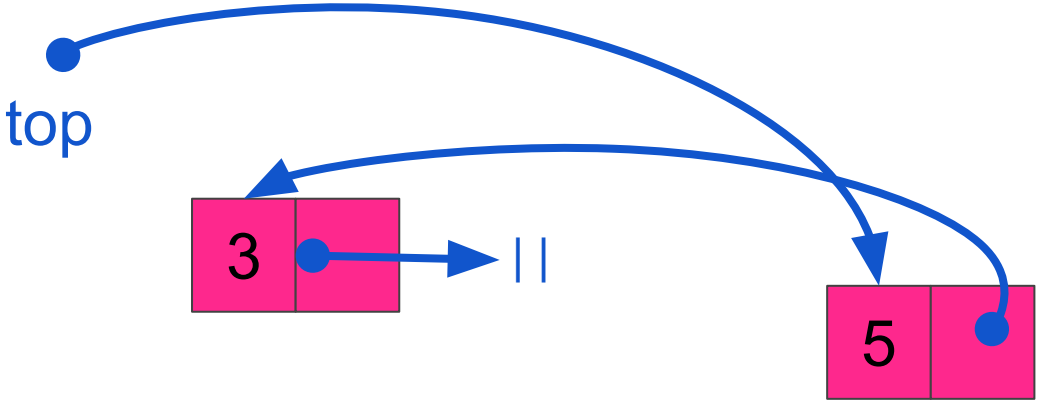


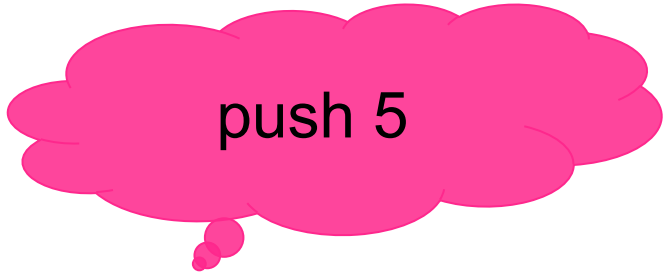
singly-linked list implementation



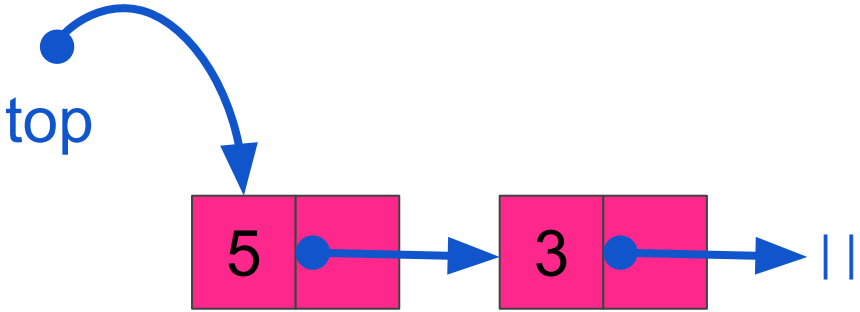


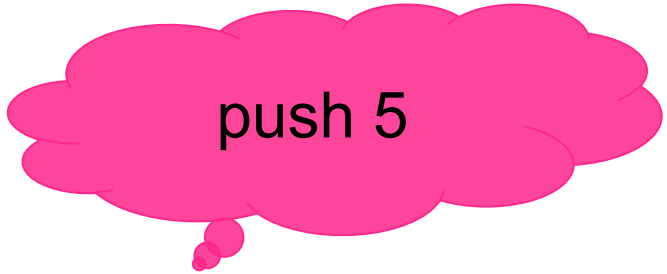
singly-linked list implementation



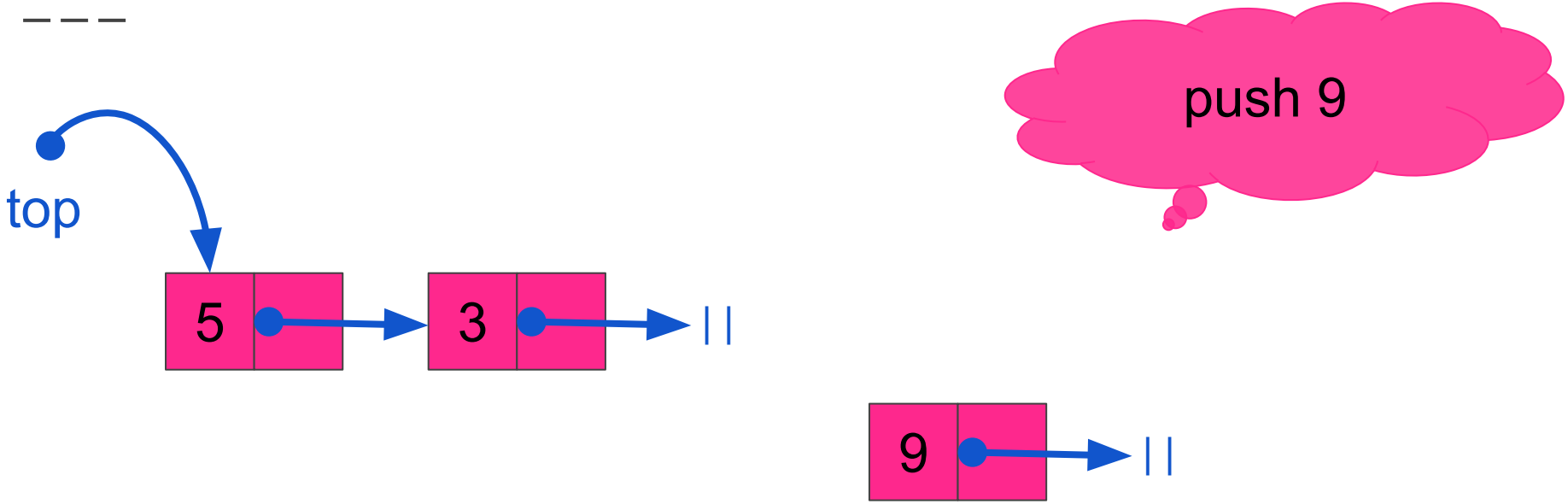


singly-linked list implementation

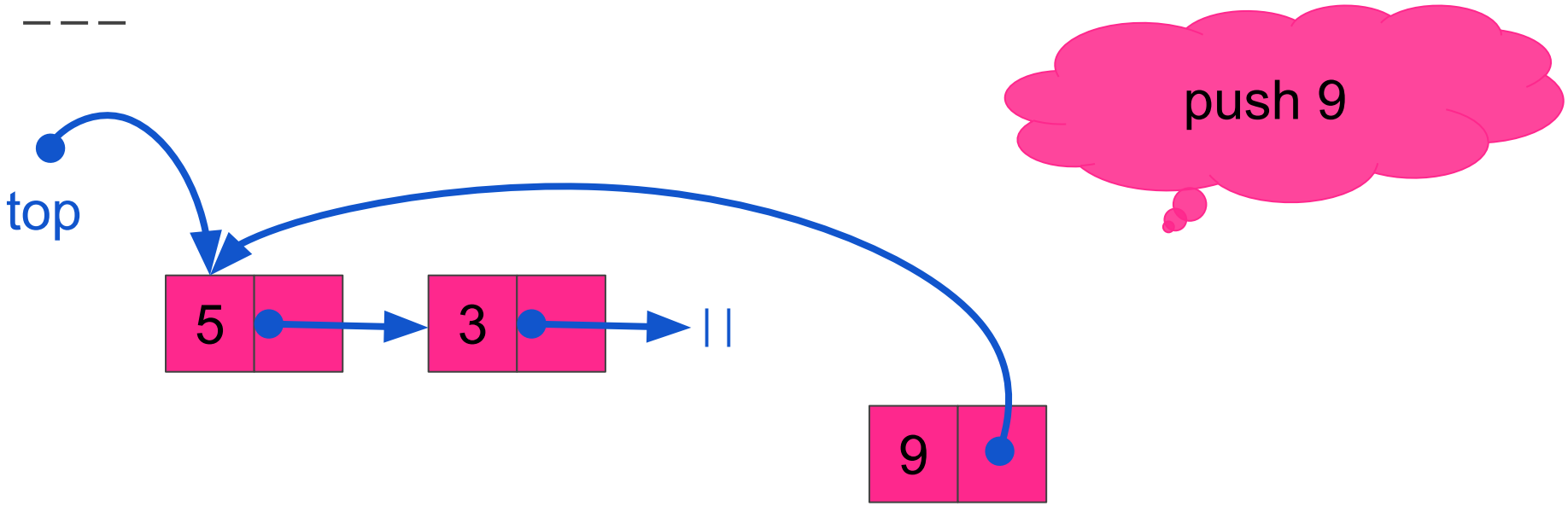




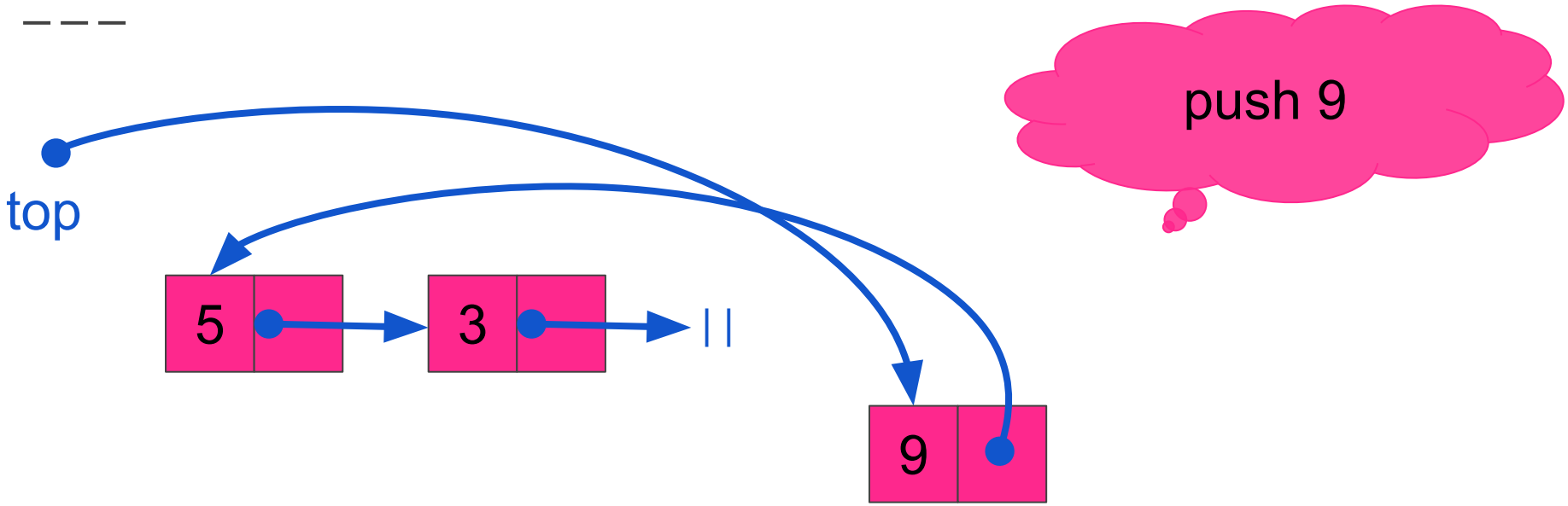
singly-linked list implementation



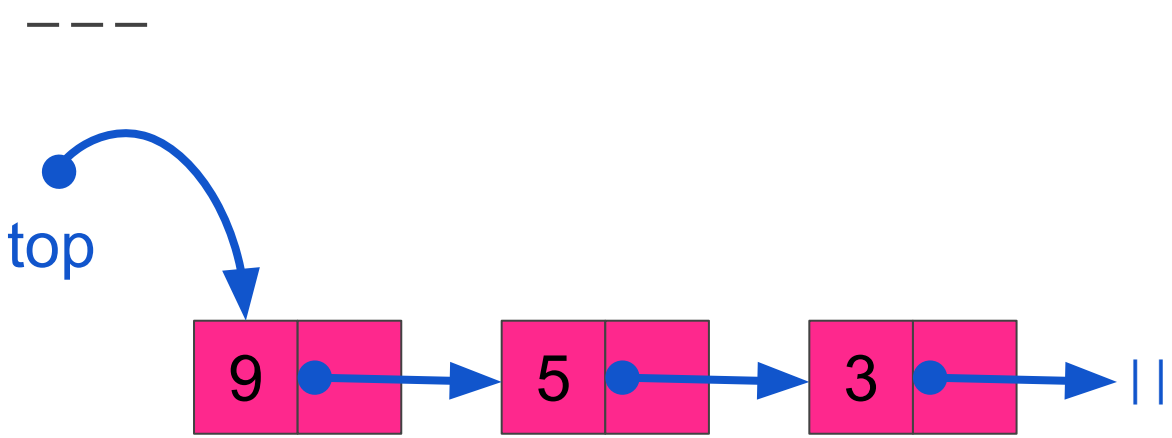
singly-linked list implementation

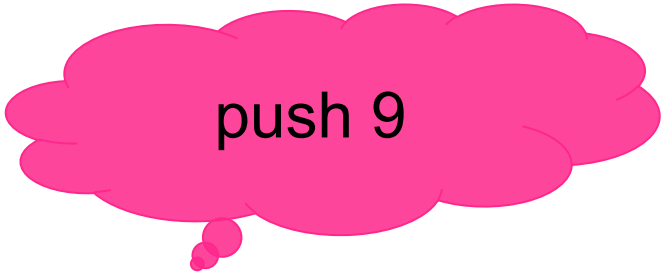


singly-linked list implementation



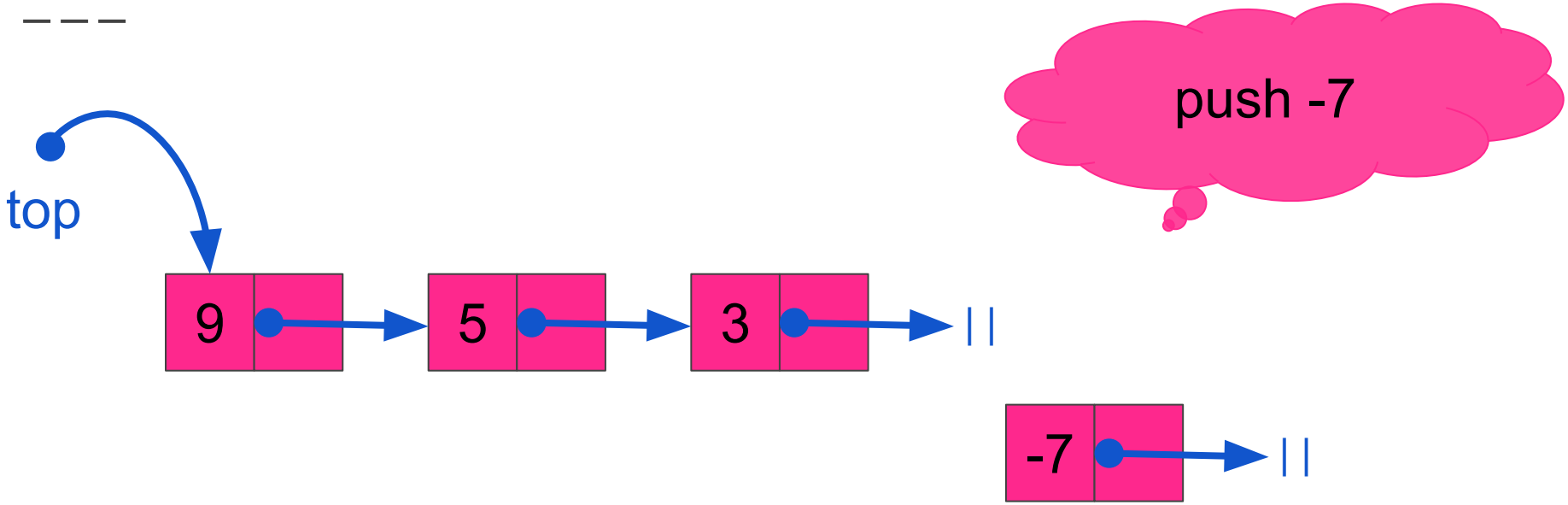
singly-linked list implementation



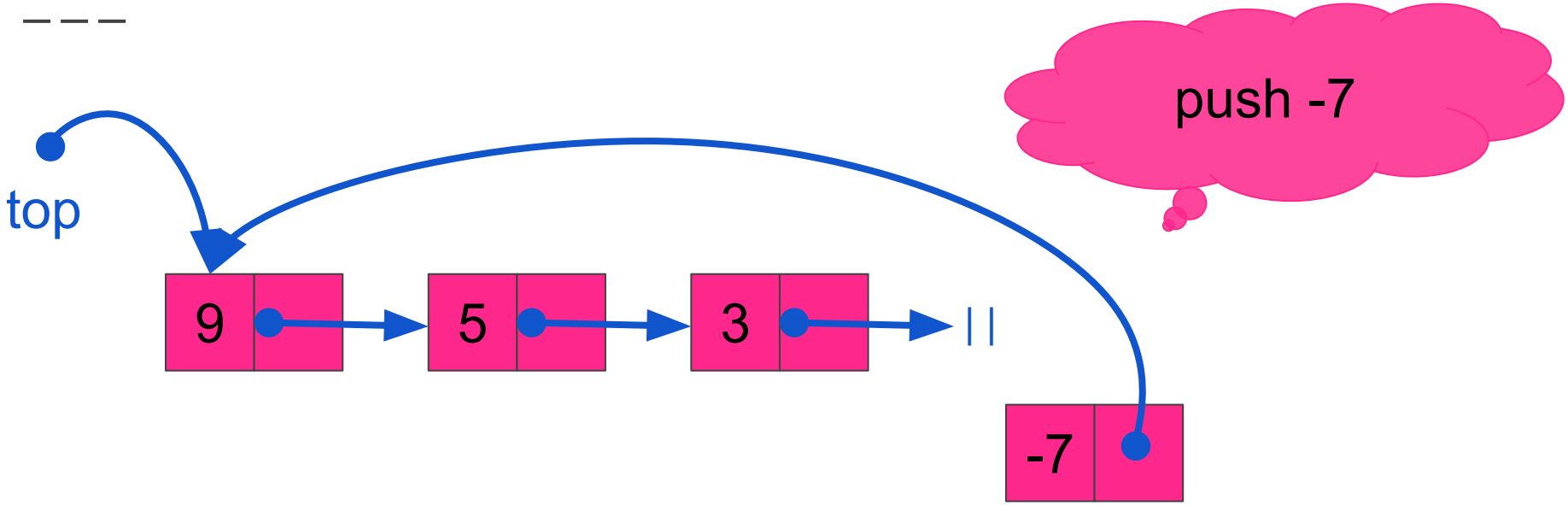




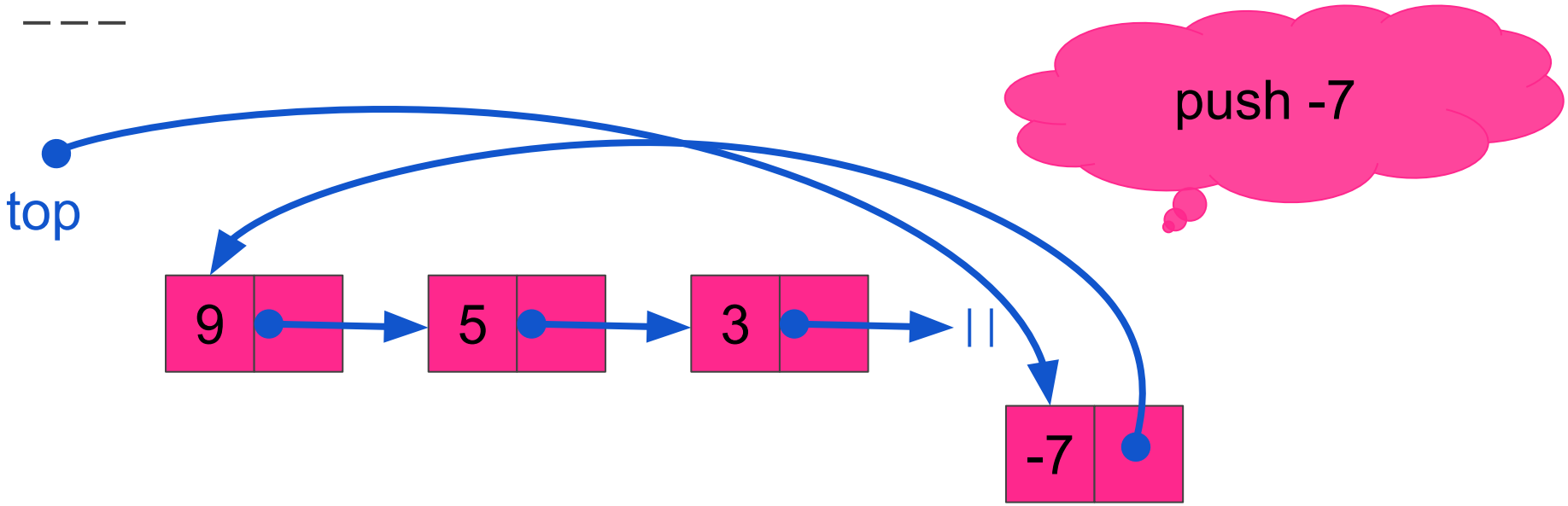
singly-linked list implementation



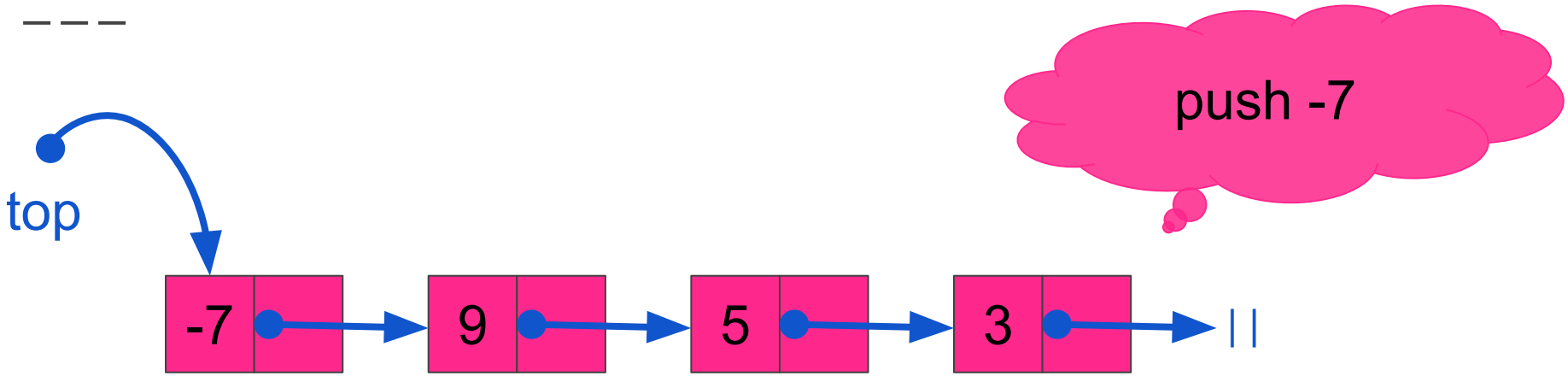
singly-linked list implementation



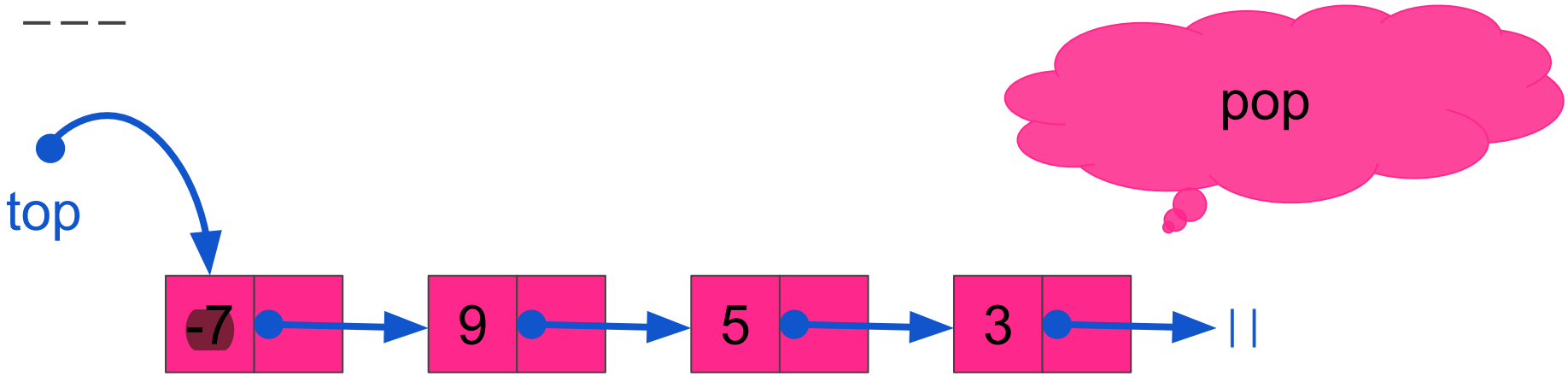
singly-linked list implementation



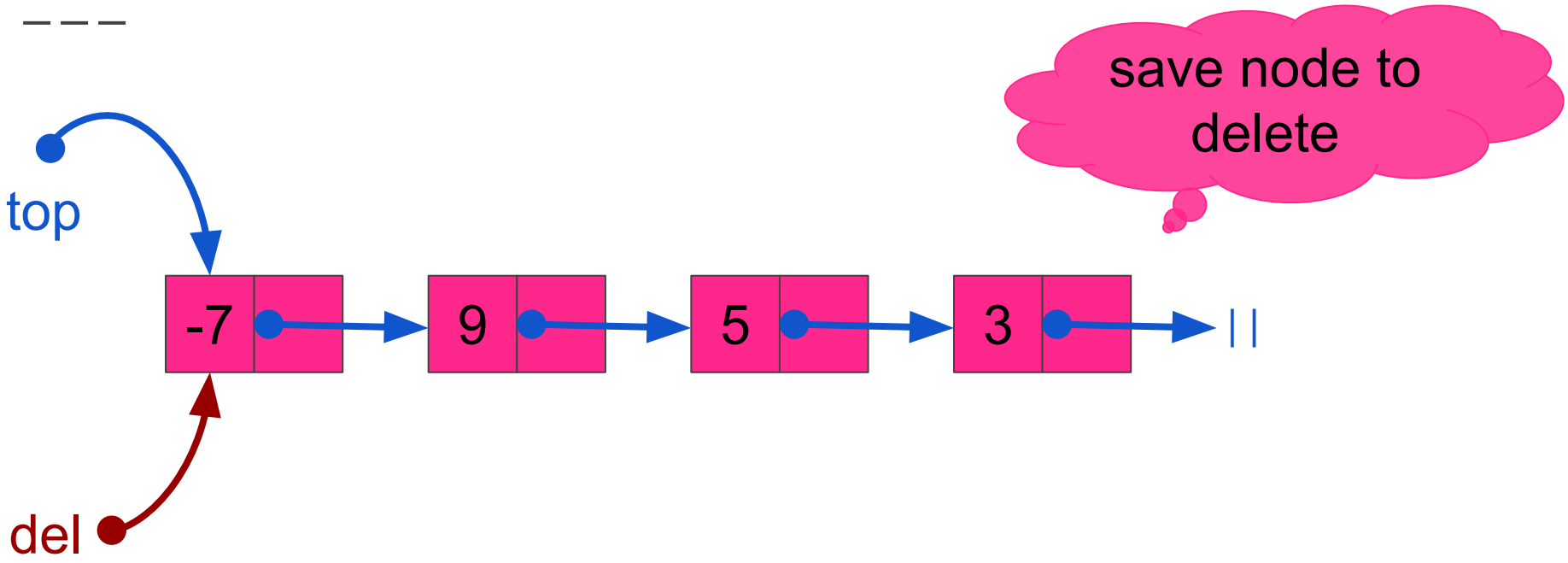
singly-linked list implementation



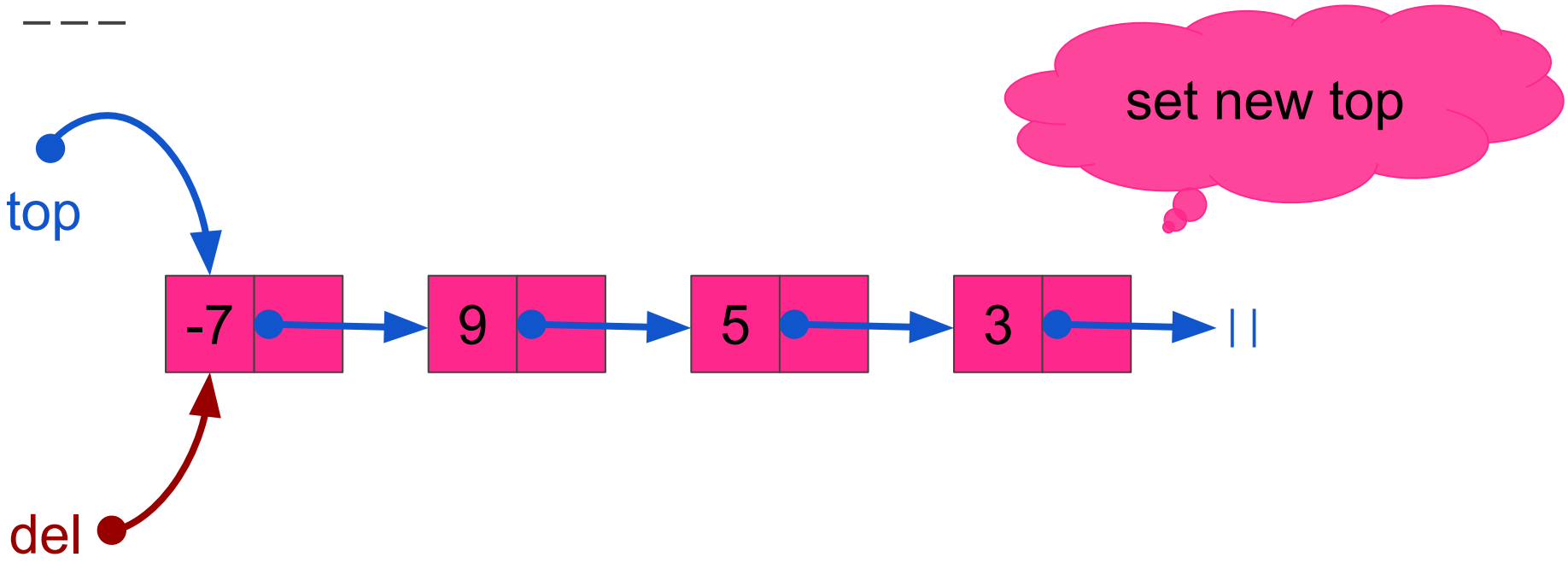
singly-linked list implementation



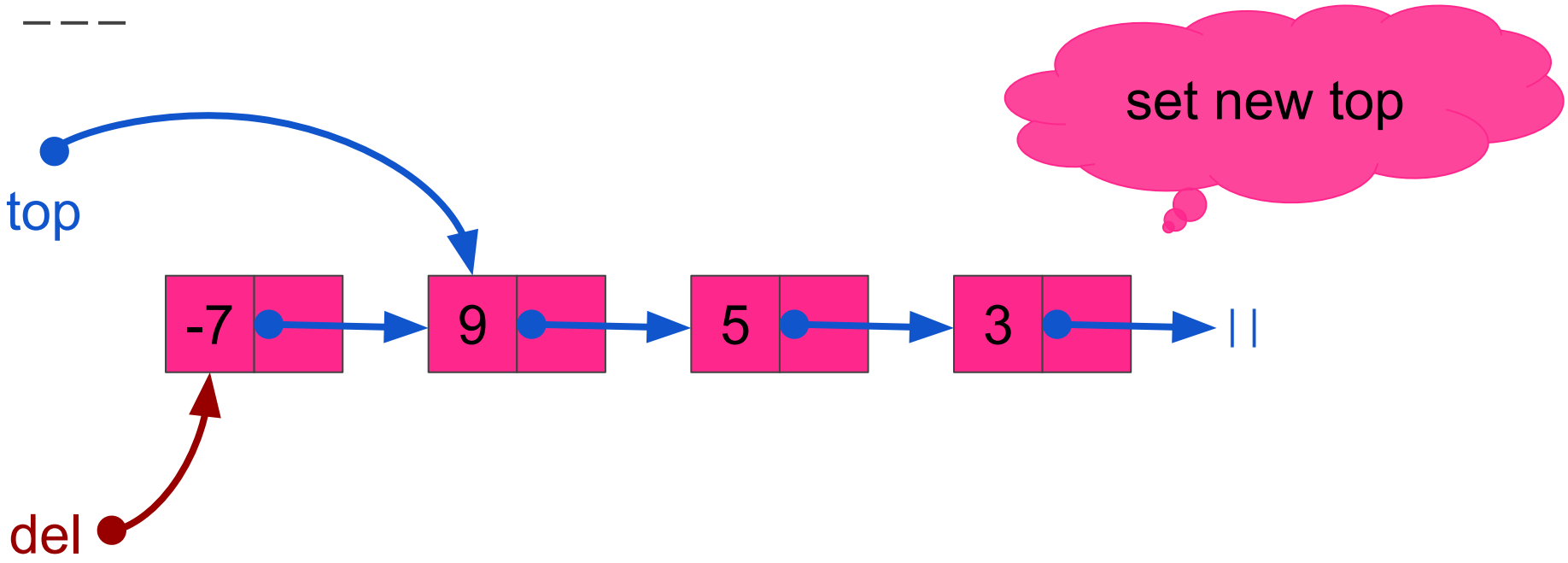
singly-linked list implementation - pop



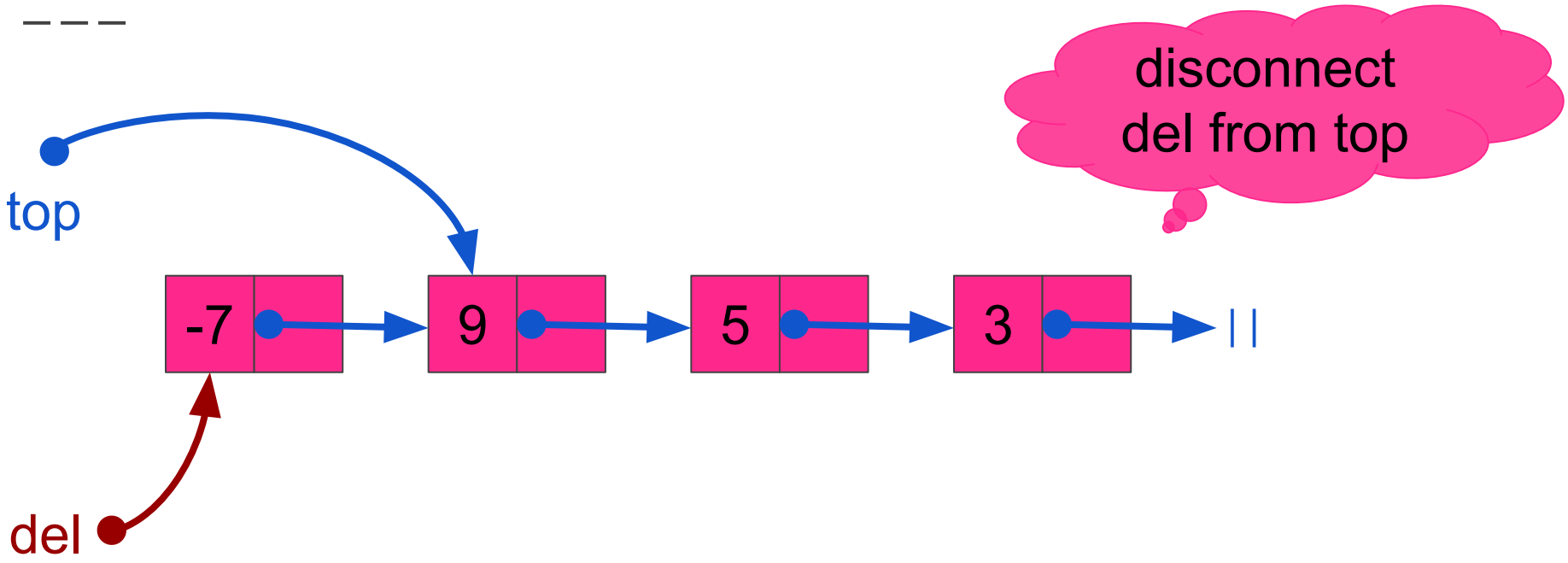
singly-linked list implementation - pop



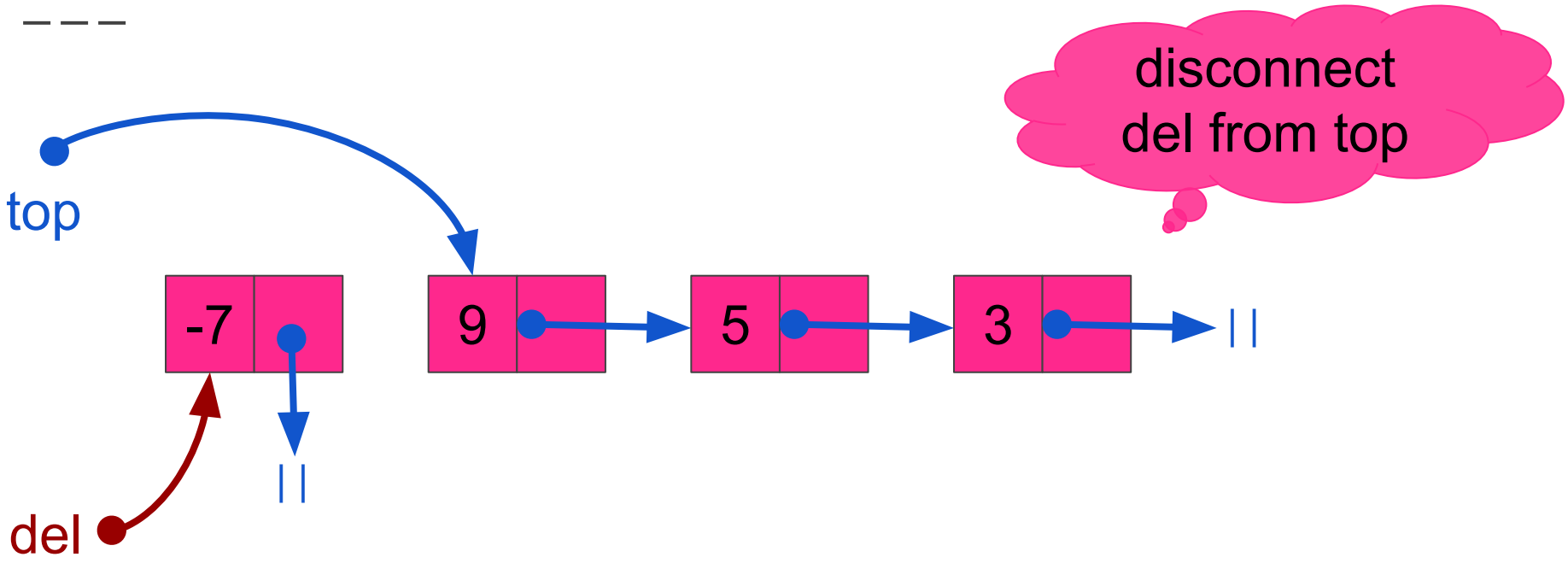
singly-linked list implementation - pop



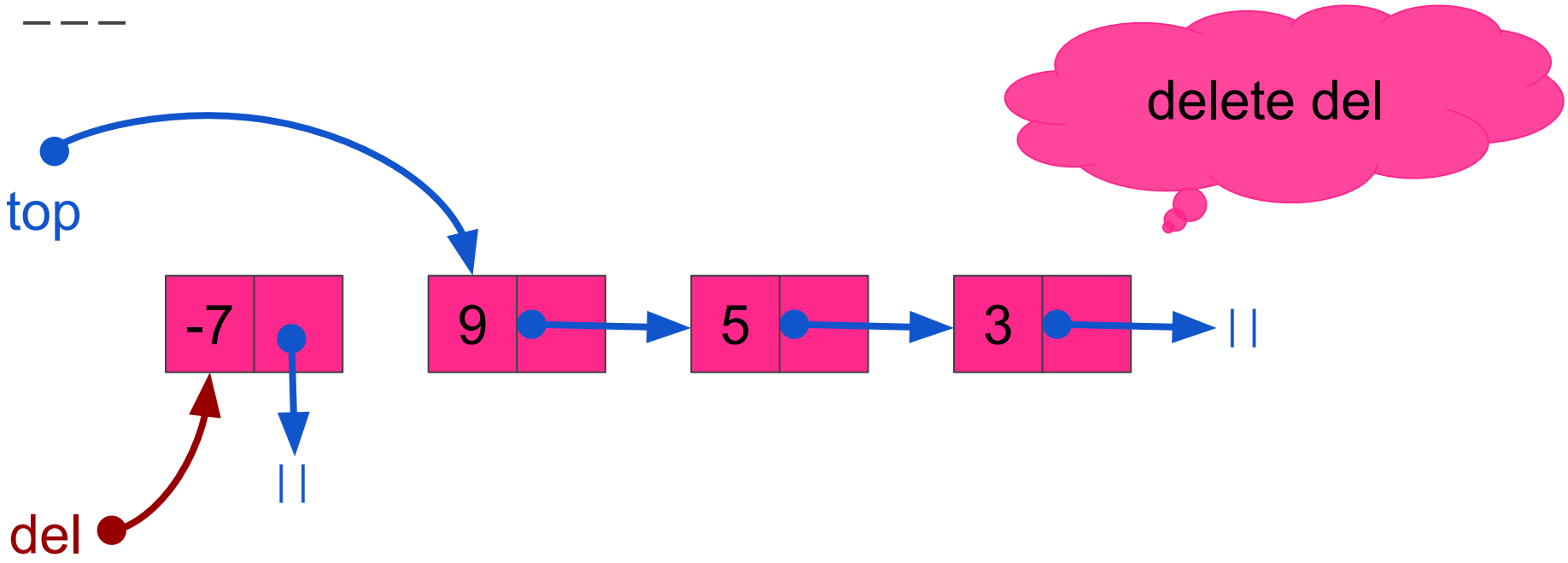
singly-linked list implementation - pop



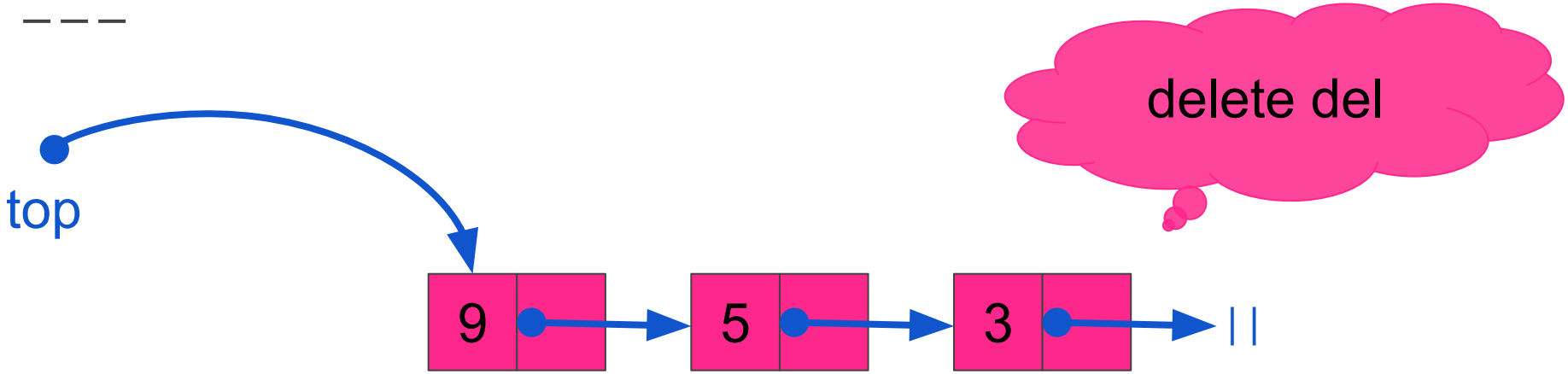
singly-linked list implementation - pop



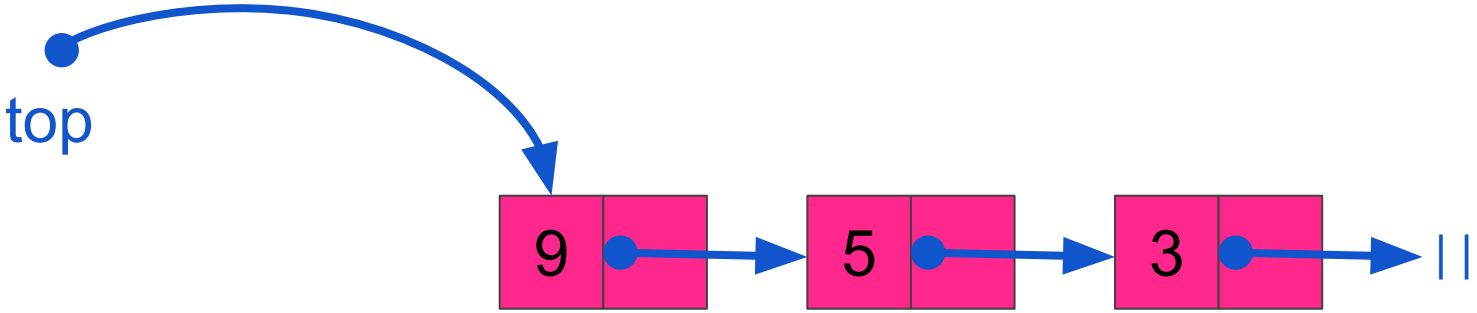
singly-linked list implementation - pop



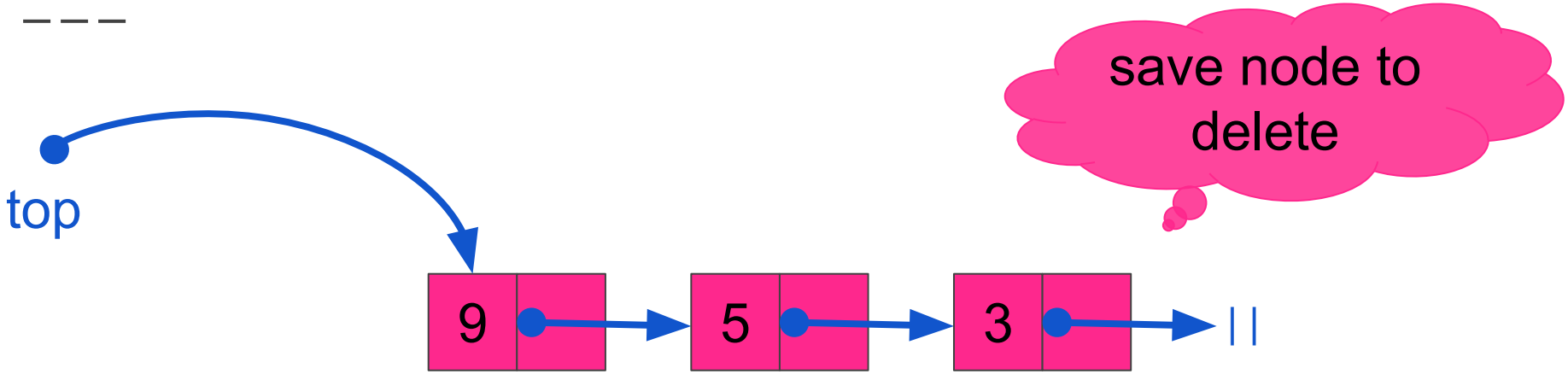
singly-linked list implementation - pop



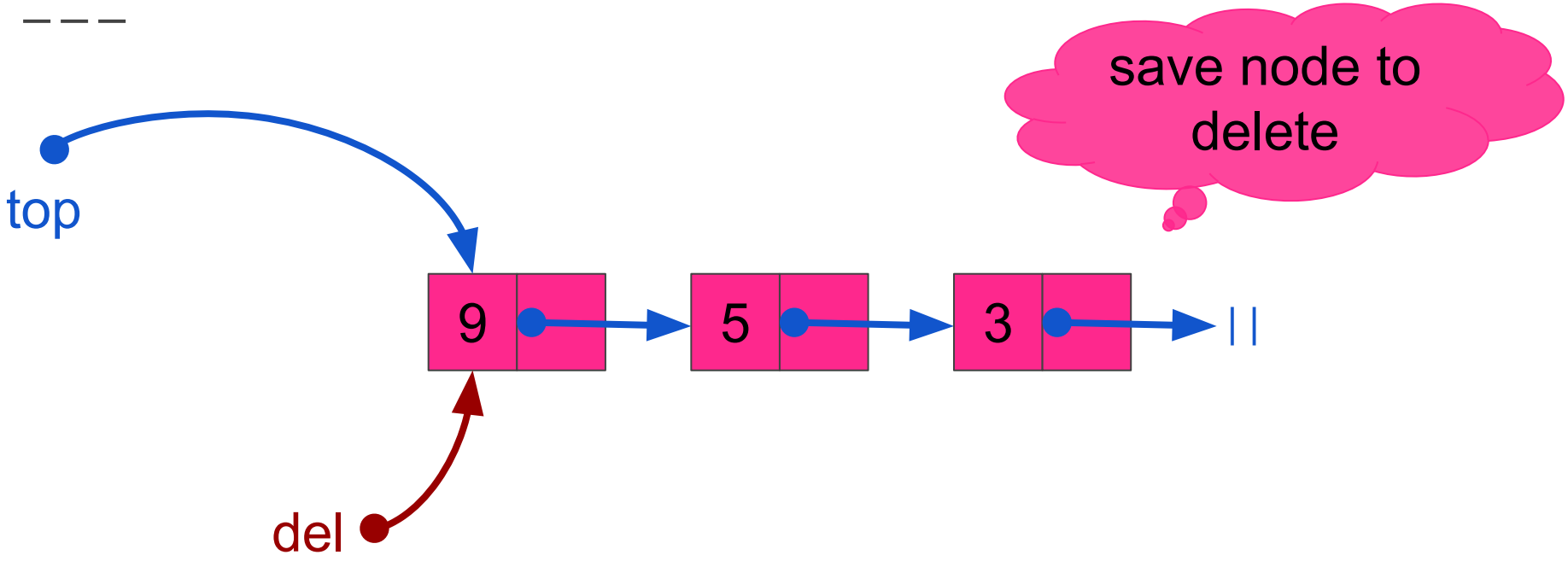
singly-linked list implementation - pop one more time



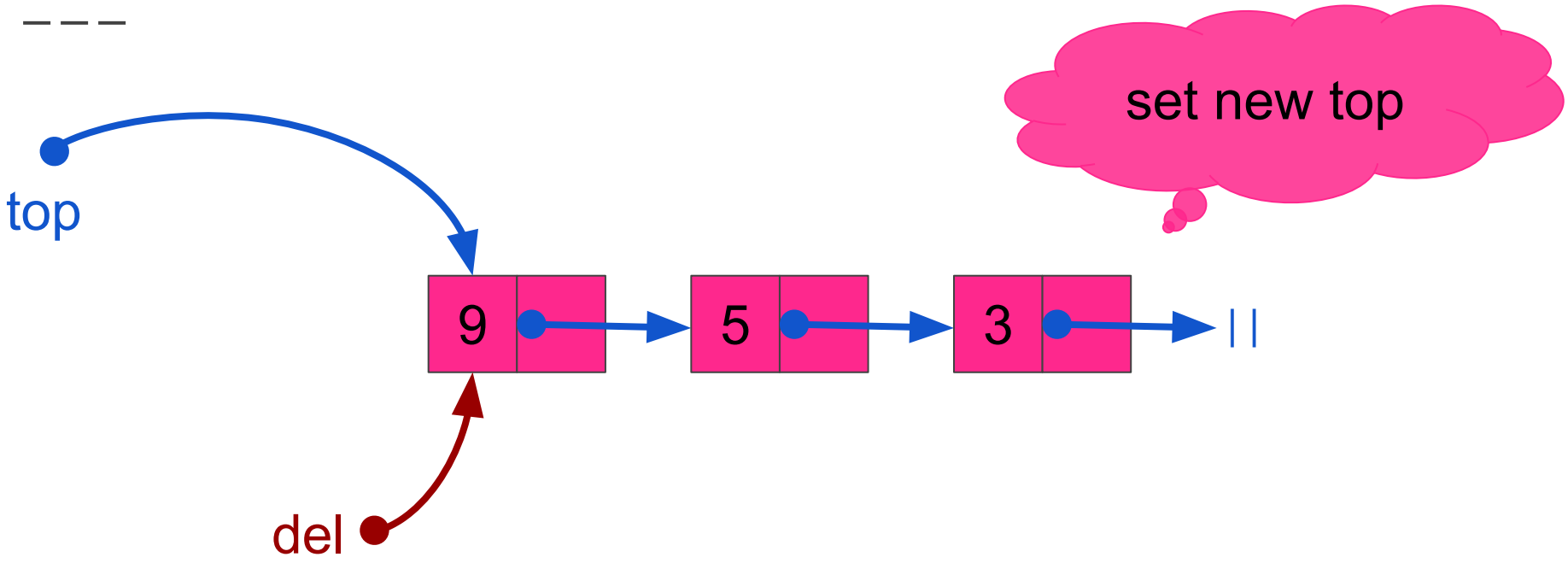
singly-linked list implementation - pop one more time



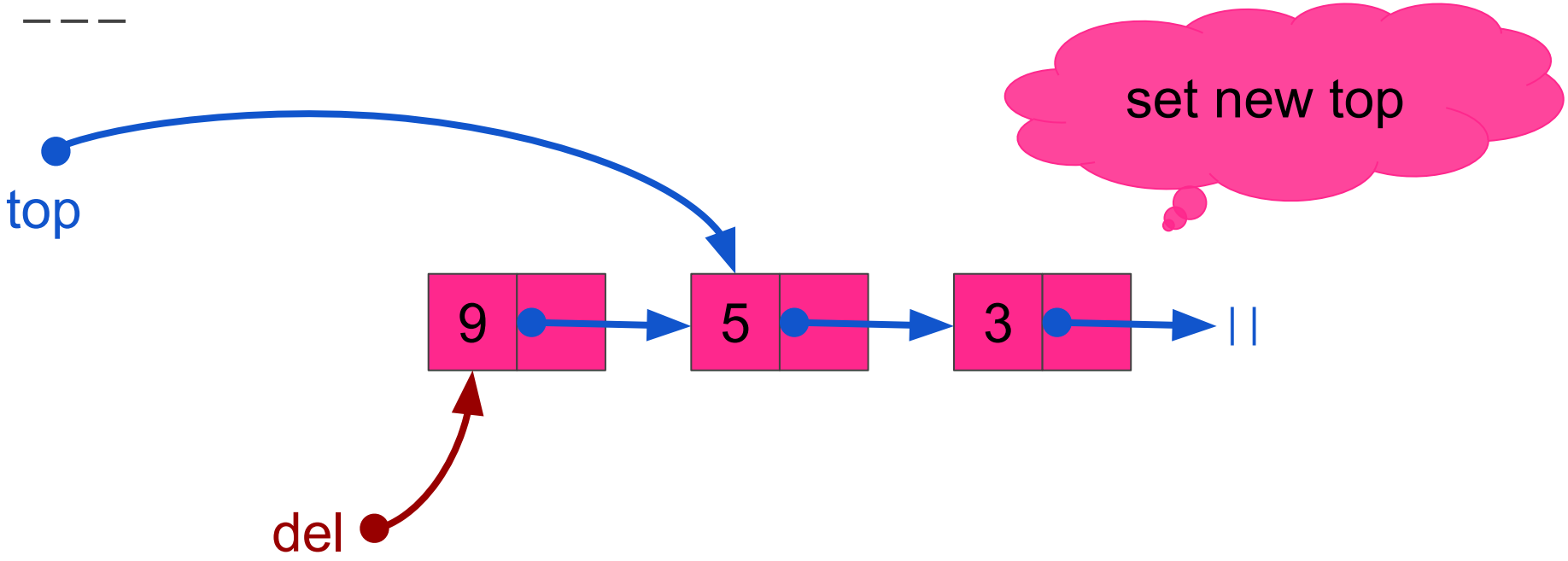
singly-linked list implementation - pop one more time



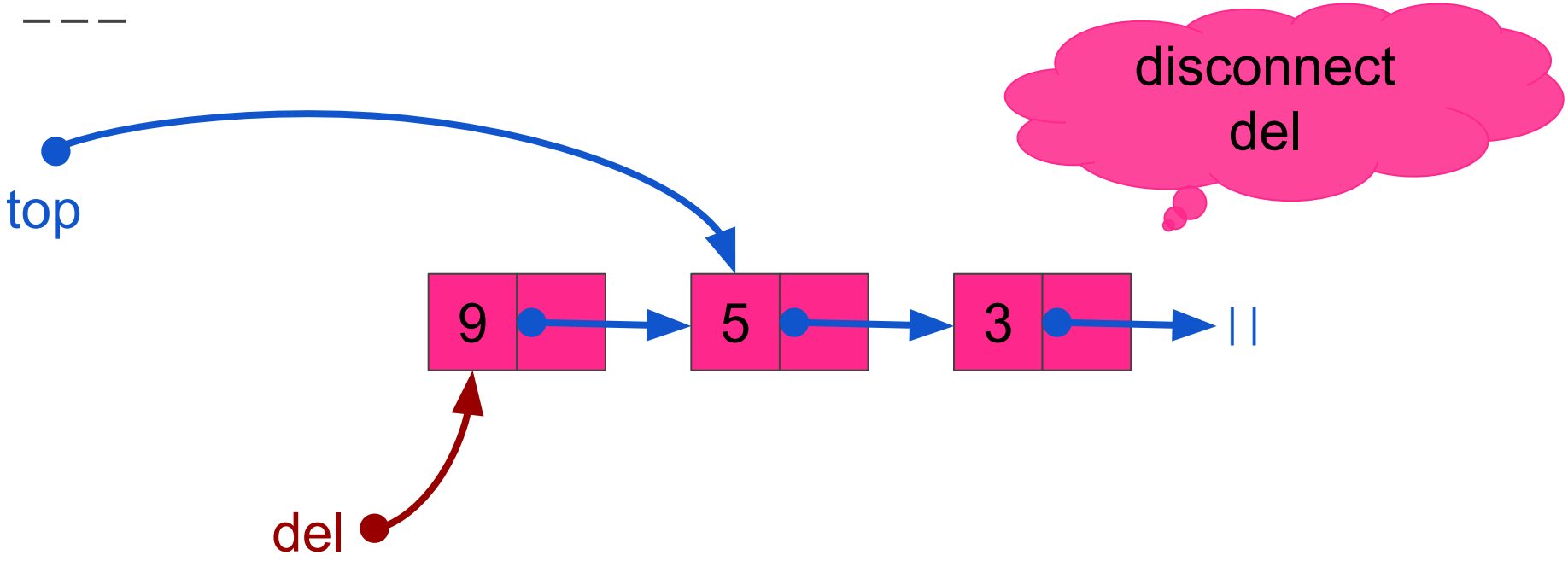
singly-linked list implementation - pop



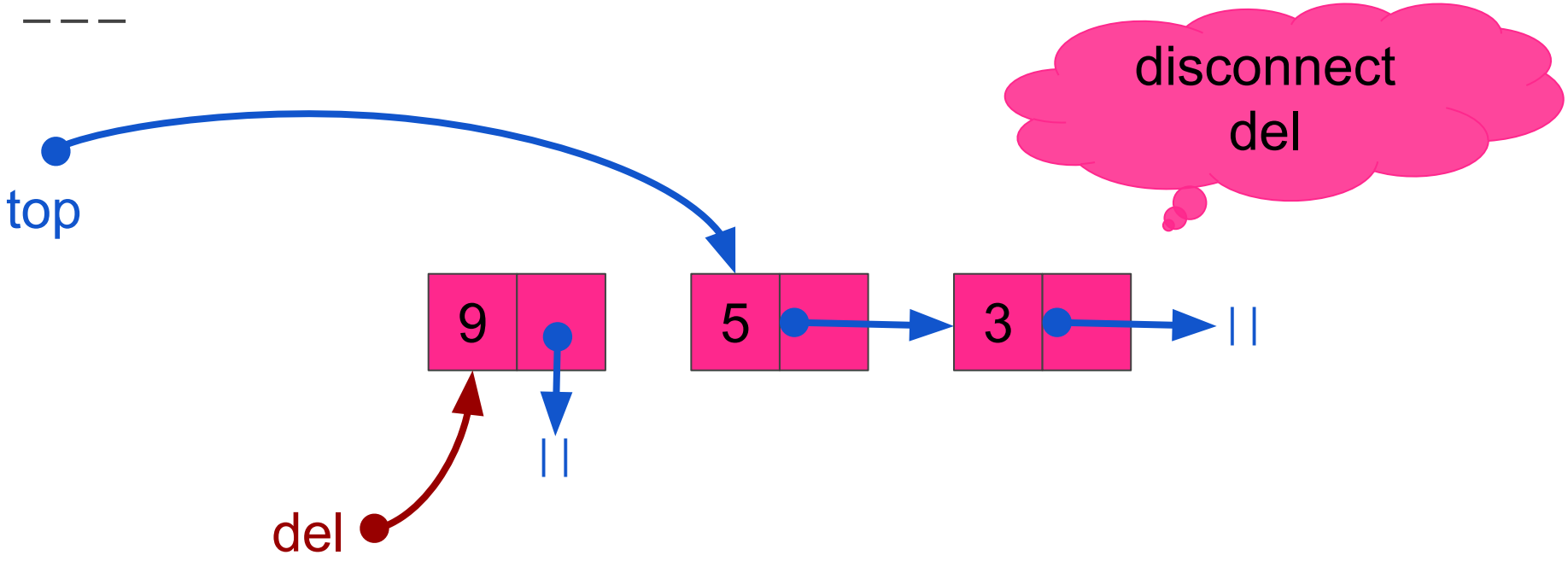
singly-linked list implementation - pop



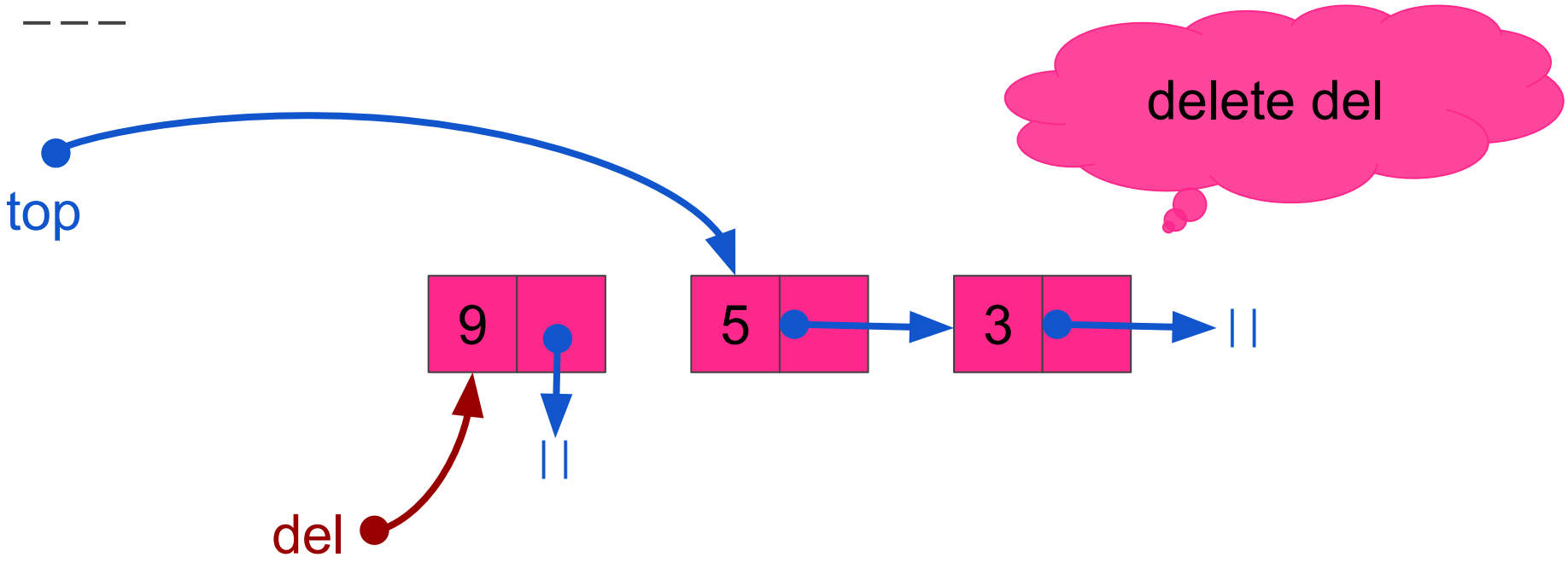
singly-linked list implementation - pop



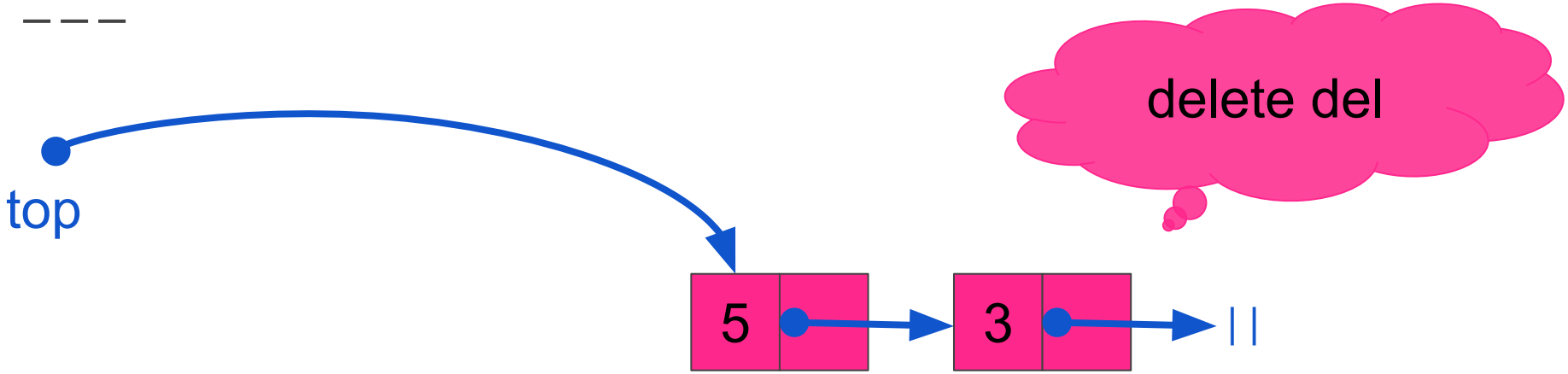
singly-linked list implementation - pop



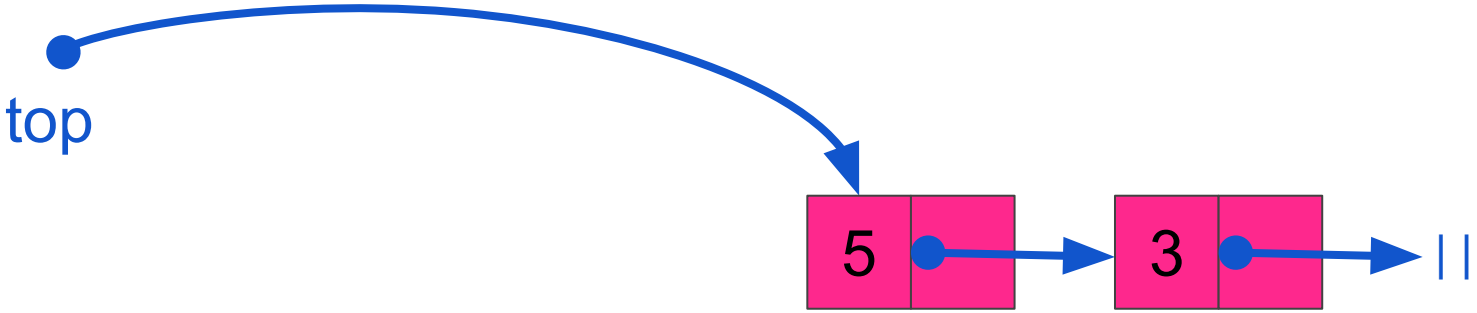
singly-linked list implementation - pop



singly-linked list implementation - pop



singly-linked list implementation



doubly-linked list implementation

– – –



operation	array	singly-linked	doubly-linked
push	O(1)	O(1)	O(1)
pop	O(1)	O(1)	O(1)
top	O(1)	O(1)	O(1)

